Sharp Poisson approximation and second-order correction for long multiplicative runs

Technical companion

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Abstract

This technical companion accompanies the article *Sharp Poisson approximation and second-order correction for long multiplicative runs*. Together the two documents give the proof of the optimal quadratic Poisson–Charlier error power 4/3, the third factorial cumulant power 41/30, and the hierarchy of explicit rational-channel approximations for critical runs of a completely multiplicative random sign function.

The companion contains the detailed arithmetic counts, fixed-source estimates, short-selection moment bounds, algebraic constructions, and conditional estimates used by the article. It also retains the regular-set reconstruction needed for unsigned positional comparison, and the further results on offsets, local blocks, effective tails, Palm identities, and deletion. Fixed-order and growing-range estimates have separate hypotheses throughout. The article states the technical interfaces and retains the main reconstruction and optimality arguments; the complete proof uses both documents.

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Scope and reading guide

This companion and the associated article, *Sharp Poisson approximation and second-order correction for long multiplicative runs*, form a single mathematical work. They share the model, notation, version number, and date. The article follows the exact-marginal proof of the optimal quadratic approximation. This companion gives the detailed proofs of the technical inputs and retains the further results on arithmetic geometry, tails, and deletion.

References beginning with T belong to this companion. Other numbered references belong to the article. The two PDFs have cross-document links when kept together under their supplied filenames. Each cited statement also has a stable printed number, so the proof can be checked without those links.

Common model and conventions. The completely multiplicative function f has independent fair signs at primes. Both signs of a constant run are counted. With

$$I_N = [N, 2N) \cap \mathbb{Z}, \quad |L - \log_2 N| \le C, \quad B = L + 1, \quad p = 2^{-L}, \quad \lambda_N = Np,$$

the event at $x \in I_N$ and its count are

$$J_x = \mathbf{1}_{\{f(x-1)=-f(x),\ f(x)=\cdots=f(x+L-1)\}}, \qquad Z_N = \sum_{x \in I_N} J_x.$$

We write $p_x = \mathbb{E}J_x$, $\mu_N = \mathbb{E}Z_N$, $\alpha_N = \operatorname{Var} Z_N - \mu_N$, $\Pi_\mu = \operatorname{Pois}(\mu)$, and $\Delta = \delta_1 - \delta_0$. Products of measures are convolutions, $\|\nu\|_1 = \sum_k |\nu\{k\}|$, and total variation for probabilities is half this norm. All unmarked logarithms are natural. Constants may depend on the fixed critical-band width C and on parameters explicitly declared fixed.

The space S_x is the image, under prime valuations modulo two, of the even selections in $V_x = \{x-1, \dots, x+L-1\}$. The exact formulation uses all x with $p_x > 0$, dimensions $d_x = \dim S_x$, and the relative nullity $\sum_{x \in I} d_x - \dim \sum_{x \in I} S_x$. The regular-set argument restricts to G_N, where $d_x = L$ and $p_x = p$. Its homogeneous notation and its restricted count are always identified when that argument is used. The raw arithmetic nullity $\rho(I) = |I|L - \dim \sum_{x \in I} S_x$ includes local kernels; it agrees with the effective nullity on G_N. Section 15 explicitly uses ρ as a local abbreviation for the effective nullity.

Technical inputs for the optimal scalar theorem. The arithmetic estimates in Sections T1 and T2 supply the split-product bounds, variable-coefficient counts, and channel budgets. The proofs associated with Sections 6, 9, and 10 establish fixed-source propagation and conditional half counts. The rigidity and triple-counting proofs then supply the inputs to Sections 13 and 14. The article states the required hypotheses at each interface and retains the reconstruction, its separately uniform tail, and the quartic lower-bound argument.

Entering a proof and returning to the article. The following local routes lead to the complete proof subsection, then to the next use in the article. A subsection and a lemma can have the same printed number; the middle column names the subsection and links to its heading.

Entry in the article	Proof in this companion	Return to the article
Proposition 9.1	Subsection T5.1: uniform moments, using the prescribed-divisor estimates above it	Corollary 9.2: short projections into the old space
Theorem 10.1	Subsection T6.3: threshold iteration, using the certificate count of Lemma T6.3	Corollary 10.2: connected hosts at fixed order
Theorem 13.2	Subsection T9.5: one curve for the whole relation space	Proposition 13.3: integer fibres, followed by the degree-uniform point count
Proposition 14.3	Subsection T10.1: all triples above the fixed channels, including exceptional pairs	Corollary 14.4: strong aligned pairs
Lemma 15.2	Subsection T11.1: two dense targets over prescribed sources	Lemma 15.3, then the primitive-mass estimates

The regular-set reconstruction and positional results. The complete earlier reconstruction is retained in Sections T14–T17 and the associated stopping proofs. Its weighted primitive and compressed-pair estimates support the unsigned positional comparison and the deletion arguments in Section T19.5. This branch is needed for those results; the scalar optimal proof is assembled directly from its stated exact-marginal inputs. Any proof below that invokes Theorem 1.1 is a consequence of that theorem, not an input to it.

Further consequences and limitations. Sections T20, T21, and T22 give refinements of the offset geometry, interval dependence, and uniform counting estimates. Section T19 treats exact local blocks, effective tails, modified counts, Palm identities, and the cost of positive forcing. Section T23 explains why fixed moments alone cannot imply a polynomial rate for a complete law. These statements retain their own quantifiers; none enlarges a fixed-order estimate to a growing range without a separate proof.

Which final sections enter an article result? The dependency map in Section 1.5 identifies the independent fixed-moment proof, the triple bound, the reconstruction, and the quartic lower bound. Within the final five sections of this companion, the roles are as follows.

Part of the companion	Role in the associated work
Positional comparison in Section T19.5	Supplies the proof of Lemma 20.1 in the article. It uses the regular-set weighted reconstruction; the scalar optimal theorem does not use it.
Other parts of Section T19	Local-block, tail, deletion, Palm, and forcing results. They are additional conclusions with their own assumptions; they are not inputs to the scalar optimal theorem.
Sections T20, T21, T22, and T23	Further geometry, interval dependence, counting refinements, and a moment obstruction. The article cites selected statements to explain scope; its scalar reconstruction does not depend on them.

Here “additional” refers to the logical role of a result, not to an assertion that it forms a separate paper. The quantized mean identity in Proposition 17.3 is proved directly from the exact marginals; it does not require the supplementary local-block asymptotics.

The source archive provides the single command `python3 check_all.py` for both manuscript builds, cross-reference checks, the French revision note, and the listed finite algebra diagnostics. The README specifies the identities and finite ranges checked. These computations do not test the uniform asymptotic estimates.

T1 Uniform arithmetic estimates

This section supplies the uniformity needed when the window length grows. The degree parameters in the determinant arguments are fixed before N tends to infinity. Dependence on such fixed parameters is permitted; dependence on a squarefree coefficient of size $N^{O(1)}$ is not.

T1.1 Split products, defects, and the regular set

Products in short intervals have a long history; see Erdős and Turk [12]. The polynomial square-root argument for a fixed even number of distinct factors appears in the proof of Munsch, Shparlinski, Sun, and Xiao [17, Lemma 3.3]. We record an explicit bound uniform in the degree, as required when the number of selected factors grows. No constant depending implicitly on that degree is used in the following lemma.

Lemma T1.1 (A split-product bound uniform in the degree)**.** *Let $d = 2k \geq 2$ and let $\gamma_1, \dots, \gamma_d$ be distinct integers with $|\gamma_i| \leq R$, $R \geq 1$. If $U \geq 2R$ is an integer and $\prod_i (U + \gamma_i)$ is a square, then*

$$U \leq (64R)^d.$$

Proof. Write $G(T) = \prod_i (T + \gamma_i)$ and

$$\phi(z) = \prod_i (1 + \gamma_i z)^{1/2} = \sum_{j \geq 0} c_j z^j, \qquad P(T) = \sum_{j=0}^{k} c_j T^{k-j}.$$

The branches equal one at the origin. Cauchy's bound on $|z| = (2R)^{-1}$ gives $|c_j| \leq (3/2)^{d/2}(2R)^j$ and, for $j \leq k$, $|c_j| \leq 2^d R^j$. Comparing coefficients in ϕ^2 inductively gives $2^{2j-1} c_j \in \mathbb{Z}$ for $j \geq 1$, so $2^d P \in \mathbb{Z}[T]$. For $U > 4R$ the tail satisfies

$$|\sqrt{G(U)} - P(U)| \leq 4\, 3^{d/2} R^{k+1}/U.$$

If this difference is nonzero, it is at least 2^{-d}, whence $U \leq (8R)^d$. If it vanishes, the nonzero integer polynomial $Q = 2^{2d}(P^2 - G)$ has root U. It is nonzero because G has simple roots. Keeping the homogeneous powers of R in the coefficient products gives

$$|[T^{d-j}](P^2 - G)| \leq (d+2) 4^d R^j \leq 8^d R^j, \qquad H(Q) \leq (32R)^d.$$

The integral-root bound gives $U \leq 1 + H(Q) \leq (64R)^d$. The range $2R \leq U \leq 4R$ already satisfies the conclusion. □

The following coefficient-uniform count is classical; see Cilleruelo and Garaev [28, Section 3, Lemma 4 and Proposition 1] and the directly comparable formulation of Tao [29, Lemma 2.10]. When ab is not a square, the substitution $(au)^2 - abv^2 = ah$ reduces it to the former lemma; the square case is a divisor count. We include a proof to keep the polynomial coefficient range explicit.

Lemma T1.2 (Uniform quadratic equation count)**.** *Fix $A_0 > 0$. Uniformly for positive integers a, b and nonzero integers h with $a, b, |h| \leq N^{A_0}$, the number of positive solutions to*

$$au^2 - bv^2 = h, \qquad u, v \leq N^{A_0},$$

is $N^{o_{A_0}(1)}$.

Proof. If ab is a square, multiply the equation by a and factor the resulting difference of squares; the divisor bound suffices. Otherwise write $ab = Ds^2$ with D squarefree and nonsquare. Each solution gives the algebraic integer $\xi = au + sv\sqrt{D}$ of norm ah in the maximal order of $\mathbb{Q}(\sqrt{D})$.

Its principal ideal has norm $|ah|$. The number of ideals of this norm is at most $\tau(|ah|)$: at a split prime there are at most $e+1$ ideals of norm ℓ^e, and at an inert or ramified prime there are no more. Elements generating the same ideal differ by a unit. A real quadratic unit greater than one is at least $(1+\sqrt{5})/2$, as follows from its integral trace and norm ± 1. The positive values of ξ arising here lie in $[1, N^{O(A_0)}]$; there are therefore $O_{A_0}(\log N)$ possibilities in each ideal class of generators. Finally $\tau(|ah|) = N^{o_{A_0}(1)}$. This argument makes the coefficient uniformity explicit and requires no bound for a fundamental unit or a class number. □

Lemma T1.3 (Local and global defect counts)**.** *Suppose $B \asymp \log N$ and $H \asymp B$, with fixed comparison constants. Any interval of diameter $O(B)$ in $[cN, CN]$ contains $O(B/\log B)$ H-defective integers. Moreover*

$$\#\{n \le CN : K_H(n) = 1\} \le N^{1/2+o(1)}, \tag{T1.1}$$

$$\#\{x \in I_N : V_x \text{ contains two } H\text{-defective integers}\} \le N^{o(1)}. \tag{T1.2}$$

All statements remain valid after fixed dilations and translations of the intervals.

Proof. For D defective integers in a short interval, form the binary matrix with their small-prime valuation columns and an additional parity row. Its kernel has redundancy at most $\pi(H)+1$. A nonzero word selects an even square product, so Lemma T1.1 gives minimum distance $\delta \gg \log N/\log B$. Choose $t \asymp \log N/\log B$ with $2t < \delta$. The Hamming bound gives

$$\sum_{j=0}^{t} \binom{D}{j} \le 2^{\pi(H)+1}.$$

If $D \ge 2t$, use $\binom{D}{t} \ge (D/t)^t$. Since $(\pi(H)+1)/t = O(1)$, this proves $D = O(B/\log B)$; the other case is immediate.

Write a defective integer uniquely as $n = a^2 c$, where c is squarefree and H-smooth. Then

$$\#\{n \le CN : K_H(n) = 1\} \le \sqrt{CN} \prod_{\ell \le H} (1 + \ell^{-1/2}) \le \sqrt{CN}\exp(O(\sqrt{H})).$$

For two distinct defective values in one window, write $x + i = c_1 a^2$, $x + j = c_2 b^2$. The offsets and the squarefree smooth parts have $B^2 4^{\pi(H)} = N^{o(1)}$ choices. All their heights are polynomial in N, and $c_1 a^2 - c_2 b^2 = i - j \ne 0$. Lemma T1.2 supplies $N^{o(1)}$ possibilities. This proves (T1.2) and the stated uniform versions. □

T1.2 Two small kernels with their smooth coefficients retained

The following form of the approximate determinant method is needed because the small-prime parts are fixed separately and may be polynomially large. It is an extension of the proof of [23, Sections 2.1–2.3], rather than an application of its fixed-coefficient statement.

Lemma T1.4 (Uniform determinant estimate)**.** *Fix $A_0, \delta, \varepsilon > 0$. Suppose $1 \le c_1, c_2 \le 3N$, $0 < |h| \le (\log N)^{A_0}$, and*

$$c_1 D^2 U \asymp c_2 E^2 V \asymp N, \quad D, E \ll \sqrt{N}, \quad c_1 c_2 (DE)^2 \ge N, \quad \log(DE), \log(UV) \ge \delta \log N.$$

The comparison constants are fixed. The number of positive solutions $d \asymp D, e \asymp E, u \asymp U, v \asymp V$ to $c_2 e^2 v - c_1 d^2 u = h$ is

$$\ll_{A_0,\delta,\varepsilon} N^{\varepsilon} \min\{(DE\mathcal{M})^{1/2} + D + E, (UV\mathcal{M})^{1/2} + U + V\}, \qquad \log \mathcal{M} = \frac{9}{8}\frac{\log(DE)\log(UV)}{\log N}. \tag{T1.3}$$

Fixed or polylogarithmic factors in the lower bound for $c_1 c_2 (DE)^2$ can be absorbed by decreasing the error margin.

Parameters in the determinant argument. Here and in the proof, height of a polynomial means the maximum absolute value of its integer coefficients. The ledger below distinguishes the fixed auxiliary degrees from the coefficients that vary with N. The symbols A, B_0, κ, L_1, T_1 are specified in Steps 2–3.

Object and dimension	Degree	Point or box scale	Coefficient height
Input hypersurface in $\mathbb{A}^4$, dimension 3	Total degree 3	(D, E, U, V), each $O(N)$	$c_1, c_2 \le 3N$; $\lvert h\rvert \le (\log N)^{A_0}$
Auxiliary curve in $\mathbb{P}^1 \times \mathbb{P}^1$, dimension 1	Bidegree $(a, b) \le (A, B_0)$, fixed	Primitive pairs of heights $L_1, T_1 \le N^C$	$\lVert G\rVert \le N^\kappa$ after the lattice changes
Eliminant curve in $\mathbb{A}^2$, when $a = 1$ or $b = 1$	Bounded in terms of A, B_0	The same primitive boxes	Resultants and nonzero right-hand sides $N^{O(\kappa)}$

All displayed height exponents, including C, are fixed, uniformly over the input boxes. The relative chart has errors $\alpha = O(M^{-1})$ and $\beta = O(|h|/N)$, with $M = N^\zeta \mathcal{M}$; no coefficient ratio is charged to the latter error. Choose the final ε first, then a sufficiently small ζ, then A, B_0 and κ, and finally the curve-count exponent ϵ' so that $(C + \kappa)\epsilon' \le \zeta$. Only after these choices does N tend to infinity. In particular, the factor $\lVert G\rVert^{\epsilon'}$ in Heath-Brown's bound is explicitly included in this budget. The coefficient uniformity in this ledger is established below; Reuss's stated theorem allows its constant to depend on the fixed shift h and does not itself assert this extension.

Proof. The proof has four steps. They identify separately the determinant scale, the uniform count on each auxiliary curve, and the two lattice summations. Only the last step sums over the short intervals.

Step 1: primitive pairs and relative normalization. If $g = \gcd(d, e)$ and $q = \gcd(u, v)$, then $g^2 q \mid h$. Divide by $g^2 q$ and replace the four dyadic scales accordingly. Both pairs are now primitive. The effective size $N/(g^2 q)$ and the hypotheses on the scales change only by polylogarithmic factors; in particular the lower bound for $c_1 c_2 (DE)^2$ has the permitted polylogarithmic loss. There are only polylogarithmically many reductions, which will be summed at the end. We keep the notation N, D, E, U, V, h for a reduced box.

Set $s = d/e$, $t = v/u$. The exact curve approximation is

$$t = (c_1/c_2)s^2 + h/(c_2 e^2 u).$$

On a relative interval of length M^{-1} write

$$s = s_0(1 + \alpha), \qquad t = t_0(1 + 2\alpha + \alpha^2 + \beta), \qquad t_0 = (c_1/c_2)s_0^2 \asymp V/U.$$

Then $|\alpha| \ll M^{-1}$ and $|\beta| \ll |h|/N$, uniformly in c_1, c_2. The relevant vertical error is relative to t_0; its bound therefore contains no additional factor c_1/c_2.

Step 2: the determinant with variable coefficients. For the auxiliary monomials $s^a t^b$, $0 \le a \le A$, $0 \le b \le B_0$, the row denominator is $e^A u^{B_0}$. Extracting $s_0^a t_0^b$ from the columns leaves the same logarithmic denominator cost as in the normalized calculation:

$$A \log E + B_0 \log U + \frac{A}{2} \log(D/E) + \frac{B_0}{2} \log(V/U) = \frac{A}{2} \log(DE) + \frac{B_0}{2} \log(UV).$$

Take $M = N^\zeta \mathcal{M}$, with a small fixed $\zeta > 0$, and use $N/|h|$ for the vertical approximation scale. The strict determinant inequality in [23, Section 2.1, Lemma 7] then holds, with degrees fixed in terms of ζ, δ. The resulting integer polynomial has coefficient height $N^{\kappa(\zeta,\delta)}$. The exponent is uniform in c_1, c_2: their sizes, and the rational column scalings, are polynomial in N, while the degrees are fixed.

To check the allowed scale, put $z = \log(DE)/\log N$ and $w = 1 - \log(c_1c_2)/(2\log N)$. Then

$$\frac{\log(UV)}{\log N} = 2w - 2z + O(1/\log N), \qquad \frac{\log \mathcal{M}}{\log N} = \frac{9}{4}z(w-z) + o(1) \le \frac{9}{16} + o(1).$$

Thus $M \le N/|h|$ for sufficiently small ζ. The two logarithmic size ratios stay in a compact subset of the positive quadrant, so the degree choices are uniform over all boxes. The parameter order is: choose ζ from the final error margin, then the auxiliary degrees and the height exponent κ, and only then the internal exponent in the curve count below.

Step 3: fibres and the count on one auxiliary curve. Apply the two unimodular lattice changes of [23, Section 2.2] to (d, e) and (v, u). Write the resulting primitive pairs as $\lambda = (\lambda_1, \lambda_2)$ and $\tau = (\tau_1, \tau_2)$, in boxes of side lengths $L_1 \ge L_2$ and $T_1 \ge T_2$, respectively. The changes of coordinates have polynomial height; enlarging κ absorbs their contribution to the heights of the original and auxiliary equations. The original equation still has bidegree $(2, 1)$.

An auxiliary factor of degree zero in (d, e) fixes only boundedly many primitive (u, v); Lemma T1.2 counts the remaining quadratic equation uniformly. For degree zero in (u, v), fix (d, e) and set $g_f = \gcd(c_1d^2, c_2e^2)$. Solvability requires $g_f \mid h$. The step of the resulting one-dimensional lattice in u is c_2e^2/g_f; the step in v is c_1d^2/g_f. Its intersection with the box has at most

$$O\left(1 + \frac{Ng_f}{c_1c_2(DE)^2}\right) \le O\left(1 + \frac{N|h|}{c_1c_2(DE)^2}\right)$$

points. This is polylogarithmic by the stated lower bound.

For an absolutely irreducible factor of bidegree (a, b), $a, b \ge 1$, [14, Lemma 2] bounds primitive solutions by $O_{a,b,\epsilon'}(X^{2/b+\epsilon'}\|G\|^{\epsilon'})$ when the first pair has height at most X; the second pair has no height restriction. Thus $b \ge 2$ gives the required bound in L_1, and $a \ge 2$ gives the bound in T_1 after exchanging the pairs. Choose ϵ' after κ so that the coefficient-height factor costs at most the chosen power $N^{O(\zeta)}$.

The missing bound when one bidegree is one follows directly by elimination. In bidegree $(1, b)$ write $\lambda_1F_1(\tau) + \lambda_2F_2(\tau) = 0$. Primitivity gives $q\lambda_1 = F_2(\tau)$, $q\lambda_2 = -F_1(\tau)$ for a nonzero integer q. The two dehomogenizations of the coprime forms F_1, F_2 give nonzero resultants R, R' of height $N^{O(\kappa)}$ and

$$q \mid R\tau_1^m, \qquad q \mid R'\tau_2^{m'}.$$

Since $(\tau_1, \tau_2) = 1$, prime by prime q divides RR'. There are $N^{o(1)}$ possibilities, uniformly in the coefficients. Substitution into the original equation gives $F_3(\tau_1, \tau_2) = hq^2$, where F_3 is homogeneous of fixed degree $2b+1$. If F_3 is zero there is no solution. Otherwise $F_3 - hq^2$ is a nonzero polynomial. Fixing the first coordinate leaves boundedly many second coordinates, except for boundedly many full fibres, each of length $O(T_2)$. As $T_2 \ll T_1$, the count is $O(T_1)$. This includes powers of a linear form and needs no uniform height-free bound for a Thue equation. Exchanging the pairs handles bidegree $(a, 1)$ with count $O(L_1)$ and right-hand side hq.

These cases prove, with arbitrarily small internal losses, the local bound of [23, Lemma 8] in both orientations. Reducible auxiliary polynomials have boundedly many factors. Rational points on an irreducible factor that is not geometrically integral lie in its bounded-degree conjugate intersections. The resulting local count is therefore uniform in all the variable coefficients.

Step 4: the two short-vector summations. We give the scale check behind [23, Section 2.3]. For the first lattice, use intervals with $s_0 = (D/E)x_3/M$, $x_3 \asymp M$. A shortest primitive vector (x_1, x_2) in a box with $L < L_1 \le 2L$ satisfies

$$x_2x_3 = (ME/D)x_1 + O(E/L), \qquad |x_1| \ll D/L, \quad |x_2| \ll E/L, \quad L \gg (DE/M)^{1/2}.$$

If $x_1x_2 \ne 0$, the integer product x_2x_3 has $O(E/L)$ choices for each of $O(D/L)$ choices of x_1. The divisor bound gives $N^{o(1)}DE/L^2$ intervals. Their local contributions sum to $N^{O(\zeta)}DE/L$;

summing the dyadic L gives $N^{O(\zeta)}(DEM)^{1/2}$. The primitive axis vectors have one coordinate equal to ± 1 and give the additional costs $D+E$.

For the other lattice choose $U = N/(c_1D^2)$ and $V = N/(c_2E^2)$ within the dyadic comparison constants. Then

$$t_0 = (c_1/c_2)(D/E)^2(x_3/M)^2 = (V/U)(x_3/M)^2.$$

Replace this by the grid point $(V/U)x_4/M$, where $x_4 = \lfloor x_3^2/M \rfloor$. The change is within the allowed interval width. Since $x_3 \asymp M$, this discretization has bounded multiplicity and $x_4 \asymp M$. In particular no factor c_1/c_2 remains. The same divisor summation gives $N^{O(\zeta)}\{(UVM)^{1/2}+U+V\}$. Finally take ζ and the internal margins sufficiently small, replace M by $N^{\zeta}\mathcal{M}$, and sum the primitive reductions. This proves (T1.3). □

Lemma T1.5 (Two small odd kernels). *Uniformly for $B \asymp \log N$, fixed $0 < \beta < 1$, and $\varepsilon > 0$,*

$$G_2(N, N^\beta) := \#\{x \in I_N : \#\{n \in V_x : K_B(n) \le N^\beta\} \ge 2\} \ll_{\beta,C,\varepsilon} N^{\max(\beta,7/12)+\varepsilon}.$$

For $7/12 \le \beta < 1$ this exponent is sharp in the sense $G_2(N, N^\beta) = N^{\beta+o(1)}$.

Proof. Two selected integers have the unique forms $n = c_1d^2u$ and $n+h = c_2e^2v$, where $1 \le h < B$, $u, v \le T = N^\beta$, and c_1, c_2 are squarefree and B-smooth. There are $N^{o(1)}$ choices for c_1, c_2, h. Fix them and dyadically subdivide d, e. With D, E fixed, $U \asymp N/(c_1D^2)$ and $V \asymp N/(c_2E^2)$. The elementary congruence count, for fixed d, e, gives in this box

$$\ll \frac{N|h|}{c_1c_2DE} + DE \ll (\log N)^{O(1)}T + DE. \tag{T1.4}$$

Indeed the common divisor of the two moduli divides h, and $DE \gg N/(\sqrt{c_1c_2}T)$. If $c_1c_2(DE)^2 < N$, then $DE < \sqrt{N}$ and (T1.4) suffices. It also suffices for $DE \le N^\delta$, with a small fixed δ. For $UV \le N^\delta$ fix u, v and apply Lemma T1.2. These edge regimes cost $N^{\max(\beta,1/2)+O(\delta)}$.

In the other boxes apply Lemma T1.4. In its notation the main exponents are

$$f_1(z,w) = z/2 + \tfrac{9}{8}z(w-z), \qquad f_2(z,w) = w - z + \tfrac{9}{8}z(w-z).$$

They meet at $z = 2w/3$; f_1 is increasing to the left of that point and f_2 decreasing to its right. Since $0 \le w \le 1 + o(1)$,

$$\min(f_1, f_2) \le w/3 + w^2/4 \le 7/12 + o(1).$$

The remaining terms satisfy $D + E \ll N^{1/2}$ and $U + V \ll T$. Sum the polylogarithmic boxes and reduce the internal margins.

For the lower bound let $D \asymp \sqrt{N/T}$ with a fixed sufficiently large comparison constant, and take coprime $d, e \in [D, 2D]$. The conditions $n \equiv 0 \pmod{d^2}$, $n+1 \equiv 0 \pmod{e^2}$ have $\gg N/D^4$ solutions in a fixed interior subinterval of $[N, 2N)$, because $\beta > 1/2$. Both remaining cofactors are at most T. The total number of incidences is $\gg D^2N/D^4 \asymp T$. A fixed n is counted at most $\tau(n)\tau(n+1) = N^{o(1)}$ times. The two neighboring integers lie in a common window, proving the claim. □

Lemma T1.6 (A large selected kernel). *For any fixed $0 < \tau < 1/2$ and $1 \le T \le 3N$, the number of separated pairs possessing a mixed square relation with a selected vertex whose B-kernel exceeds T is at most*

$$N^{3/2+o(1)}T^{-1/2+\tau}.$$

Also the number of windows containing one vertex with kernel at most T is at most $N^{1/2+o(1)}T^{1/2}$.

Proof. Write $n = a^2cr$, with c squarefree and B-smooth and $r = K_B(n)$. The one-kernel assertion follows from

$$\sum_{\substack{n\le 3N\\ K_B(n)\le T}} 1 \le \sqrt{3N}\prod_{\ell\le B}(1+\ell^{-1/2})\sum_{r\le T} r^{-1/2} \le N^{1/2+o(1)}T^{1/2},$$

with a factor B for its placements. For the first assertion anchor a selected value $n = x + i$ with $K_B(n) > T$. Every prime of its kernel must occur at one of the B offsets of the other window. There are at most $B^{\omega(r)}$ assignments; each specifies a residue class modulo r. Since $r \le 3N$, there are $O(N/r)$ possible other starts, including $r > N$. Summation over anchors is bounded by

$$BN^{3/2}\prod_{\ell\le B}(1+\ell^{-1/2})\sum_{\substack{r>T\ \text{squarefree}\\ \ell\mid r\Rightarrow \ell>B}} B^{\omega(r)}r^{-3/2}.$$

Rankin's inequality bounds the last sum by

$$T^{-1/2+\tau}\prod_{\ell>B}(1+B\ell^{-1-\tau}) = T^{-1/2+\tau}\exp(O_\tau(B^{1-\tau})).$$

Both products and the placement factor are $N^{o(1)}$. □

T1.3 Multiquadratic surfaces and the curves that can survive

We use two uniform determinant estimates. A geometrically integral affine surface of degree d in $\mathbb{A}^3$ has, outside $O_{d,\varepsilon}(H^{2/\sqrt{d}+\varepsilon})$ integral points of height at most H, a covering by $O_{d,\varepsilon}(H^{1/\sqrt{d}+\varepsilon})$ curves of bounded degree [24, Lemma 2]. An integral curve of degree e in fixed affine space has $O_{e,n,\varepsilon}(H^{1/e+\varepsilon})$ integral points in the same box [21, Theorem A]. These constants do not depend on the defining coefficients. For a curve irreducible over $\mathbb{Q}$ but not geometrically integral, its rational points lie in the intersection of distinct conjugate components and have bounded cardinality.

Consider equations

$$w_\nu^2 = d_\nu P_\nu(X)Q_\nu(Y),\quad P_\nu(X) = \prod_{i\in I_\nu}(X+i),\quad Q_\nu(Y) = \prod_{j\in J_\nu}(Y+j),\quad 1\le\nu\le k, \tag{T1.5}$$

where I_ν, J_ν are nonempty, disjoint within each list, and $|I_\nu| + |J_\nu| \le K$. The offsets are $O(B)$ and the nonzero coefficients d_ν are squarefree and $O(B)$-smooth. Constants k, K are fixed.

Lemma T1.7 (The uniform geometric counts)**.** *Fix $A_0, K, \varepsilon > 0$. For sufficiently large fixed k, a system of the form (T1.5), with arbitrary nonzero integer coefficients $|d_\nu| \le N^{A_0}$, has $O_{A_0,K,\varepsilon}(N^{1/2+\varepsilon})$ integral hosts $X, Y \asymp N$ at which the even-selection square-class maps on the two actual sets of selected values are injective and their images are in direct sum. If every Q_ν has degree one, the bound is $O_{A_0,K,\varepsilon}(N^\varepsilon)$. If the degrees of the P_ν vary, select sufficiently many equations of one of their finitely many possible degrees before applying the assertion. These estimates are uniform in the coefficients and do not require their smoothness; any enumeration of such coefficients is a separate cost. They remain valid after other fixed centers have been specialized into the d_ν.*

For one variable, a bounded-degree system of split-product square equations with polynomial-height coefficients has $N^{o(1)}$ possible integer values as soon as its disjoint root sets include at least two distinct selected offsets. This includes either one equation of degree at least two or two equations with disjoint nonempty root sets.

The following graph consequences hold, with the number of component equations chosen sufficiently large after the stated fixed parameters.

1. *For any $\eta, \varepsilon > 0$, pairs of separated nonaligned logarithmic windows with at least ηB nontrivial unpinned components number $O_{\eta,C,\varepsilon}(N^{1/2+\varepsilon})$.*

2. *For every fixed integer $a \geq 1$ and $\varepsilon > 0$, there is k_a such that k_a components having a vertices in one window and one in the other have $O_{a,C,\varepsilon}(N^{\varepsilon})$ dyadic hosts. For $a = 1$ aligned hosts are excluded; for $a \geq 2$ no such exclusion is required.*

3. *After one start is fixed, a component with at least two vertices in the other window has $N^{o(1)}$ possible partners, uniformly for a fixed bound on the component size and polynomial coefficient heights.*

Parameters in the geometric counts. The external estimates are applied in the following embeddings. Write $m = \binom{D+2}{2} + k - 1$ for the ambient dimension of the affine Veronese chart, and $q = (a + 1 + A_0)/2$ in the degree-$(a, 1)$ case.

External count	Dimension and degree	Point height	Coefficient height
Surface covering, Steps 1–4	Surface in $\mathbb{A}^3$; $2^k \leq d_{\mathrm{im}}$, $d_{\mathrm{im}} = O_{k,K}(1)$	$H \ll N^D$	Polynomial in N; no coefficient-height dependence in the covering constant
Lifted curves, Steps 2–4	Curves in $\mathbb{A}^m$; $2D \leq e = O_{D,k,K}(1)$	$H \ll N^D$	No bound on the coefficients of the covering curves is required
Degree-$(a, 1)$ surface covering, Steps 5–6	Surface in $\mathbb{A}^3$; $2^k \leq d_{\mathrm{im}}$, $d_{\mathrm{im}} = O_{k,a}(1)$	$H \ll N^q$	Polynomial in N; the same coefficient-uniform covering estimate
Degree-$(a, 1)$ lifted curves, Steps 5–6	Curves in $\mathbb{A}^{k+2}$; $E + 1 \leq e = O_{k,a}(1)$	$H \ll N^q$	No coefficient-height bound is required by the curve estimate

The curve rows concern components that can contain the hosts being counted; Steps 3 and 5–6 exclude the lower degrees. The input coefficients satisfy $|d_\nu| \leq N^{A_0}$, and the offsets are $O(\log N)$. Thus the defining equations and the projected surface equations have polynomial height with fixed exponents, while the external constants depend only on the fixed dimensions, degrees and internal error margins. For the first bound choose A_0, K, ε, then $D \geq (K + A_0)/2$, then k, and finally the internal margin in the surface and curve counts. For the degree-$(a, 1)$ bound choose q, then E with $q/(E + 1) < \varepsilon/8$, then k satisfying the two conditions in Step 6, and finally that internal margin. All these choices precede $N \to \infty$. The count for one system is uniform over arbitrary coefficients in the stated height range; the $N^{o(1)}$ enumeration in the graph application uses smoothness separately.

Proof. We first count one fixed system, keeping the coefficient height visible; only afterwards do we sum the patterns arising from the graph. We will retain a sufficiently large fixed number k of equations, chosen after the height parameter D below. All degrees and ambient dimensions below depend on that number and on the fixed parameters, not on the values of the coefficients.

Step 1: geometric integrality and the finite affine projection. Over an algebraically closed field of characteristic zero, the square classes on the right of (T1.5) are independent: the valuation along a factor $X + i$ detects exactly its own equation. The coordinate ring is free of rank 2^k over the polynomial ring in X, Y, and its generic localization is a field. It is therefore integral.

The affine projection $(X, Y, w) \mapsto (X, Y, \sum_\nu w_\nu)$ is finite and birational onto its image. Finiteness follows from integrality over $\mathbb{Q}[X, Y]$. Birationality follows because no nontrivial sign change fixes the sum: the individual radicals are independent basis elements of the multiquadratic extension. The image surface in $\mathbb{A}^3$ has degree at least 2^k and degree bounded in terms of k, K. The projection need not extend across the projective boundary; only its affine finiteness is used when curves are lifted.

Step 2: projective degrees and the height of integral points. Choose an integer $D \geq (K + A_0)/2$ before choosing k, and homogenize in the weighted ring

$$\mathbb{Q}[X, Y, Z, W_1, \ldots, W_k] / \left(W_\nu^2 - d_\nu Z^{2D-|I_\nu|-|J_\nu|} \prod_{i \in I_\nu} (X + iZ) \prod_{j \in J_\nu} (Y + jZ) \right),$$

with $\deg X = \deg Y = \deg Z = 1$, $\deg W_\nu = D$. Its Dth Veronese ring is generated in degree one by the degree-D monomials in X, Y, Z and the W_ν. This embeds an integral projective surface finite of degree 2^k over $\mathbb{P}^2$. The hyperplane bundle is the pullback of $\mathcal{O}_{\mathbb{P}^2}(D)$, so its degree is $2^k D^2$ and every integral curve C satisfies

$$\deg C = D \deg \sigma(C) \, [\mathbb{Q}(C) : \mathbb{Q}(\sigma(C))]. \tag{T1.6}$$

Here σ is the finite map to $\mathbb{P}^2$. At an integral point in the dyadic region, $|w_\nu| \ll N^{(K+A_0)/2}$. In the chart $Z = 1$, the Veronese coordinates are integers of height $H \ll N^D$, including the W_ν. The same bound holds for the affine projection in Step 1. Thus both applications of the determinant estimates use one common height bound.

Step 3: excluding the curves of smallest possible degree. A curve of degree D maps with degree one to a line. A vertical or horizontal line meeting the positive region is impossible: a nonconstant product of distinct linear factors in the other variable would have to be a rational square. On an oblique line, square products force every root on one side to match exactly one root on the other. Two component equations give two distinct exact units, hence a positive rational alignment of height $O(B)$. Thus a nonaligned point is on no curve of degree D; by (T1.6), its curves have degree at least $2D$.

In the direct-sum formulation the exceptional degree-D curves are excluded for a more intrinsic reason: their two exact units give nonempty even selections, two values on each side, with equal square classes. Injectivity makes both classes nonzero, contradicting the direct sum. Thus the argument does not require absence of geometric alignment when the actual spaces are already in direct sum.

Step 4: the surface covering and the first graph consequence. Apply the surface covering, lift its curves by the finite affine map, and take their closures in this embedding. Their numbers of components and degrees are bounded in terms of the fixed parameters by Bézout. Pila's estimate and the degree lower bound $2D$ give, per pattern,

$$O\big(N^{2D2^{-k/2}+\varepsilon'} + N^{1/2+D2^{-k/2}+\varepsilon'}\big).$$

Choose k large enough after D, ε and reduce the internal margin ε'. This proves the per-system bound.

For the graph consequence, smoothness gives $|d_\nu| \leq N^{A_0}$ for a fixed A_0. At least half the components in a population $c \geq \eta B$ have size at most $K = \lceil 4/\eta \rceil$. Select k of them. The offsets and smooth coefficients have $B^{O_K(k)} 2^{k\pi(O(B))} = N^{o(1)}$ patterns. Choosing k large after D, ε, and summing these patterns, proves the first assertion.

Step 5: reduced fibres for equations of type $(a, 1)$. For the stronger degree-one assertion we exclude all curves of a fixed degree at most E, before making k large. For this part use the ordinary affine embedding with coordinates $X, Y, w_1, \ldots, w_k$. Write the equations as $w_\nu^2 = d_\nu P_\nu(X)(Y + j_\nu)$ with $\deg P_\nu = a$. On the normalization of an integral curve of degree $e \leq E$, the coordinate functions X, Y have degrees $m_X, m_Y \leq e$ and the genus is at most $(e-1)(e-2)/2$. Neither function can be constant: otherwise the many distinct roots in the other variable would all be branch values, contradicting Riemann–Hurwitz. Discard equations having a branch value among their selected roots, or a finite fibre meeting the pole divisor of the other coordinate. At most

$$(2g - 2 + 2m_X) + (2g - 2 + 2m_Y) + m_X + m_Y \leq 2e^2$$

equations are lost. For each remaining equation all finite fibres are reduced and avoid the poles. Parity of divisors gives

$$F_Y(-j_\nu) = \sum_{i \in I_\nu} F_X(-i), \qquad m_Y = a m_X.$$

In fact

$$\operatorname{div}\left(\frac{P_\nu(X)}{Y + j_\nu}\right) = F_Y(\infty) - a F_X(\infty),$$

independently of ν. For two good equations the quotients therefore differ by a nonzero constant. Eliminating this common quotient gives $Y = R(X)$ with $\deg R \le a$. Since $m_Y = (\deg R) m_X = a m_X$, the degree is exactly a.

Step 6: interpolation on the offsets and exclusion of low-degree curves. If $k \ge 4(E+1)^2$, there are at least three good equations. Their $3a$ distinct grid points $(-i, -j_\nu)$ lie on the graph of R: the reduced-fibre identity shows that R is finite and equals $-j_\nu$ at each selected $-i$. These grid points determine R. Choose $2a+1$ of them. Rational interpolation, using maximal minors of the integer evaluation matrix, gives coprime integer polynomials P, Q of degrees at most a, coefficients $B^{O_a(1)}$, and $R = P/Q$. The kernel of this interpolation system is one-dimensional: two solutions would have cross-product of degree at most $2a$ vanishing at $2a+1$ distinct abscissae. Since the reduced rational function has degree exactly a, no nonconstant polynomial multiple of its coprime numerator and denominator still has both degrees at most a. This makes the coefficient uniformity explicit: the interpolation data are the offsets, so their height bound is independent of the d_ν.

If Q is nonconstant, the nonzero integer resultant $\operatorname{Res}(P, Q)$ has size $B^{O_a(1)}$ and $Q(X)$ divides it at an integral point. Thus $|X| \le B^{O_a(1)}$. If Q is constant and $a \ge 2$, then $|Y| \gg |X|^a / B^{O_a(1)}$, impossible for $X, Y \asymp N$. If $a = 1$ and Q is constant, the graph is a rational line; its grid points give an exact positive alignment. Thus every remaining dyadic nonaligned point lies on a curve of degree greater than E.

In the ordinary affine embedding the height is $H \ll N^q$ for a fixed $q = (a + 1 + A_0)/2$. Choose E so that $q/(E+1) < \varepsilon/8$, then $k \ge 4(E+1)^2$ so that $2q 2^{-k/2} < \varepsilon/8$. The same finite projection, surface covering and curve estimate now give N^ε per fixed coefficient pattern. For $a = 1$, its exceptional line is also excluded under the direct-sum hypothesis: two grid matches give the forbidden even selections on the two actual blocks. Hence the degree-one per-system estimate is proved with arbitrary polynomial-height coefficients. In the graph application one sums the $N^{o(1)}$ smooth patterns, proving the second graph assertion, including $a = 1$.

Step 7: fixing one center and reducing to a quadratic equation. For the last assertion fix the first start and the component data. The other start satisfies $\prod_{j \in J}(Y + j) = e v^2$, with fixed squarefree $e \le N^{O(1)}$ and $|J| \ge 2$. Outside primes dividing $e \prod_{i \ne j}(i - j)$, every selected factor has even valuation. Fixing the squarefree parts of two factors costs $2^{O(\omega(e \prod_{i \ne j}(i-j)))} = N^{o(1)}$ for fixed $|J|$. Their difference is a nonzero fixed offset, so Lemma T1.2 gives $N^{o(1)}$ starts. This is a bounded-height argument; no height-free theorem for all integral points is being invoked. The one-variable assertion with two separate equations follows in exactly the same localization: take one selected factor from each equation and put the prime divisors of both coefficients and both within-equation discriminants into the exceptional prime set. Outside it each chosen factor has even valuation. The two distinct offsets again give one quadratic equation counted by Lemma T1.2. □

Lemma T1.8 (The dense weighted sector)**.** *For every fixed $\eta, \varepsilon > 0$,*

$$\sum_{\substack{x, y \in I_N,\ |x-y| > L \\ (x,y) \text{ nonaligned},\ c(x,y) \ge \eta B}} (2^{\rho(x,y)} - 1) \ll_{C,\eta,\varepsilon} N^{1+\varepsilon}.$$

The same bound holds for the full value kernel, without block-parity rows.

Proof. Let t_2, t_3 count nontrivial unpinned components with two and three vertices. If either exceeds a sufficiently large fixed threshold, Lemma T1.7(2), in both orientations for size three, gives N^{ε} hosts. Their weight is at most $2^{B+o(B)} = N^{1+o(1)}$, since each other component needs two vertices and the number of defective vertices is $o(B)$. Otherwise $t_2, t_3 = O(1)$ and the vertex budget gives $4c - 2t_2 - t_3 \le 2B$, hence $c \le B/2 + O(1)$. Lemma T1.7(1) gives $N^{1/2+\varepsilon}$ hosts, each of weight at most $2^{B/2+o(B)} = N^{1/2+o(1)}$. Reduce the internal margins in both estimates. □

Lemma T1.9 (Uniform residual rank on an aligned pair)**.** *For separated aligned windows with reduced height $b = \max(a, b) \le B^3$, let u be the number of exact units. Then*

$$\rho(x, y) \le (u - 1)_+ + o(B) \le L/b + o(B),$$

uniformly over the channels. For fixed channel height and a fixed enlarged prime threshold proportional to B, the residual is $O(B/\log B)$.

Proof. Normalize the two windows to bV_x and aV_y, and merge equal values. All remaining distinct integers lie in an interval of diameter $O(B^4)$ centered at an integer at least N. Remove prime rows $\ell \le B$ and rows with $\ell \mid ab$. Every other prime row meets at most two distinct normalized vertices, since it occurred at most once in each original window. Its equations again leave one variable per unpinned component. Let c_K count these components of size at most a fixed K.

For these components, the removed prime rows and one total-parity row form a binary matrix of redundancy at most $s = \pi(B) + \omega(ab) + 1 = O(B/\log B)$. A nonzero kernel word gives an even square product of distinct normalized integers. By Lemma T1.1, it uses at least $\delta \gg \log N/\log B$ vertices, and hence at least δ/K components. The Hamming argument in Lemma T1.3 yields $c_K = O_{C,K}(B/\log B)$. Components larger than K number at most $2B/K$. Since K can be chosen arbitrarily large before N, the dimension of the value kernel on the distinct normalized set is $o(B)$.

The kernel of the merging map on the original relative selections is exactly the even-selection space of the u exact units, of dimension $(u-1)_+$. Its image lies in the value kernel just bounded, proving the first assertion. For fixed height the normalized interval has diameter $O(B)$. Private primes above a sufficiently large fixed multiple of B leave only $O(B/\log B)$ defective normalized vertices, directly by Lemma T1.3; this gives the last assertion. □

T1.4 Asymmetric extensions and homogeneous seeds

Lemma T1.10 (An asymmetric pair profile)**.** *Let U_X be any subset of $X + [-DB, DB]$, with D fixed and $X \asymp N$, and let V_z be a new window. Choose H a fixed sufficiently large multiple of B. Let $\delta^*(X, z)$ be the dimension of the even selections of V_z whose prime valuations above H belong to the value space of U_X. Then*

$$\sum_{\substack{X, z \asymp N \\ (X,z) \text{ nonaligned}}} (2^{\delta^*(X,z)} - 1) \ll_{C,D,\varepsilon} N^{31/24+\varepsilon}.$$

Adding any small-prime bits to the old data does not change the estimate.

Proof. For the graph above H, let D_z be the number of new defective vertices and let c count nontrivial unpinned components. Projection to the new block gives $\delta^* \le \min(L, D_z + c)$, with $D_z = o(B)$. A nonzero even selection uses at least two new vertices. If it uses no old vertex, the new window has two defects and contributes $N^{1+o(1)}$ hosts after the old parameter is chosen. Otherwise it uses at least one old and two new vertices. Set $T = N^{5/12}$. If all their kernels are at most T, Lemmas T1.5 and T1.6 give

$$N^{7/12+\varepsilon'} N^{1/2+5/24+o(1)} = N^{31/24+\varepsilon'+o(1)}.$$

If one kernel exceeds T, the same CRT anchor proof gives $N^{3/2+o(1)}T^{-1/2+\tau} = N^{31/24+(5/12)\tau+o(1)}$. All these arguments concern the large-prime valuations only, so they do not impose parity on the old selection. They are unchanged at the fixed enlarged threshold H.

For $c < \eta B$, the weight is at most $2^{\eta B+o(B)}$, and the support bound suffices. For $c \geq \eta B$, the proof of Lemma T1.7(1) applies to the unequal fixed widths and gives $N^{1/2+\varepsilon'}$ hosts. Fix a large integer K. Components with one new vertex and more than K old vertices number $O_D(B/K)$. A sufficiently large fixed population of any other type $(a, 1)$, $1 \leq a \leq K$, has only $N^{\varepsilon'}$ hosts by Lemma T1.7(2). On this exceptional set use $2^{\delta^*} \leq 2^L \ll N$. Else the number c_1 of one-new-vertex components is $O_D(B/K) + O_K(1)$, and $2c - c_1 \leq B$. Thus $\delta^* \leq B/2 + O_D(B/K) + o(B)$ and the remaining dense mass is $N^{1+O_D(1/K)+\varepsilon'}$. Choose K, then η, τ, ε' after the required margin. Finally shrinking the old set only decreases δ^*, and prime bits at most H have already been forgotten. □

Lemma T1.11 (Aligned loss above a rational pair)**.** *Fix a geometrically good pair in channel $a : b = 2 : 3$ or $3 : 4$, regardless of its affine compatibility. Among targets z with $|z - x|, |z - y| \geq L$ and distinct from the two sources, there are $B^{O(1)}$ aligned targets per source. Except for $B^{O(1)}$ source pairs at dyadic endpoints, their additional relative rank loss modulo the old value space is at most*

$$L/3 + o(L) \quad (a : b = 2 : 3), \qquad L/2 + o(L) \quad (a : b = 3 : 4).$$

The result allows the prime bits of a, b in the old information.

Proof. Represent the source by $x = at - i_0$, $y = bt - j_0$ and put $X = abt$. The true old normalized values form a subset of $X + [-bB, bB]$. Applying the residual argument of Lemma T1.9 to a fixed-width enclosing interval and the new interval leaves $o(B)$ dimensions after exact units are removed. If an exact unit of the enlarged graph uses an old value absent from the true set, its new nondefective vertex is pinned in the true graph. Consequently the remaining loss is at most the number of new vertices matched to actual old values, plus $o(B)$.

Write an aligned target as $z = (ct + h)/d$, with c, d positive coprime, $c, d \leq B^{O(1)}$ and $h/d = O(B)$. A match to the first old family requires $h + dk \equiv 0 \pmod{c/(c, a)}$; the second family requires the same condition modulo $c/(c, b)$. Their intersection is the class modulo c, because $(a, b) = 1$. Thus the union has at most

$$B\frac{(c, a) + (c, b) - 1}{c} + O(1) \tag{T1.7}$$

new offsets. Conditions on the true intervals can only reduce this count.

For $2 : 3$, the source range is $N/2 - O(B) \leq t \leq 2N/3 + O(B)$, so $3/2 \leq c/d \leq 4$ apart from endpoint errors of order B/N. A distinct rational of polynomial height cannot cross either endpoint by such a small amount. For $c \geq 7$ the density in (T1.7) is at most $1/3$ (check $7 \leq c \leq 12$, then bound the numerator by four). For $c = 5$ it is $1/5$. The cases $c/d = 2, 3$ physically reproduce one old family. Their supports intersect the new support in at most one integer, because the starts are at distance at least L. The other family supplies at most $B/3 + O(1)$ units, counted in its actual window of width B. The only remaining possibilities with $c \leq 6$ are $c/d = 4, 3/2$; these force $t = N/2 + O(B)$ or $t = 2N/3 + O(B)$. They account for only $B^{O(1)}$ source pairs.

For $3 : 4$, one has $N/3 - O(B) \leq t \leq N/2 + O(B)$ and $2 \leq c/d \leq 6$. The density is at most $1/2$ for $c \geq 9$ (check $9 \leq c \leq 12$, then use numerator at most six), and also for $c = 5, 7, 8$. The physical cases $c/d = 3, 4$ leave at most one unit in one old family and $B/4 + O(1)$ in the other. The remaining fractions $2, 6$ force $t = N/2 + O(B)$ or $N/3 + O(B)$ and are endpoint sources. Finally the canonical slopes and displacements have polynomially many choices in B, proving the target count. □

Proposition T1.12 (Homogeneous seeds for fixed extensions)**.** *Let $\mathcal{A}_{\mathrm{raw}}$ be the geometric good-edge family obtained by omitting the compatibility condition in the definition of $\mathcal{E}_N$. For a finite*

set $I \subset G_N$ put

$$W(I) = p^{|I|}2^{\rho(I)}.$$

All configurations in the following sums have distinct, pairwise separated starts. For every $\varepsilon > 0$:

1. *The sum of $W(\{x, y\})$ over dependent pairs outside $\mathcal{A}_{\mathrm{raw}}$ is $O_{C,\varepsilon}(N^{-4/5+\varepsilon})$. Moreover $\sum_{e\in\mathcal{A}_{\mathrm{raw}}} W(e) \ll_C LN^{-2/3}$.*
2. *For a source e in channel $a : b$, sum over targets z for which $S_z \cap \sum_{x\in e} S_x \neq \{0\}$. Then*

$$\sum_{e,z} W(e \cup \{z\}) \ll_{C,\varepsilon} \begin{cases} N^{-4/3+\varepsilon}, & a : b = 2 : 3, \\ N^{-5/4+\varepsilon}, & a : b = 3 : 4. \end{cases}$$

3. *For two distinct geometric good edges, whose union has three or four starts,*

$$\sum_{e\neq f} W(e \cup f) \ll_{C,\varepsilon} N^{-5/4+\varepsilon} + (LN^{-2/3})^2.$$

These are rank bounds without any affine compatibility restriction.

Proof. The homogeneous indicators $H_x = \mathbf{1}_{\{f(x-1)=\cdots=f(x+L-1)\}}$ have zero affine target and therefore $\mathbb{E}\prod_{x\in I} H_x = W(I)$ exactly. The first assertion follows from the positive profile and the same exceptional-channel estimates used in Theorem 3.3. On a dependent pair $2^\rho \leq 2(2^\rho - 1)$, so the support contribution is already controlled by that profile.

For a dependent extension let $\delta = \dim(S_z \cap \sum_{x\in e} S_x) > 0$. Then $W(e \cup \{z\}) = W(e)p2^\delta$, with $2^\delta \leq 2(2^\delta - 1)$. Lemma T1.10 gives the nonaligned mass $N^{-3+1/b+31/24+\varepsilon} \leq N^{-11/8+\varepsilon}$. Lemma T1.11 and $NB^{O(1)}$ aligned hosts give $N^{-4/3+\varepsilon}$ or $N^{-5/4+\varepsilon}$. Endpoint sources cost at most $N^{-5/3+o(1)}$. No contact was dropped on a compatibility argument: all starts in this proposition are separated.

Two edges sharing a vertex have $NB^{O(1)}$ hosts. The same three-window rank bound gives their mass $N^{-5/4+\varepsilon}$. For disjoint edges first treat nonaligned compressed blocks. Write $r_f = |U_f| - 1 \geq 5L/3 - O(1)$ and let δ^* be the loss of the even-selection rank on U_f after the old normalized values and small primes have been exposed. The excess above $W(e)W(f)$ is at most $W(e)W(f)(2^{\delta^*} - 1)$. The weak-component support proof of Lemma T1.10 still applies to two subsets of fixed-width logarithmic intervals: it needs two selected new values and one selected old value. It gives $N^{31/24+\varepsilon}$ hosts. In this sector $W(e)W(f) \leq N^{-10/3+o(1)}$, so its mass is $N^{-49/24+\varepsilon}$.

In the dense sector, the multiquadratic argument gives $N^{1/2+\varepsilon}$ hosts. A fixed large population of components with one new vertex and boundedly many old vertices has only N^ε hosts, by Lemma T1.7(2). On that exceptional population use $W(f)2^{\delta^*} \leq 1$ and $W(e) \leq N^{-5/3+o(1)}$. Otherwise all but $O(B/K) + O_K(1)$ components use two new vertices. Hence

$$\delta^* \leq |U_f|/2 + O(B/K) + o(B), \qquad W(f)2^{\delta^*} \leq N^{-5/6+O(1/K)+o(1)}.$$

The remaining dense mass is $N^{-2+\varepsilon}$. The entire nonaligned excess is therefore $N^{-5/3+\varepsilon}$.

Aligned compressed pairs, including overlapping envelopes, have $NB^{O(1)}$ hosts. Away from endpoint sources their first three original windows have nullity at most $3L/4 + o(L)$, by Lemma T1.11. Adding the fourth loses at most L further ranks. Their mass is at most $N\,N^{-4+7/4+o(1)} = N^{-5/4+o(1)}$. The endpoint source contribution is again $N^{-5/3+o(1)}$. Finally the sum of all baseline products is bounded by $(\sum_{e\in\mathcal{A}_{\mathrm{raw}}} W(e))^2$. All estimates were made with homogeneous ranks, proving the assertion independently of the original phases. □

T1.5 Support and branching at a fixed number of blocks

Lemma T1.13 (Small-kernel counts for fixed-width blocks)**.** *Let $U_X \subset X + [-DB, DB]$ with D fixed and $X \asymp N$, and let $H \asymp B$ exceed the diameter. Uniformly for $1 \le T \le CN$, the number of translations containing one value with $K_H(n) \le T$ is at most $N^{1/2+o(1)}T^{1/2}$. For fixed $0 < \theta < 1$, the number containing two such distinct values with $T = N^\theta$ is $O_{C,D,\theta,\varepsilon}(N^{\max(\theta,7/12)+\varepsilon})$. In particular it is $O(N^{\theta+\varepsilon})$ for $\theta \ge 7/12$.*

Proof. Enlarge the selected set to its interval. The proofs of Lemmas T1.6 and T1.5 use only a fixed logarithmic width and a prime threshold of that order. In particular $2^{\pi(H)} = N^{o(1)}$, the smooth parts have fixed polynomial height, and all changes of offset count are powers of B. □

Proposition T1.14 (A relative relation meeting several blocks)**.** *Fix integers $r \ge s \ge 2$. For r separated logarithmic windows, the number of tuples having a relative square relation whose projection is nonzero in at least s blocks is*

$$O_{C,r,\varepsilon}\left(N^{r-2s/(2s+1)+\varepsilon}\right).$$

The same assertion holds for injective even-selection spaces on subsets of fixed-width logarithmic intervals, including the compressed good edges, with only $B^{O_r(1)}$ choices of fixed offsets and fixed dilation factors.

Proof. Put $T = N^{\beta_s}$ with $\beta_s = (2s-1)/(2s+1) > 7/12$. If every selected kernel is at most T, each of the s nonzero even projections selects at least two distinct values. By Lemma T1.13, the number of hosts is

$$N^{r-s}N^{s\beta_s+\varepsilon'} = N^{r-2s/(2s+1)+\varepsilon'}.$$

Otherwise anchor a selected value of kernel $K > T$. Assign every prime of K to one of the other $(r-1)O(B)$ offsets where it must reappear. For a fixed assignment let K_j be the product assigned to block j. These products are pairwise coprime, at most $K \ll N$, and their CRT cost is $O_r(N^{r-1}/\prod_j K_j) = O_r(N^{r-1}/K)$. The Rankin calculation from Lemma T1.6, with $(r-1)O(B)$ in place of B, gives

$$N^{r-1/2+o(1)}T^{-1/2+\tau} = N^{r-2s/(2s+1)+\beta_s\tau+o(1)}.$$

Choose τ and all internal margins after ε. The argument uses actual distinct values in each selected set; enlarging its containing interval only increases the count. Injectivity ensures that a nonzero compressed projection still gives a nonempty even selection. □

Proposition T1.15 (A fixed number of exposed source windows)**.** *Fix $j \ge 1$ and source starts $x_1, \dots, x_j \in I_N$. Outside a deterministic set of $N^{1/2+o_{C,j}(1)}$ possible starts, the new relative vector is uniform and independent of every value in the old windows. An explicit bound for the exceptional set is*

$$j(2L+1) + B\sqrt{3N}\exp(2\sqrt{m_{j,N}}), \qquad m_{j,N} = \left\lceil \pi(B) + \frac{jB\log(3N)}{\log B} \right\rceil.$$

Fixed dilations, subsets, and fixed enlarged logarithmic windows satisfy the same $N^{1/2+o(1)}$ conclusion.

Proof. Let $\mathcal{P}$ contain all primes at most B and all primes dividing an old value. It has at most $m_{j,N}$ members. The number of integers at most $3N$ whose squarefree part is supported on $\mathcal{P}$ is at most

$$\sqrt{3N}\prod_{\ell\in\mathcal{P}}(1+\ell^{-1/2}) \le \sqrt{3N}\exp(2\sqrt{|\mathcal{P}|}),$$

because the ith distinct prime is at least i. Discard overlapping starts and windows containing two such values; the stated estimate follows even by charging each such window to one of its values

and multiplying by B. If a nonzero new relative combination belonged to the old value space, its even selection would contain at least two integers. Every large odd prime of each selected integer is private inside the new window, so it would have to divide an old value. Both selected integers would have squarefree part supported on $\mathcal{P}$, contrary to the retained set. The new rank therefore increases by exactly L over the old value space, which proves the asserted conditional uniformity. For fixed j, $m_{j,N} = O_{C,j}((\log N)^2/\log\log N)$. □

T1.6 Products grouped by their ternary support

Proposition T1.16 (A ternary group budget)**.** *Let $U_i \subset X_i + [-DB, DB]$, $X_i \asymp N$, be three sets with injective even-selection square-class maps whose images are pairwise in direct sum. The width constant and the number of offset patterns are fixed. Choose H greater than the three diameters. For an offset triple $u = (u_1, u_2, u_3)$ let*

$$G_u = \prod_{\substack{\ell > H \text{ prime} \\ v_\ell(X_i+u_i) \text{ odd } (i=1,2,3)}} \ell.$$

Retain the groups greater than one and write their decreasing values $G_{(1)} > G_{(2)} > \cdots$. Put $\mathcal{G}_k = \prod_{i=1}^{k} G_{(i)}$, or zero if fewer than k exist. For every fixed $k \geq 3$, $\varepsilon > 0$, and $1 \leq T \leq N$,

$$\#\{\textit{tuples} : \mathcal{G}_k > T\} \ll_{C,D,k,\varepsilon} N^{\varepsilon}(N^3/T^2 + N^{6/5}). \tag{T1.8}$$

Proof. A prime larger than H has only one occurrence in each block, so the groups are pairwise coprime and $G_u \mid \gcd(X_1 + u_1, X_2 + u_2, X_3 + u_3)$. Two groups satisfy $G_u G_v \ll NB$. If they share an occurrence this follows from divisibility of that integer. Otherwise choose two blocks in which the offsets differ. Their product divides a determinant $nm' - n'm$ of size $O(NB)$. Its vanishing would give a nonzero relative relation on those two blocks, contrary to their direct sum.

Three groups with product greater than a sufficiently large constant times NB^2 force their three value vectors to have zero determinant. The corresponding offset triples cannot be collinear. Indeed, writing $u_j = u_1 + t_j d$, with d a primitive integer vector and t_j integers, in the collinear case, all three coprime groups divide $d \times (X + u_1)$. If this vector is nonzero, its size $O(NB)$ gives a contradiction. If it is zero, two positive value vectors are proportional with distinct coordinates in every block, again giving a forbidden pair relation. Thus the centers lie on a genuine plane

$$((u_2 - u_1) \times (u_3 - u_1)) \cdot X = -((u_2 - u_1) \times (u_3 - u_1)) \cdot u_1.$$

There are only $B^{O(1)}$ such planes, determined by the offsets alone.

For fixed k group first by the squarefree product $Q = \mathcal{G}_k$. The ordered factorizations cost $d_k(Q) \ll_{k,\varepsilon'} Q^{\varepsilon'}$, and the supports cost B^{3k}. CRT fixes one residue class modulo Q for each center. Summing $O((N/Q + 1)^3)$ gives, for $1 \leq T \leq N \leq R \leq N^{O(1)}$,

$$\#\{T < \mathcal{G}_k \leq R\} \ll_{k,\varepsilon} N^{\varepsilon}(N^3/T^2 + R). \tag{T1.9}$$

For $\mathcal{G}_3 > R \gg NB^2$, use its forced plane. The two largest groups have product $P > R^{2/3}$ and $P \ll NB$. Two free coordinates are fixed by CRT modulo P; the plane determines the third. Summing over the two positive group sizes and their offsets gives

$$\#\{\mathcal{G}_3 > R\} \ll_{\varepsilon} N^{\varepsilon}(N^2 R^{-2/3} + N). \tag{T1.10}$$

Here one uses $\sum_{ab>A}(ab)^{-2} \ll \log(2A)/A$, $\sum_{ab\leq U}(ab)^{-1} \ll \log^2(2U)$, and $\#\{ab \leq U\} \ll U\log(2U)$. The third numerical group is not enumerated again: it certifies the plane and the lower bound for P.

For $k = 3$, take $R = N^{6/5}$ in (T1.9)–(T1.10). For induction on fixed k, the region $T < \mathcal{G}_k \leq N^{6/5}$ is already covered by (T1.9). Beyond it, $\mathcal{G}_{k-1} > N^{6(k-1)/(5k)}$. At $k = 4$ the preceding N^3/T^2 term is $N^{6/5}$; at $k = 5$ it is smaller; for $k \geq 6$ the threshold is at least N and the preceding tail at N suffices. Only a fixed number of induction steps is taken. This proves (T1.8). □

T1.7 The probability of two active good edges

Lemma T1.17 (A positive second moment for good edges)**.** *Let* $\Xi_N = \sum_{e\in\mathcal{E}_N}\prod_{x\in e} J_x$*. Then*

$$\mathbb{E}(\Xi_N)_2 = \beta_N^2 + o_C(\beta_N^2) \ll_C L^2 N^{-4/3}.$$

In particular $0 \le \beta_N - \mathbb{P}(\Xi_N \ge 1) \le \frac{1}{2}\mathbb{E}(\Xi_N)_2$*.*

Proof. We only need a strict gap below $2L/3$ for the nullity of three aligned windows containing a good source. Here is a direct bound that avoids a complete classification of the optimal three-window configuration. Keep the target parametrization and congruences from Lemma T1.11. If its fraction c/d is physical, two of the three parameter intervals have the same coefficient and disjoint interiors, up to one endpoint. Their intersections with the third interval have total length at most that third interval. Thus the full three-window nullity is at most $L/2 + o(L)$ for $2:3$, and $L/3 + o(L)$ for $3:4$. This bound includes the original source nullity. The residual on this fixed-width normalized union is $o(L)$ by the private-prime defect estimate, just as in the fixed-height case of Lemma T1.9.

For a nonphysical target above $2:3$, the additional rank loss is at most $3L/10 + o(L)$. Indeed the density $((c,2)+(c,3)-1)/c$ already suffices except for $c = 9, 12$ and the physical or endpoint fractions. For $c \ge 14$ it is at most $2/7$; $c = 5, 7, 8, 10, 11, 13$ also satisfy the desired bound directly. For $c = 9$, the allowed reduced denominators are $4, 5$. Count the new class modulo nine everywhere, and the remainder of the class modulo three inside the true three-family parameter interval of length $3B/d$. This gives

$$B/9 + (3B/d)(1/3 - 1/9) + O(1) = B/9 + 2B/(3d) + O(1) \le 5B/18 + O(1) < 3B/10 + O(1).$$

Counting only the three-family units would miss possible matches outside that actual interval; the union count retains them. For $c = 12$ the possible denominators are $5, 7$. At $d = 7$ the two true old intervals supply at most $2B/7 + O(1)$ units. At $d = 5$ count the entire new class modulo six and the part of the class modulo four outside it. The latter lies in a true parameter interval of length $4B/5$, giving

$$B/6 + (4B/5)(1/4 - 1/12) + O(1) = 3B/10 + O(1).$$

The full nullity is therefore at most $(1/3 + 3/10)L + o(L) = 19L/30 + o(L)$.

Above $3:4$, the cruder additional bound $2L/5 + o(L)$ is enough. The density test proves it for $c \ge 15$ and for $c = 5, 7, 9, 10, 11, 13, 14$. The only other nonphysical interior fractions are $8/3$ and $12/5$. For $8/3$, count the class modulo eight everywhere and the remainder of the class modulo two inside the true four-family parameter interval of length $2B/3$. The bound is

$$B/8 + (2B/3)(1/2 - 1/8) + O(1) = 3B/8 + O(1).$$

For $12/5$, each true old family supplies at most $B/5 + O(1)$ units. Thus the full nullity is at most $(1/4 + 2/5)L + o(L) = 13L/20 + o(L)$. The finitely many endpoint fractions concern only $B^{O(1)}$ sources, as before. Since $19/30 < 13/20 < 2/3$, the bound $13L/20 + o(L)$ works for both channels outside this exception.

For two distinct good edges, configurations sharing a vertex and aligned configurations with four vertices have $NB^{O(1)}$ hosts. In the four-vertex case adding the last window loses at most L ranks. The joint probabilities in both cases therefore total at most $N^{-27/20+o(1)}$. Endpoint sources cost $N^{-5/3+o(1)}$. Nonaligned compressed edges have total absolute covariance $O_{C,\varepsilon}(N^{-5/3+\varepsilon})$ by the two-block argument in Proposition T1.12; that argument also applies to compatible affine sources. Cross-contacts at distance less than L have zero joint probability. Their baseline products, those

of adjacent edges, and the diagonal $\sum_e q_e^2$ have total $N^{-7/3+o(1)}$, since the geometric edge degree is $O(B)$. Contacts at distance exactly L were included in the physical bounds above. Consequently

$$\mathbb{E}(\Xi_N)_2 = \beta_N^2 + O_{C,\varepsilon}(N^{-27/20+\varepsilon}).$$

Take $\varepsilon < 1/60$ and use $\beta_N \asymp LN^{-2/3}$. The final inequality is the elementary pointwise bound $(m-1)_+ \le m(m-1)/2$ for integers $m \ge 0$. □

T2 Coefficient budgets, coordinate restrictions, and fixed channels

The first two tools below do not depend on the critical-window scaling. They control, respectively, the small-prime coefficients of selected values and the repetition of coordinate forms in a binary relation space. We then give the extension estimates for any fixed rational channel. In that part, the channel and every finite family of channels are fixed before N tends to infinity.

T2.1 A deterministic budget for the small-prime part

For a positive integer n, write

$$u_B(n) = \frac{\mathrm{sf}(n)}{K_B(n)} = \prod_{\substack{\ell \le B \text{ prime} \\ v_\ell(n) \text{ odd}}} \ell.$$

Lemma T2.1 (A small-prime coefficient budget)**.** *Let V be any interval of $B \ge 2$ consecutive positive integers. Then*

$$\sum_{n \in V} \log u_B(n) \le B \log B + O(B),$$

with an absolute constant independent of the position of V. For every fixed $A > 0$, at most

$$(A^{-1} + O_A(1/\log B))B$$

of its values satisfy $u_B(n) > B^A$.

Proof. Dropping the parity condition gives

$$\sum_{n \in V} \log u_B(n) \le \sum_{\ell \le B} \left(\left\lfloor \frac{B}{\ell} \right\rfloor + 1 \right) \log \ell \le \log(B!) + \sum_{\ell \le B} \log \ell.$$

Here the sums are over primes. Legendre's formula gives the second inequality, and the elementary Chebyshev bound makes the last sum $O(B)$. Finally $\log(B!) \le B \log B$. Each value counted in the last assertion contributes more than $A \log B$ to the nonnegative sum. □

T2.2 Regularising coordinate forms by independent restrictions

Lemma T2.2 (Coordinate regularisation)**.** *Let $R \subset \mathbb{F}_2^V$ have dimension r, put $n = |V|$, and let $\ell_v \in R^*$ be the coordinate forms. Suppose that s distinct nonzero forms occur among them. For integers $K, d \ge 1$, there is $D \subset V$ such that*

$$R' = \{x \in R : x_v = 0 \text{ for every } v \in D\}, \qquad \operatorname{codim}_R R' = |D| \le \frac{n}{K+1} + \frac{s-r}{d+1},$$

and the restricted coordinate forms have the following properties:

1. *every nonzero form has multiplicity at most K;*
2. *every nonzero form ℓ has at most d unordered decompositions $\ell = a + b$, where a, b are distinct nonzero coordinate forms present on R'.*

Coordinates whose restricted forms vanish may then be removed without any additional loss of dimension.

Proof. The coordinate forms span R^*, and their restrictions continue to span the dual of every subspace encountered below. At an intermediate stage let S be the set of distinct nonzero forms and let r_0 be their rank. Setting a present nonzero form ℓ equal to zero reduces the dimension by one and replaces the forms by their images modulo $\langle \ell \rangle$. The form ℓ disappears. All other collisions are exactly the disjoint unordered pairs $\{a, a + \ell\} \subset S$. If there are $t(\ell)$ such pairs, then

$$|S_{\text{new}}| = |S| - 1 - t(\ell), \qquad (|S| - r_0)_{\text{new}} = |S| - r_0 - t(\ell).$$

Thus the nonnegative potential $|S| - r_0$ never increases.

If a form has multiplicity greater than K, set it equal to zero. This permanently kills at least $K + 1$ previously nonzero coordinates. There are at most $n/(K + 1)$ such steps, since these sets of killed coordinates are disjoint. When the multiplicity condition holds but a form has more than d decompositions, set that form equal to zero instead. Such a step decreases the potential by at least $d + 1$, so there are at most $(s - r)/(d + 1)$ of these steps. Each step decreases the dimension, and the procedure therefore ends with both required properties.

At every step select one coordinate carrying the form just killed. Its restriction was nonzero before that step, so its original coordinate form is independent of the previously selected forms. The selected coordinates constitute D, and consequently $\operatorname{codim}_R R' = |D|$. □

Corollary T2.3 (Constraints meeting three blocks)**.** *Suppose V is partitioned into three blocks and each row of a family of binary constraints annihilating R has at most one occurrence in each block. Apply Lemma T2.2 and delete the coordinates whose forms on R' are zero. Every remaining row has weight zero, two, or three. A row of weight two joins equal nonzero forms. A row of weight three has three distinct nonzero forms a, b, c with $a + b + c = 0$.*

The connected components formed by the rows of weight two have at most K vertices. After repeated rows with the same support are identified, the number of ternary rows incident to any vertex is at most $2dK^2$.

Proof. The sum of the forms on a row is zero, since that row annihilates R'. A row of weight one is therefore impossible. At weight two its forms are equal. At weight three a repeated form would force the third form to be zero, which has already been excluded. Binary components stay inside a class of equal forms, of size at most K. At a vertex of form ℓ, a ternary row chooses one of at most d decompositions $\ell = a + b$, then the other two coordinates from classes of size at most K. Allowing their two assignments to the remaining blocks gives the asserted $2dK^2$ bound. □

Remark T2.4 (Order of restrictions)*.* If some coordinates must first be killed for an arithmetic reason, do this before applying the regularisation lemma. A further restriction after regularisation can identify previously distinct forms and destroy both bounds. The lemma is an existence statement about a subspace; its proof does not enumerate possible relation spaces or histories of restrictions.

T2.3 A finite rank bound for every fixed channel

Fix coprime integers a, b with

$$b \geq 3, \qquad b/2 < a < b.$$

Let $\mathcal{G}_{a:b}$ be the symmetric family of ordered separated pairs in this channel with at least two exact units and with a private prime coordinate on every distinct normalized value in $bV_x \cup aV_y$, above a fixed threshold $H_{a,b} = C_{a,b}B$. Choose the constant large enough for the fixed normalized widths. For $\alpha = (x, y)$ put

$$q_\alpha = \mathbb{P}(J_x = J_y = 1), \qquad M_{a:b} = \sum_{\alpha \in \mathcal{G}_{a:b}} q_\alpha.$$

Pairs of probability zero remain in the family and are omitted from conditional expressions. The private-prime condition makes the relative kernel exactly the canonical even-selection kernel.

Lemma T2.5 (Mass of a fixed channel)**.** *With $S_{a,b}(L)$ as defined before (3.9), and $w_{a,b} = 2(2a - b)/(ab)$, one has*

$$M_{a:b} = p^2 N w_{a,b} S_{a,b}(L) + O_{a,b,C,\varepsilon}(N^{-1+\varepsilon}).$$

In particular $M_{a:b} \asymp_{a,b,C} LN^{-1+1/b}$.

Proof. In either orientation the canonical parameter satisfies $N/a \le t < 2N/b$, up to fixed multiples of B at the endpoints. The two orientations therefore supply $Nw_{a,b}$ translations per choice of the within-window placements. The diameter and sign calculation giving (3.9) applies for this fixed a, b.

Removing the normalized defective pairs costs $N^{-3/2+1/b+o(1)}$: there are $N^{1/2+o(1)}$ such hosts, their nullity is at most $L/b + o(L)$, and their weights include p^2. Endpoint errors are smaller. On the retained dependent pairs the signed kernel contribution is $q_\alpha - p^2$. Adding back p^2 costs $O_{a,b}(p^2 NB)$, because each start has $O_{a,b}(B)$ possible partners. This proves the displayed formula. Its leading term is positive and of order $LN^{-1+1/b}$, by the finite identity, uniformly in the critical band. □

Define the finite, explicitly computable number

$$\gamma_{a:b} = \max\left(\left\{\frac{1}{b}\right\} \cup \left\{\frac{(c,a)+(c,b)-1}{c} : \begin{array}{l} 1 \le c \le b(a+b-1),\ d \ge 1,\ (c,d)=1, \\ b/2 < c/d < 2a, \quad c \nmid a, \quad c \nmid b \end{array}\right\}\right). \tag{T2.1}$$

All variables in the second set are integers.

Lemma T2.6 (Additional rank in a fixed channel)**.** *One has $\gamma_{a:b} \le 2/3$. Apart from $B^{O_{a,b}(1)}$ source pairs at the dyadic endpoints, every aligned target $z \ne x, y$ compatible with a source $(x, y) \in \mathcal{G}_{a:b}$ satisfies*

$$\delta_{x,y}(z) := L - \dim \frac{S_x + S_y + S_z}{S_x + S_y} \le \gamma_{a:b} L + o_{a,b,C}(L).$$

This definition includes any internal rank deficiency of the new window. The bound remains true when the prime signs of a, b are included in the old information.

Proof. Write $x = at - i_0$, $y = bt - j_0$, choosing the lower orientation. As in Lemma T1.11, the normalized old values lie in a fixed-width interval about abt. An aligned target has the form $z = (ct + h)/d$, with $(c, d) = 1$, numerator, denominator and offsets of polynomial size in B. The dyadic positions require $b/2 \le c/d \le 2a$, up to $B^{O(1)}/N$. Separation of rational fractions excludes strict violations of this interval once N is large. Either equality forces $at = N + B^{O(1)}$ or $bt = 2N + B^{O(1)}$; these give the stated exceptional sources.

For the interior fractions, a new offset k matched to a true old value satisfies one of

$$h + dk \equiv 0 \pmod{c/(c,a)}, \qquad h + dk \equiv 0 \pmod{c/(c,b)}.$$

Their intersection has modulus c. The union contains at most

$$B\,\frac{(c,a)+(c,b)-1}{c}+O(1)$$

new positions. The same residual argument as in Lemma T1.11 leaves only $o(L)$ further dimensions. Matching to a value in an enclosing interval but absent from the true old set contributes no new freedom.

Suppose $c\nmid a$ and $c\nmid b$, and put $g=(c,a)$, $h_1=(c,b)$. These numbers are coprime and $gh_1\mid c$. If one equals one, the other is a proper divisor of c, giving density at most $1/2$. Otherwise $g,h_1\ge 2$ and

$$\frac{g+h_1-1}{c}\le\frac{1}{g}+\frac{1}{h_1}-\frac{1}{gh_1}\le\frac{2}{3}.$$

The last maximum occurs at the smallest coprime pair $\{2,3\}$. When $c>b(a+b-1)$, the density is less than $1/b$. All other nonphysical fractions are included in (T2.1).

If $c\mid a$, a proper divisor has $c\le a/2<b/2$, which is impossible here; thus $c=a,d=1$. Similarly $c\mid b$ leaves only $c=b,d=1$. These are the physical fractions reproducing one old family. Compatibility gives distance at least L from that source, so the two true windows share at most one value. Matches to the other old family have steps a,b in intervals of width B, and number at most $B/b+O(1)$. This proves the rank bound in these cases as well. □

Proposition T2.7 (Palm mean for a fixed channel)**.** *Put*

$$\eta_{a:b}=\min\{17/24,1-\gamma_{a:b}\}\ge 1/3.$$

Then, with the sum over $z\in I_N\setminus\{x,y\}$,

$$R_{a:b}:=\frac{1}{M_{a:b}}\sum_{\alpha=(x,y)\in\mathcal{G}_{a:b}}q_\alpha\sum_z\big(\mathbb{P}(J_z=1\mid J_x=J_y=1)-p\big)_+\ll_{a,b,C,\varepsilon}N^{-\eta_{a:b}+\varepsilon}.$$

Moreover

$$\frac{1}{M_{a:b}}\sum_{\alpha=(x,y)\in\mathcal{G}_{a:b}}q_\alpha\mathbb{E}[Z_N-2\mid J_x=J_y=1]=\lambda+O_{a,b,C,\varepsilon}\big(N^{-\min(\eta_{a:b},1/2)+\varepsilon}\big).$$

These bounds are not asserted to be optimal, or uniform in growing channel height.

Proof. The source probability is $O_{a,b,C}(p^2 2^{L/b})$. The nonaligned asymmetric estimate in Lemma T1.10, with the fixed larger widths and $B^{O_{a,b}(1)}$ patterns, bounds the total positive excess by

$$p^3 2^{L/b}N^{31/24+\varepsilon'}.$$

Division by $M_{a:b}\asymp NLp^2 2^{L/b}$ gives $N^{-17/24+\varepsilon}$.

Each source has $B^{O_{a,b}(1)}$ aligned targets. Lemma T2.6 bounds their positive conditional excess by $pN^{\gamma_{a:b}+o(1)}$ per target. The exceptional endpoint sources have total weight $B^{O(1)}p^2 2^{L/b}$. Even bounding every conditional probability by one, their aligned contribution after division by $M_{a:b}$ is $N^{-1+o(1)}$. This proves the first assertion, including all new windows.

For the mean, Proposition T1.15 with two exposed source windows leaves at most $N^{1/2+o(1)}$ exceptional targets for each source. Outside them the new relative vector is uniform and independent of the old values, so its conditional probability is exactly p. The total negative deviation is therefore at most $pN^{1/2+o(1)}$. Combining it with the positive estimate and the removal of the two sources, which costs $2p$, proves the assertion. □

T2.4 The ternary mass generated by a finite family of pairs

Corollary T2.8 (Triples containing a good pair)**.** *Let $\mathcal{F}$ be a nonempty finite fixed family of admissible channels, and set*

$$M_{\mathcal{F}} = \sum_{a:b\in\mathcal{F}} M_{a:b}, \qquad \nu_{\mathcal{F}} = \min\{1/2, \min_{a:b\in\mathcal{F}} \eta_{a:b}\}.$$

The sum of $q(\{x,y,z\})$ over all ordered triples of distinct starts containing at least one good pair from this family is

$$3\lambda M_{\mathcal{F}} + O_{\mathcal{F},C,\varepsilon}(M_{\mathcal{F}} N^{-\nu_{\mathcal{F}}+\varepsilon}).$$

In particular, for fixed $Q \geq 5$, the triples containing a good pair of height $5 \leq b \leq Q$ have total mass $O_{Q,C,\varepsilon}(N^{-4/5+\varepsilon})$.

Proof. Choose which two positions of an ordered triple form the designated pair. For each of the three choices, Proposition T2.7 gives $\lambda M_{\mathcal{F}} + O(M_{\mathcal{F}} N^{-\nu_{\mathcal{F}}+\varepsilon})$. The pair orientations are already included in $M_{\mathcal{F}}$.

This designated-pair sum counts a triple once for each good pair that it contains. If the third window is independent of all old values of the designated pair, it cannot form a dependent good pair with either source. Every overcount therefore has its third start in the exceptional set of Proposition T1.15. For each designated source, the conditional mass of this set is at most

$$pN^{1/2+o(1)} + \sum_z \big(\mathbb{P}(J_z = 1 \mid J_x = J_y = 1) - p\big)_+.$$

Its weighted average is $O_{\mathcal{F},C,\varepsilon}(N^{-\nu_{\mathcal{F}}+\varepsilon})$. This bounds the entire overcount, since a triple has at most three pairs. The finite channel families are disjoint for large N, by the separation of distinct reduced slopes. Finally, Lemma T2.5 gives $M_{\mathcal{F}} \ll_{Q,C} LN^{-4/5}$ for heights between five and Q. □

The last corollary concerns the part of the third moment containing a good pair. It makes no assertion about the primitive remainder or about a family whose heights grow with N.

T3 Pair profiles and the regular-set estimates

This section supplies the estimates used in Section 3. The regular-set transfer is used for the explicit channel formulas and for the positional comparison. The scalar reconstruction with exact marginals does not discard these exceptional windows.

T3.1 Regular-set transfer

Proof of Proposition 3.1. The one-window estimates proved in Lemma T1.3 give $m_B(x) = O_C(B/\log B)$ uniformly, and only $N^{o(1)}$ windows contain two B-defective integers. A nonzero relative relation is an even nonempty selection. Private prime coordinates therefore force it to use at least two defects. Its dimension is at most $m_B(x)$. Consequently

$$\sum_{x\in I_N} (2^{L-\dim S_x} - 1) = N^{o(1)}.$$

Fourier inversion proves (3.3). The sum of the probabilities of active discarded starts is at most $pN^{o(1)}$; the coupling using the original prime signs proves the total-variation assertion. On G_N the marginal is exactly p, giving both assertions about the mean.

For the last statement, let $D = \{Z_N \neq Z_N^\circ\}$. Since $0 \leq Z_N^\circ \leq Z_N$, Hölder's inequality gives, for any fixed $s > 1$,

$$\mathbb{E}[Z_N^j \mathbf{1}_D] \leq (\mathbb{E} Z_N^{js})^{1/s} \mathbb{P}(D)^{1-1/s}.$$

Choose a sufficiently large integer s, and then use the fixed-moment bound. This last step is used only after that bound has been established; a small coupling error alone would not justify a variance transfer. □

T3.2 Sparse hosts and weighted residual mass

Proof of Proposition 3.2. A relative relation that is internal to one window contributes at most $N^{1+o(1)}$ hosts, by Proposition 3.1. Otherwise it selects at least two vertices in each window. Put $T = N^{3/5}$. If all selected kernels are at most T, each window contains two small kernels. The estimate

$$\#\{x \in I_N : \#\{n \in V_x : K_B(n) \le N^{\beta}\} \ge 2\} \ll_{C,\beta,\varepsilon} N^{\max(\beta,7/12)+\varepsilon} \tag{T3.1}$$

proved in Lemma T1.5 gives $N^{6/5+\varepsilon}$ choices. If one selected kernel exceeds T, Lemma T1.6 gives $N^{3/2+o(1)}T^{-1/2+\tau}$, for every fixed $\tau > 0$. Choose τ sufficiently small.

For the weighted assertion, first restrict to $c(x, y) < \eta B$. Equation (3.5) bounds the weight by $2^{\eta B+o(B)}$; choose η after the desired error exponent. The complementary sector has total weight $O_{C,\eta,\varepsilon}(N^{1+\varepsilon})$ by Lemma T1.8. The latter estimate uses the component equations themselves, rather than just their support: many small components put the pair on a multiquadratic surface, and its low-degree curves are either aligned or absent from the dyadic region. □

T3.3 The complete pair profile

Proof of Theorem 3.3. For the channel $2 : 3$, either orientation permits $N/2 \le t < 2N/3$, up to $O(B)$ endpoint changes. For $3 : 4$ the corresponding interval is $N/3 \le t < N/2$. Both intervals have length $N/6 + O(B)$. The two orientations therefore give $N/3$ times the finite sums above. Translations near a boundary contribute $O(B^3 2^{L/b})$.

For either fixed channel choose H a sufficiently large fixed multiple of B and normalize the values to $bV_x \cup aV_y$. Its diameter is $O(B)$. If its distinct values are all nondefective at level H, each has a private prime and the only relations before normalization are the even selections of exact units. Lemma T1.3 shows that the exceptional channel pairs number $N^{1/2+o(1)}$. Lemma T1.9 bounds their nullity by $L/b + o(L)$. Replacing the rational kernel by the full arithmetic kernel consequently costs at most $N^{1/2+1/b+o(1)}$, for either the positive or the signed sum.

There are $NB^{O(1)}$ aligned hosts altogether. The slope of height two forces both starts within $O(B)$ of the dyadic endpoints; its mass is $N^{1/2+o(1)}$. Height one has no exact unit in separated windows. The only reduced slopes of heights three and four that occur in the interior are $2 : 3$ and $3 : 4$. Every remaining channel has height at least five and nullity at most $L/5 + o(L)$, again by Lemma T1.9; its total contribution is $N^{6/5+o(1)}$. Proposition 3.2 controls the nonaligned sector. The inequality following (3.1) permits exactly the same error estimates for the signed sum.

At distances $1, \dots, L-1$, two run starts are incompatible. At distance L, their edge systems form a tree on an interval of diameter $2L$; Lemma T1.3, at threshold $2L + 1$, bounds its nullity by $o(L)$. Hence the entire contact band changes the second factorial moment by $O(N^{-1+o(1)})$. Multiplying (3.13) by p^2, and using $\mu_N - \lambda = O(N^{-1+o(1)})$, proves (3.14). Finally $2^{(L-r)/b} = 2^{-r/b}(N/\lambda)^{1/b}$ in (3.10)–(3.11) gives the two equivalents. □

T3.4 The good-edge mass

Proof of Corollary 3.4. Each start has $O(B)$ possible partners in a fixed channel. The rational kernel has dimension one less than the number of exact units when that number is positive, giving the rank bound. All pairs excluded from the good family have precisely the error mass estimated in the proof of Theorem 3.3, together with the contact band. For good compatible pairs the centered contribution is $q_e - p^2$. Since $p^2|\mathcal{E}_N| = O(B/N)$, replacing it by q_e has the stated cost. The asymptotic follows from (3.15). □

T3.5 A further window after a good edge

The following channel-specific estimate is used in the first reconstruction and in the positional comparison. Its good-edge notation is that of Section 3.

Proposition T3.1 (Positive Palm cost of a good edge). *For $a : b \in \{2 : 3, 3 : 4\}$ let $\mathcal{E}_{a,b}$ be its good edges and put $\beta_{a,b} = \sum_{e\in\mathcal{E}_{a,b}} q_e$. Then*

$$\frac{1}{\beta_{2,3}} \sum_{\{x,y\}\in\mathcal{E}_{2,3}} q_{xy} \sum_{z\in G_N\setminus\{x,y\}} (\mathbb{P}(J_z = 1 \mid J_x = J_y = 1) - p)_+ \ll_{C,\varepsilon} N^{-2/3+\varepsilon}, \tag{T3.2}$$

$$\frac{1}{\beta_{3,4}} \sum_{\{x,y\}\in\mathcal{E}_{3,4}} q_{xy} \sum_{z\in G_N\setminus\{x,y\}} (\mathbb{P}(J_z = 1 \mid J_x = J_y = 1) - p)_+ \ll_{C,\varepsilon} N^{-1/2+\varepsilon}. \tag{T3.3}$$

Consequently the unnormalised positive two-block masses are $O_{C,\varepsilon}(N^{-4/3+\varepsilon})$ and $O_{C,\varepsilon}(N^{-5/4+\varepsilon})$, respectively.

Proof. We give the two geometric estimates needed for this bound in Lemma T1.10 and Lemma T1.11. Compress the source pair using Lemma 3.5. Away from alignment with this source, the asymmetric profile gives $N^{31/24+\varepsilon}$ for the sum of $2^\delta - 1$, where δ is the loss of new rank. There are $B^{O(1)}$ source offset patterns and $q_{xy} \ll p^2 2^{L/b}$. Multiplication by the additional factor p and division by $\beta_{a,b} \asymp LN^{-1+1/b}$ give $N^{-17/24+\varepsilon}$.

For aligned targets there are $B^{O(1)}$ possibilities per source. Outside $B^{O(1)}$ source pairs at the dyadic endpoints, Lemma T1.11 bounds the new rank loss by $L/3 + o(L)$ for $2 : 3$ and $L/2 + o(L)$ for $3 : 4$. Thus their positive conditional cost is at most $pB^{O(1)}2^{L/3+o(L)}$ or $pB^{O(1)}2^{L/2+o(L)}$. The exceptional source mass is $N^{-2+1/b+o(1)}$ and each has only $B^{O(1)}$ aligned targets, so after normalization it contributes $N^{-1+o(1)}$. This proves both estimates. The source mass itself is $N^{-1+1/b+o(1)}$, which gives the final assertions. □

T4 Fixed-source geometry and the independent moment proof

The statements of the fixed-source estimates and the fixed-moment theorem are in Section 6. This section supplies the geometric count and the positive classification used in their proofs. The original window estimates below use the regular-set convention $p = 2^{-L}$; Section 2 gives their extension to effective spaces. The moment proof is independent of either reconstruction of the law.

T4.1 Uniformly few targets of large rank loss

We first record the one-parameter geometric estimate in the precise form needed here.

Lemma T4.1 (Several disjoint square equations). *Fix $K, A, D_0 > 0$ and $\varepsilon > 0$. There is a fixed integer j such that the following holds. Let $P_1, \dots, P_j$ be nonconstant products of distinct linear factors $Z + u$, of degrees at most K, with $|u| \le D_0 B$. Suppose their root lists are pairwise disjoint, and let $0 < |D_\nu| \le N^A$ be integers. Then*

$$\#\{z \asymp N : w_\nu^2 = D_\nu P_\nu(z) \ (1 \le \nu \le j) \text{ for some integers } w_\nu\} \ll_{K,A,D_0,\varepsilon} N^\varepsilon.$$

The constants are uniform in the displayed coefficients and offsets.

Proof. The square classes of the functions $D_\nu P_\nu(Z)$ are geometrically independent: a root of P_ν has odd valuation in that class and even valuation in every other class. Thus the resulting curve is geometrically integral and has degree 2^j over the Z-line.

Choose an integer $D \geq (K+A)/2$. Give T, S degree one and each W_ν degree D, and use the equations

$$W_\nu^2 = D_\nu S^{2D-\deg P_\nu} \prod_{u \in P_\nu} (T + uS).$$

The graded ring is free of rank 2^j over $\mathbb{Q}[T,S]$; its generic localization is a field. Its D-th Veronese ring is generated in degree one by the monomials in T, S of degree D and the W_ν. The corresponding projective curve has degree $D2^j$, since its hyperplane bundle is the pullback of $\mathcal{O}_{\mathbb{P}^1}(D)$. On $S = 1$, the coordinates are integral and have height $O_{K,D}(N^D)$.

Pila's uniform curve estimate [21, Theorem A] bounds their number by

$$O_{K,A,j,\varepsilon'}\big((N^D)^{1/(D2^j)+\varepsilon'}\big) = O(N^{2^{-j}+D\varepsilon'}).$$

Choose j and then ε' to put this exponent below ε. Both choices precede N. □

Lemma T4.2 (A product certificate with fixed sources)**.** *Fix $s, k \geq 1$. Uniformly over sources $x_1, \dots, x_s \in I_N$, the number of centres $z \in I_N$ for which there are pairwise coprime $G_1, \dots, G_k > 1$ satisfying*

$$G_i \mid x_{a_i} + u_i, \qquad G_i \mid z + w_i, \qquad \prod_i G_i > N, \quad u_i, w_i \in \{-1, \dots, L-1\},$$

is $O_{s,k,C,\varepsilon}(N^\varepsilon)$ for every $\varepsilon > 0$.

Proof. There are $(sB)^k B^k = N^{o(1)}$ choices of positions. For fixed positions, the product $Q = \prod_i G_i$ divides $\prod_i (x_{a_i} + u_i) \leq (3N)^k$. Its divisors and their ordered factorizations into k factors have $N^{o(1)}$ possibilities. Coprimality fixes one residue class for z modulo $Q > N$, hence at most one centre in I_N. This counts the certificate without requiring the groups to be fixed in advance. □

T4.2 Proof of Proposition 6.2

Proof. Take $H = D_0 B$, with fixed $D_0 > 1$. A prime exceeding H has at most one occurrence of odd valuation in each window. Apply Lemma 6.1 to $R(\boldsymbol{x}, z)$, with $d = s$, choosing t so that the loss of dimension is at most $\eta L/4$. The surviving target projection has dimension at least $3\eta L/4$. Nonzero labels have multiplicity at most $K = t - 1$.

For each prime $q > H$ incident to an active target coordinate whose label has odd multiplicity in the q-row, choose a minimal circuit of distinct labels containing this label. Such a circuit exists: a minimal zero-sum subset containing the label cannot contain a proper zero-sum subset, because it could be removed. Choose incident coordinates for its other labels. They all belong to sources, because q has only one target occurrence. Group primes according to the selected physical support. These groups are pairwise coprime. The number of supports through a fixed target coordinate is at most

$$D = \max\left(1, \sum_{j=2}^{s} j! t^{j-1} K^j\right).$$

Choose the integer j in Lemma T4.1 for degree at most K and coefficient height N^{A_0+2K+2}, where $\prod_{q \leq H} q \leq N^{A_0}$. Choose k large enough that $\theta = KD/k \leq 1$ and $j\theta < \varepsilon/4$. If the product of the k largest nontrivial groups exceeds N, Lemma T4.2 applies. It remains to count the complementary case.

Discard from selection the label classes meeting the target positions of these k groups. At most k classes are discarded; no further restriction of the relation space is made. Every group incident to a remaining target class has size at most $N^{1/k}$. For such a class A, write

$$M_A = \prod_{v \in A \text{ at sources}} n_v, \qquad P_A(Z) = \prod_{v \in A \text{ at target}} (Z + w_v).$$

Its squarefree product $d_A = \mathrm{sf}(M_A P_A(z))$ can be factored as

$$d_A = d_{0,A} d_{1,A} e_A, \qquad d_{0,A} \mid \prod_{q \le H} q, \quad d_{1,A} \mid M_A, \quad 1 \le e_A \le N^{\theta}.$$

Indeed, every remaining prime with odd multiplicity and not dividing M_A lies in one of at most KD incident groups. The choices of $d_{0,A}$ and $d_{1,A}$ are $N^{o(1)}$, uniformly for fixed source positions.

There are at least $3\eta L/4 - k$ available classes meeting the target. Select j of them. Their target polynomials have simple disjoint roots. After choosing their positions and coefficients, the equations

$$w_\nu^2 = d_{A_\nu} M_{A_\nu} P_{A_\nu}(z)$$

are counted by Lemma T4.1. The positions cost $B^{O(1)}$, and the coefficients cost $N^{j\theta+o(1)}$. The stated choices, with the error margin split between these factors, give (6.3). Only the selected positions and coefficients have been enumerated, not the possible relation spaces or their regularizations. □

T4.3 Primitive families at a fixed order

A family is *primitive* if it has a nonzero relative relation and every proper subfamily has none. Its relation space projects injectively to each block. For original windows its nullity is consequently at most L. For the homogeneous prescription define

$$W(I) = p^{|I|} 2^{\rho(I)}.$$

This is an actual joint probability, regardless of the compatibility of the run-start prescriptions. Those have joint probability either $W(I)$ or zero.

Proposition T4.3 (Fixed primitive mass)**.** *For every fixed $r \ge 3$ and $\varepsilon > 0$,*

$$\sum_{\substack{(x_1,\dots,x_r)\ \text{primitive}\\ \text{distinct, separated}}} p^r 2^{\rho(x_1,\dots,x_r)} \ll_{r,C,\varepsilon} N^{-4/5+\varepsilon}. \tag{T4.1}$$

In particular the same estimate holds for the affine joint probabilities of the original events.

Proof. We give the rank stratification, including the order of the parameter choices. Fix a small $\eta > 0$, and first consider $\rho \ge \eta L$. With $H = D_0 B$, regularize the relation space using Lemma 6.1, $d = r - 1$, and codimension at most $\eta L/4$. For each prime $q > H$, pair equal active labels in its row. The remaining distinct labels sum to zero; partition them into minimal circuits. Each circuit has between three and r coordinates in distinct blocks.

The graph of the binary pairs has components of size at most $K = t - 1$. A component contained in a single block is a singleton, because its edges always join distinct blocks. For each physical circuit support u, let G_u be the product of primes assigned to it. The number of such supports through a coordinate is at most

$$D = \sum_{j=2}^{r-1} j! t^{j-1} K^j.$$

A prime can belong to several groups, but its selected circuit supports are disjoint.

Put $M = \binom{r}{3}$, and let $G_{(1)}, \dots$ be the groups in decreasing order. If the product of the k largest groups exceeds N^M, assign each support to one triple of its blocks. One triple receives groups of product greater than N. Those groups are coprime: two disjoint supports for the same prime cannot both meet the same three blocks. Fuse groups with the same triple of offsets. The ternary budget of Proposition T1.16 counts the three centres by $N^{6/5+\varepsilon}$. For completeness, the cases with only one or two fused groups have an even smaller count. Write $Q > N$ for their product.

With one group, $Q \ll N$ because it divides one value in each of the three blocks. With two groups, $Q \ll NB$ by the product bound at the start of the proof of Proposition T1.16; a vanishing determinant would give the pair relation excluded by primitivity. For fixed offsets and a fixed factorization of Q, the coprime groups impose one residue class modulo Q on each center. Since $Q > N$, CRT leaves $O(1)$ choices for each center in its interval. Summing over $N < Q \ll NB$, their factorizations into at most two factors, and the $B^{O(1)}$ offsets gives $N^{1+o(1)}$ hosts, below the required $N^{6/5+\varepsilon}$. Fixed dilations and compressed blocks only change the fixed constants in these bounds; their direct-sum condition excludes the same zero determinant. The remaining centres are free. Thus this sector has

$$O_{r,k,C,\varepsilon}(N^{r-9/5+\varepsilon}) \quad \text{hosts.} \tag{T4.2}$$

In the complementary sector, omit from selection all binary components meeting those k groups. At most rk components are omitted, and the subspace is not restricted again. Every remaining group is at most $N^{M/k}$. For each selectable binary component A,

$$K_H\left(\prod_{v\in A} n_v\right) \le N^{\theta}, \qquad \theta = \frac{MKD}{k}. \tag{T4.3}$$

Every prime occurring oddly in this product must belong to a circuit group incident to A; binary edges have both endpoints in the component. There are at most KD incident groups. The squarefree coefficient of the product consequently has $N^{\theta+o(1)}$ possibilities for fixed positions.

Every block meets at least $\dim R'$ binary components, because its projection of R' is injective. Their types $(a_1, \dots, a_r)$, with $\sum a_i \le K$, form a fixed finite list. If one transverse type occurs sufficiently many times, choose a fixed number of its components. Fix all but two of their centres and apply the two-parameter square-equation estimate of Lemma T1.7. Primitivity excludes the pair relations needed to remove the small curves. This gives $N^{r-3/2+\varepsilon}$ hosts. The number of equations is chosen for the increased coefficient height after the other centres are fixed. Then k is chosen large enough to absorb all costs from θ and the offsets.

If no transverse type occurs sufficiently often, only boundedly many transverse components remain. Select one singleton in each block. The one-value small-kernel bound of Lemma T1.13 gives

$$N^{r/2+r\theta/2+o(1)} \ll N^{r-3/2+\varepsilon}, \qquad r \ge 3.$$

Together with (T4.2), this proves

$$\#\{\text{primitive hosts} : \rho \ge \eta L\} \ll N^{r-3/2+\varepsilon}. \tag{T4.4}$$

If $\rho > (1/2+\delta)L$, regularize instead with loss at most $\delta L/4$. A sufficiently repeated transverse type with a singleton in one of its blocks is counted by the stronger, degree-one case of Lemma T1.7, giving $N^{r-2+\varepsilon}$ hosts. Otherwise, let d_i be the number of selectable singleton components in block i. All other components use at least two of its B coordinates, apart from boundedly many exceptions. Hence

$$\dim R' - rk \le d_i + \frac{B-d_i}{2} + O(1), \qquad d_i \ge \frac{3\delta L}{2} - O(1).$$

Choose two singleton components in each block. The uniform two-value Pell estimate gives $N^{2r\theta+o(1)} \ll N^{r-2+\varepsilon}$ hosts. Restoring the group tail proves

$$\#\{\text{primitive hosts} : \rho > (1/2+\delta)L\} \ll N^{r-9/5+\varepsilon}. \tag{T4.5}$$

It remains to multiply these counts by their joint weights. For $0 < \rho < \eta L$, the support estimate of Proposition T1.14 gives mass

$$N^{r-2r/(2r+1)+\varepsilon} N^{-r+\eta} = N^{-2r/(2r+1)+\eta+\varepsilon}.$$

For $\eta L \le \rho \le (1/2+\delta)L$, (T4.4) gives $N^{-1+\delta+\varepsilon}$. For the remaining sector, $\rho \le L$ and (T4.5) give $N^{-4/5+\varepsilon}$. Choose $\eta < 2/35$, $\delta < 1/5$, and reduce all intermediate error margins. This proves (T4.1). □

The proof is homogeneous throughout. It therefore bounds incompatible configurations as well as compatible ones. This feature will be needed for centered Fourier coefficients.

Proposition T4.4 (Fixed primitive mass after pair compression)**.** *Fix a channel $a : b \in \{2 : 3, 3 : 4\}$, and put $\kappa_b = 2 - 1/b$. For every fixed $r \geq 3$, the total homogeneous mass of primitive families consisting of one exactly compressed pair in this channel and $r - 1$ further original windows is*

$$O_{r,C,\varepsilon}(N^{-\kappa_b+1/5+\varepsilon}). \tag{T4.6}$$

For a compressed pair and just one further original window the corresponding masses are, respectively, $O_\varepsilon(N^{-4/3+\varepsilon})$ and $O_\varepsilon(N^{-5/4+\varepsilon})$. The affine positive masses satisfy the same bounds.

Proof. For a fixed compression motif, the compressed block is the even-selection space on a genuine set of at most $2B$ values in an interval of width $O(B)$. Its rank r_e satisfies $r_e \geq \kappa_b L - O(1)$. There are $B^{O(1)}$ motifs. Primitivity after compression makes projection to every remaining original block injective, so its additional nullity is still at most L. Its homogeneous weight is

$$2^{-r_e-(r-1)L+\rho} \ll_C N^{-\kappa_b-r+1+\rho/L}.$$

The group argument and the high-nullity count in the preceding proof use only fixed numbers of true values, width $O(B)$, and pairwise direct relative spaces. They therefore apply to the compressed blocks. In the very-high-nullity case, the singleton budget is used only on the $r - 1$ original blocks; the compressed centre is left free. The Pell count becomes

$$N^{1+2(r-1)\theta+o(1)} \ll N^{r-2+\varepsilon}, \qquad r \geq 3.$$

The three nullity sectors have masses

$$N^{-\kappa_b+1/(2r+1)+\eta+\varepsilon}, \quad N^{-\kappa_b+\delta+\varepsilon}, \quad N^{-\kappa_b+1/5+\varepsilon}.$$

Choose η, δ small and sum the motifs.

The two-block case is the positive extension estimate in Proposition T1.12. For affine compatible extensions it also follows directly from the positive Palm bounds: if the additional rank loss is an integer $d \geq 1$,

$$p2^d \leq 2(p2^d - p)_+.$$

The ordered pair mass times its mean positive Palm excess is $O(N^{-4/3+\varepsilon})$ in $2 : 3$ and $O(N^{-5/4+\varepsilon})$ in $3 : 4$. □

T4.4 Proof of Theorem 6.4

Proof. Expand $\mathbb{E}(Z_N)_r$ over ordered distinct centres. We describe all configurations removed before evaluating the two principal classes.

Local defects and contacts have mass $N^{-1+\varepsilon}$ after propagation. A dependent separated pair outside the two retained channels has mass $N^{-4/5+\varepsilon}$, also after propagation. Tuples containing a primitive original subfamily of size at least three have that same bound, by Proposition T4.3. The positive mass of two good pairs is $O(N^{-5/4+\varepsilon} + a_N^2)$, including pairs sharing a vertex, by Proposition T1.12; propagation again preserves the required precision.

There remains the case of one good pair with an additional relation. Compress this pair exactly. A minimal dependent subfamily of the compressed system has at least two blocks and contains the compressed block, unless an original primitive family has already been removed. Apply Proposition T4.4 to this seed and then propagate. Each of its exponents exceeds $4/5$. This exhausts the dependent residual configurations: after their removal the spaces are either all direct, or one good pair is direct from all remaining singletons.

The support estimate of Proposition T1.14, together with the $O_r(N^{r-1}B)$ contact tuples, shows that the number of full-rank tuples is

$$N^r + O_{r,C,\varepsilon}(N^{r-4/5+\varepsilon}).$$

Their joint probability is exactly p^r, and their contribution is $\lambda_N^r + O(N^{-4/5+\varepsilon})$.

Let A_N be the ordered positive mass of good pairs. The pair calculation gives

$$A_N = a_N + O_{C,\varepsilon}(N^{-4/5+\varepsilon}).$$

Fix one such pair. After any fixed number of already chosen windows, at most $N^{1/2+o(1)}$ target centres are dependent on their relative space, by Proposition T1.15. Thus the number of ordered direct extensions by $r-2$ windows is

$$N^{r-2}(1 + O_{r,C}(N^{-1/2+o(1)})),$$

uniformly in the pair. Their joint weight is $q(e)p^{r-2}$. Placing the ordered pair in two of the r positions gives contribution

$$\binom{r}{2}\lambda_N^{r-2}A_N + O_{r,C}(A_N N^{-1/2+o(1)}).$$

Since $A_N = O_C(LN^{-2/3})$, the last error is smaller than the stated remainder. The two principal classes give the formula with (λ_N, a_N). The mean transfer $\mu_N - \lambda_N = O(N^{-1+\varepsilon})$ and $\alpha_N - a_N = O(N^{-1+\varepsilon})$ give (6.6).

The Stirling-number identity $Z^r = \sum_{j=0}^{r} \left\{ {r \atop j} \right\} (Z)_j$ proves (6.7) and boundedness. Finally, to any fixed formal order, the factorial generating function equals

$$e^{\mu_N t}\left(1 + \frac{\alpha_N t^2}{2}\right) + O_{\mathrm{coeff}}(N^{-4/5+\varepsilon}).$$

Taking its formal logarithm is legitimate at that fixed order. Terms quadratic in α_N are $O(N^{-4/3+o(1)})$, which proves the cumulant claim. □

T5 Short split products: kernels and moment enumeration

This section proves the short-support moment input from Section 9. The degree is allowed to grow as a sufficiently small multiple of $\log\log N$; all estimates below retain that dependence.

Lemma T5.1 (A quadratic kernel with a prescribed divisor)**.** *Let $1 \le h \le 2DB$, where D is fixed, and $n \asymp N$. Suppose $q = \mathrm{sf}(n(n+h)) \asymp N^{\beta}$ and a fixed squarefree integer e divides q. Put $\delta = \log e / \log N$. Uniformly in these data, the number of n is at most*

$$N^{c_2(\beta,\delta)+o(1)}, \qquad c_2(\beta,\delta) = \min\{\beta - \delta, \max(\beta/2 - \delta, 1 - \beta/2)\} \le \tfrac{1}{2} + \tfrac{\beta}{4} - \tfrac{\delta}{2}.$$

Every $o(1)$ in this lemma is uniform and may be replaced by an arbitrary fixed positive exponent.

Proof. For a fixed q, the equation $n(n+h) = qv^2$ becomes $(2n+h)^2 - 4qv^2 = h^2$. Lemma T1.2 gives $N^{o(1)}$ solutions, uniformly for the polynomial heights in question. When $q = 1$, factor the difference of squares instead. Summing over the $O(1 + N^{\beta}/e)$ possible kernels gives the first exponent.

For the other estimate, $v \asymp N^{1-\beta/2}$ and $ev^2 \mid n(n+h)$. If $\rho_h(M) = \#\{a \bmod M : a(a+h) \equiv 0 \pmod M\}$, then

$$\rho_h(\ell^a) \le 2\ell^{\min(v_\ell(h), \lfloor a/2 \rfloor)}, \qquad \rho_h(M) \le h\, 2^{\omega(M)}.$$

For $a \le 2v_\ell(h)$, a root is divisible by $\ell^{\lceil a/2\rceil}$; for larger a, the two separated roots give the displayed bound. The Chinese remainder theorem proves the second inequality. Here $M = ev^2 \le N^{2+o(1)}$, so summing residue classes gives

$$N^{o(1)} \sum_{v \asymp N^{1-\beta/2}} \left(\frac{N}{ev^2} + 1 \right) \le N^{o(1)} \{ N^{\beta/2-\delta} + N^{1-\beta/2} \}.$$

If $\beta - \delta \ge 1$, the maximum is $\beta/2 - \delta$, which is bounded by the affine expression since $\beta \le 2 + o(1)$. Otherwise the maximum is $1 - \beta/2$, and the minimum of this number and $\beta - \delta$ is at most their average. □

Lemma T5.2 (Kernels of a split product)**.** *Let*

$$P_A(X) = \prod_{a \in A} (X + a), \qquad A \subset [-DB, DB] \cap \mathbb{Z}, \qquad 2 \le d = |A| \le c \log\log N,$$

where d is even and $D, c > 0$ are fixed. For a fixed squarefree e, the number of $n \asymp N$ with $q = \mathrm{sf}(P_A(n)) \asymp N^\beta$ and $e \mid q$ is at most

$$N^{1/2+\beta/(2d)-\delta/d+c\log 2+o(1)}, \qquad \delta = \frac{\log e}{\log N}. \tag{T5.1}$$

For a fixed kernel q, the fibre has at most $N^{2c\log 2+o(1)}$ elements. All estimates are uniform in A.

Proof. Pair the d linear factors into $m = d/2$ quadratics Q_j, put $q_j = \mathrm{sf}(Q_j(n))$, and write

$$D_A = \prod_{\substack{a,b \in A \\ a<b}} |a - b|.$$

Its logarithm is $O_D(d^2 \log B) = o(\log N)$. Outside D_A, the q_j are pairwise coprime, and

$$\frac{\prod_j q_j}{q} \mid D_A^2. \tag{T5.2}$$

Indeed, a prime occurring in s of the q_j divides at least s distinct linear factors. Its exponent in the quotient is $2\lfloor s/2 \rfloor$, whereas its exponent in D_A^2 is at least $2\binom{s}{2}$.

Remove the primes of D_A from e, obtaining $e' = e/\gcd(e, \mathrm{rad}\, D_A)$. There is a factorization $e' = \prod_j e_j$, with $e_j \mid q_j$, such that the factors are pairwise coprime. Set $\beta_j = \log q_j / \log N$ and $\delta_j = \log e_j / \log N$. Then

$$\sum_j \beta_j = \beta + o(1), \qquad \sum_j \delta_j = \delta + o(1).$$

For at least one j, $\beta_j - 2\delta_j \le (\beta - 2\delta)/m + o(1)$. Lemma T5.1 bounds its quadratic count by $N^{1/2+\beta/(2d)-\delta/d+o(1)}$. We enumerate only that j, the divisor $e_j \mid e'$, and its dyadic size. This costs at most $m\tau(e')O(\log N)$, not the number of all factorizations of e'. Since $e' \le (C_D N)^d$, the product of its distinct prime factors is at least $\omega(e')!$. Stirling's estimate therefore gives

$$\omega(e') \le (c + o(1)) \log N, \qquad \tau(e') \le N^{c\log 2+o(1)}.$$

This proves (T5.1).

For a fixed q, choose two distinct offsets a, b. Both $\mathrm{sf}(n + a)$ and $\mathrm{sf}(n + b)$ divide $\mathrm{rad}(qD_A)$, so there are at most $4^{\omega(qD_A)} \le N^{2c\log 2+o(1)}$ choices. The effective coefficients u, v satisfy $u, v \ll_D N$. For each such choice, $us^2 - vt^2 = a - b \ne 0$ has $N^{o(1)}$ solutions by Lemma T1.2, including its factorizable case. Thus Pell is applied to coefficients of height $O(N)$, even though q itself may be as large as $N^{O(d)}$. □

T5.1 Proof of Proposition 9.1

Proof. The moment counts the tuples for which $\prod_i P_A(n_i)$ is a square. Write $q_i = \mathrm{sf}(P_A(n_i))$. There is a unique representation

$$q_i = \prod_{\substack{S\subseteq\{1,\dots,v\}\\ |S|\geq 2 \text{ even},\ i\in S}} d_S$$

with pairwise coprime squarefree d_S. Fix dyadic size classes for these numbers, and put $\alpha_S = \log d_S/\log N$ and $\beta_i = \sum_{S\ni i}\alpha_S$. There are $N^{o(1)}$ size profiles, since v is fixed.

Expose $v-1$ of the integers in a chosen order. Before exposing i, the product of the d_S whose sets meet an earlier index is a prescribed divisor $e_i \mid q_i$. If its exponent is δ_i, Lemma T5.2 gives the cost

$$N^{1/2+\beta_i/(2d)-\delta_i/d+c\log 2+o(1)}.$$

After choosing n_i, assign the new prime factors of q_i/e_i to the at most 2^{v-1} possible sets S. Their number is at most $N^{c(v-1)\log 2+o(1)}$; inconsistent assignments are discarded. Every S has at least two members, so the last kernel is now fixed and its fibre costs $N^{2c\log 2+o(1)}$. The enumeration costs in total at most $N^{c[v(v-1)+2]\log 2+o(1)}$.

For the remaining exponent, average over orders. A set S of size a has coefficient

$$\frac{1}{d}\left(1-\frac{a}{2}+\frac{1}{2}\mathbf{1}_{\{\text{last index}\in S\}}\right)$$

in $\sum_{i\neq\text{last}}(\beta_i/(2d)-\delta_i/d)$. Its average is $d^{-1}(1-a/2+a/(2v))$, which is $1/(dv)$ for $a=2$ and nonpositive for $a\geq 4$. As $2\sum_{|S|=2}\alpha_S \leq \sum_i \beta_i \leq vd+o(1)$, some order has total exponent at most $v/2+o(1)$. The order can depend on the size profile; its $v!$ choices do not affect the estimate. This proves (9.1). □

T6 Conditional half-counts and the certificate iteration

We prove Theorem 10.1 with its stated uniformity in arbitrary fixed source blocks. Blocks and effective spaces have the meaning specified in Section 10; in particular, the nonzero projections are required in the individual new spaces, not in their quotients by the old space.

T6.1 Small kernels and smooth moduli

Choose $H = C_D B$ larger than every block diameter, and set

$$K_H(n) = \prod_{\substack{\ell>H\\ v_\ell(n)\text{ odd}}} \ell.$$

For $q>0$, let $\mathcal{X}_q$ be the set of centres containing distinct values n_1, n_2 with $K_H(n_1)K_H(n_2)\leq N^q$.

Lemma T6.1 (Two small kernels)**.** *For every fixed $q>0$ and $\varepsilon>0$, $\#\mathcal{X}_q \ll_{q,D,C,\varepsilon} N^{q+\varepsilon}$.*

Proof. Write $n_i = c_i k_i a_i^2$, where $k_i = K_H(n_i)$ and c_i is squarefree and supported on primes at most H. The c_i have $2^{2\pi(H)} = N^{o(1)}$ choices, while the pairs $k_1k_2\leq N^q$ have $N^{q+o(1)}$ choices. Fix the two distinct offsets. Their nonzero difference h gives

$$c_1k_1a_1^2 - c_2k_2a_2^2 = h.$$

The effective coefficients are $O_D(N)$. Lemma T1.2 gives $N^{o(1)}$ solutions, including the factorizable case, and each solution determines the centre. The $O_D(B^2)$ offsets are absorbed. □

For $Y = (\log N)^A$, with fixed $A \geq 2$, write

$$n = s_Y(n) a_Y(n)^2 K_Y(n).$$

Here $s_Y(n)$ is the entire part supported on primes at most Y; $a_Y(n)$ contains the halves of the valuations at larger primes; and $K_Y(n)$ is their odd squarefree part.

Lemma T6.2 (A large smooth modulus)**.** *Fix t, $c > 0$, and $A \geq 2$. There is a deterministic set $\mathcal{M}_Y$ of centres, of cardinality $O(N^{1-1/A+\varepsilon})$, with the following property. If a selection has more than $c \log\log N$ values, its centre lies outside $\mathcal{X}_{1/2}$, and it contains fewer than t values with $K_Y(n) > N^{1/5}$, then its centre belongs to $\mathcal{M}_Y$.*

Proof. Outside $\mathcal{X}_{1/2}$, at most one value satisfies $a_Y(n)^2 > N^{3/4+1/100}$: two such values would give

$$K_H(n_1) K_H(n_2) \leq \frac{O_D(N^2)}{a_Y(n_1)^2 a_Y(n_2)^2} < N^{1/2}.$$

Remove this possible value and the at most $t-1$ large kernels. For sufficiently large N, select 26 remaining values. Their smooth parts satisfy

$$s_i \gg_D N^{1-3/4-1/100-1/5} = N^{1/25}.$$

Since $\gcd(s_i, s_j) \leq |n_i - n_j| = O_D(B)$,

$$\operatorname{lcm}(s_1, \dots, s_{26}) \geq \frac{\prod_i s_i}{\prod_{i<j} \gcd(s_i, s_j)} \geq N^{26/25 - o(1)} > C_1 N,$$

where $C_1 N$ exceeds the length of the centre interval. Multiply its prime factors, one at a time and with multiplicity, until the first passage above $C_1 N$. The resulting smooth modulus satisfies

$$C_1 N < D_0 \leq C_1 N Y, \qquad D_0 \mid \operatorname{lcm}(s_1, \dots, s_{26}).$$

For fixed $1 - 1/A < \sigma < 1$, Rankin's elementary bound counts such moduli by

$$(C_1 N Y)^\sigma \prod_{\ell \leq Y} (1 - \ell^{-\sigma})^{-1} = N^{\sigma + o(1)}.$$

Indeed the logarithm of the product is $O_\sigma(Y^{1-\sigma}) = o(\log N)$. Assign each maximal prime power of D_0 to one of the 26 offsets on which it occurs. This costs $26^{\omega(D_0)} = N^{o(1)}$. The Chinese remainder theorem determines at most one centre in its interval, since $D_0 > C_1 N$. The offsets themselves have $B^{26} = N^{o(1)}$ choices. Choose σ sufficiently close to $1 - 1/A$. □

T6.2 Several anchors force a congruence certificate

Lemma T6.3 (Incoming and outgoing certificate counts)**.** *Suppose a relation in (10.1) selects t values in one new block whose kernels exceed N^b, where $b = 1/5$ and $tb > r + s - 1$. Some other old or new block receives a congruence certificate whose modulus exceeds a fixed multiple of NB. For fixed r, s, t, A, the directed graph of these source–receiver certificates has both incoming and outgoing degrees at most $N^{1/A+o(1)}$.*

Proof. Represent the old vector by a selection of actual old values; parity in the old blocks is not required. Together with the new selections this is a square product. Primes above Y cannot occur in two distinct values of one block. The t anchor kernels are therefore pairwise coprime, and their product Q satisfies $Q > N^{tb}$, $Q \ll_D N^t$. Assign each prime of Q to one of its other odd occurrences. The $r + s - 1$ receiver moduli have product Q. For sufficiently large N, at least one exceeds $C_2 NB$.

In that modulus keep the primes only up to its first passage above C_2NB. The retained list has at most

$$k = \left\lfloor \frac{\log(C_2NB)}{\log Y} \right\rfloor + 1$$

members. For a fixed source, the t offsets and a subset of the $O_t(\log N/\log Y)$ available primes cost $N^{o(1)}$. Assigning their destinations among $O_D(B)$ receiver positions costs at most $B^k = N^{1/A+o(1)}$. The congruences determine the receiver.

For a fixed receiver, its values contain $R = O_D(B \log N/\log Y)$ primes above Y. Choose at most k of them and assign each to an anchor: the number of codes is at most

$$B^t \sum_{j\le k} \binom{R}{j} t^j = N^{1/A+o(1)}.$$

When $R \ge 2k$, use $(k+1)(etR/k)^k$; when $R < 2k$, use $(1+t)^R = N^{o(1)}$. The retained modulus again exceeds the source interval. Forgetting all other primes enlarges the count and does not remove the original full-support relation from the remaining blocks. □

T6.3 The iteration

Proof of Theorem 10.1. We induct strongly on r, simultaneously for every fixed number s of sources. Put $T_0 = 1$. The case $r = 1, s = 0$ is empty, since its sole projection is required to be nonzero. For all other cases, choose

$$t = 5(r + s - 1) + 1.$$

First choose a small fixed step $\delta > 0$ within the final error margin. Let

$$x_0 = 0, \qquad x_j = \min\{\tfrac{1}{2}, x_{j-1} + \delta\}, \qquad A_j = x_j^{-1}, \qquad Y_j = (\log N)^{A_j}.$$

There are $J = \lceil 1/(2\delta)\rceil$ stages. All these numbers are fixed before N; choose the internal margins after J.

Corollary 9.2 counts lists possessing a witness with all new selections short. For every remaining list choose one witness, without enumerating it. It has at least one long new selection. At the start of stage j, every surviving long new source outside $\mathcal{X}_{1/2}$ belongs to a set of size $N^{1-x_{j-1}+\varepsilon'}$. This is trivial for $j = 1$.

If such a source selects t large K_{Y_j} kernels, Lemma T6.3 supplies a receiver. When the receiver is new, count the source–receiver pair by $N^{1-x_{j-1}+x_j+O(\varepsilon')}$. The other $r-2$ projections are still nonzero in their own spaces; their sum belongs to the old space enlarged by these two fixed blocks. The induction hypothesis counts them by $N^{(r-2)/2+\varepsilon'}$. The total is

$$N^{r/2+x_j-x_{j-1}+O(\varepsilon')} \le N^{r/2+\delta+O(\varepsilon')}.$$

If the receiver is old, it is already fixed. The incoming degree counts the new source by $N^{x_j+O(\varepsilon')}$. Adjoin that source to the old space and apply induction to the $r-1$ remaining targets. The cost is at most

$$N^{x_j+(r-1)/2+O(\varepsilon')} \le N^{r/2+O(\varepsilon')}.$$

Only a fixed number of choices of roles is involved.

Remove all these saturated cases. Every surviving long selection outside $\mathcal{X}_{1/2}$ has fewer than t large kernels, and Lemma T6.2 places its centre in a set of size $N^{1-x_j+\varepsilon'}$. This proves the stage invariant. At the final stage every long source lies in $\mathcal{X}_{1/2} \cup \mathcal{M}_{Y_J}$, whose size is $N^{1/2+\varepsilon'}$. Fix one such centre and use the induction hypothesis for the other $r-1$ targets. This costs $N^{1/2+\varepsilon'}N^{(r-1)/2+\varepsilon'}$.

Every application of induction uses fewer targets, although the fixed number of old sources can increase. Thus the simultaneous induction is legitimate. Choosing δ and then all internal margins within ε proves the theorem. □

T7 Pair partitions and the cost of one dependent target

The signed triple identities in Section 11 use one quantitative input: the independent target deficit following a pair. We first prove the finite partition needed for that estimate.

Lemma T7.1 (A finite partition of the pair activity)**.** *For each fixed $\eta > 0$, dependent pairs can be partitioned into good pairs from a finite fixed set of rational channels, a weak class, and an exceptional class, with*

$$A_{\mathrm{w}} \ll_{C,\eta,\varepsilon} N^{-1+\eta+\varepsilon}, \qquad A_{\mathrm{s}} \ll_{C,\eta,\varepsilon} N^{-1+\varepsilon}.$$

A good pair in a channel $a : b$, $b \geq 3$, has

$$|c(e)| \ll_{a,b,C} N^{-(2-1/b)+o(1)}$$

and an exact compression into a block of fixed logarithmic width. Moreover $A_2 \ll_{C,\varepsilon} N^{-2/3+\varepsilon}$.

Proof. Choose Q with $1/(Q+1) < \eta/2$, and retain the channels $3 \leq b \leq Q$, $b/2 < a < b$, $(a,b) = 1$. Lemma T1.9 bounds the rank in all larger aligned channels by $L/(Q+1) + o(L)$. Put these pairs, nonaligned pairs of rank at most a sufficiently small fixed fraction of ηL, and affinely incompatible pairs in the weak class. Their coefficients have magnitude at most $N^{-2+\eta+o(1)}$. Corollary 10.3 counts their dependent hosts by $N^{1+\varepsilon'}$, proving the weak bound.

Nonaligned pairs of macroscopic rank have total activity $O(N^{-1+\varepsilon'})$ by Lemma T1.8. The local defects contribute only bounded factors under (2.2). Pairs meeting the irregular set have the same bound: its marginal mass is $N^{-1+o(1)}$, by Proposition 3.1, and fixed-source propagation (2.9) bounds their joint mass. The independent products have the same cost. At each retained fixed height, normalized defective pairs cost $N^{-3/2+1/b+o(1)}$, as in the proof of Lemma T2.5. Endpoint height-two pairs and non-strict height-one contacts cost $N^{-1+o(1)}$, by the same rank and endpoint estimates. These are the exceptional class. The remaining pairs have the private coordinates defining the good fixed-channel compression. Its rank is at most $L/b + O(1)$, giving the stated individual weight.

The finite channel sums are $O(N^{-2/3+\varepsilon'})$, by Lemma T2.5. Together with the other two classes, for a sufficiently small fixed η, this also proves the assertion about A_2. □

T7.1 Proof of Proposition 11.1

Proof. After any fixed pair, Theorem 10.1 with one target counts at most $N^{1/2+\varepsilon'}$ centres. Their total independent mass is at most $N^{-1/2+\varepsilon'}$. The weak and exceptional pair classes thus cost $N^{-3/2+\eta+O(\varepsilon')}$.

For a good pair in a fixed channel, multiply its two windows by the channel coefficients and merge identical values. An even selection in each original window is multiplied by a square, so its square class is unchanged. Merging cancels repeated occurrences and preserves total even parity. Consequently every nonzero class in $(S_x + S_y) \cap S_z$ is represented by a nonempty even selection in the normalized block and in the target. The normalized block has fixed logarithmic width. Corollary 10.3 counts these two centres by $N^{1+\varepsilon'}$. The offsets and their inverse normalization have $B^{O_{a,b}(1)}$ multiplicity. Each host carries weight at most $N^{-(2-1/b)-1+o(1)}$, so its contribution is

$$N^{1+\varepsilon'} N^{-(2-1/b)-1+o(1)} \ll N^{-5/3+2\varepsilon'}.$$

Sum over the finite channels, then choose η and ε' within the final margin. □

T8 Polynomial confinement and dense arithmetic profiles

We use the block and alignment conventions of Section 12. The group argument first treats primitive triples and nonaligned pairs, then proves the two triple interfaces stated there.

Lemma T8.1 (Polynomial confinement of common divisors). *Fix $\tau > 0$, $0 < \delta < 1/2$, and $C_0 > 0$. Let $x, y \asymp N$, and let pairwise coprime integers $g_a > N^\tau$ satisfy*

$$g_a \mid x + i_a, \quad g_a \mid y + j_a, \qquad |i_a|, |j_a| \le C_0 B.$$

Suppose no line through $(-x, -y)$ contains two distinct points among the (i_a, j_a). Then their number, with multiplicity, is $O_{\tau,\delta,C_0}(B^{1/2+\delta}) = o(B)$. The same estimate holds after points on exact alignments have been removed.

Proof. Choose a fixed degree d such that, for $M = (d+1)(d+2)/2$, $(M-1)\tau > d$. Let $v_d(s,t)$ list the M monomials of total degree at most d, including 1. If M of the vectors $v_d(i_a, j_a)$ form a basis, replace each row of that basis in turn by $v_d(-x, -y)$. Subtracting the new row from the other rows shows that the resulting determinant is divisible by the product of their $M-1$ pairwise coprime divisors. Its absolute value is at most

$$C_d N^d B^{d(M-1)} < N^{(M-1)\tau}.$$

All the determinants vanish. The vector $v_d(-x, -y)$ would then belong to the intersection of the M coordinate hyperplanes of this basis, which is $\{0\}$. Its constant coordinate is one, a contradiction.

Thus a nonzero polynomial $F(S,T)$ of degree at most d vanishes on all the offset points. For a geometrically integral component of degree at least two, Pila's uniform estimate [21, Theorem A] gives $O_d(B^{1/2+\delta})$ integer points of height $O(B)$. A rationally irreducible component that is not geometrically integral has only $O_d(1)$ rational points, by intersecting two conjugate components.

A line through at least two distinct grid points has a primitive equation $aS + bT = c$ with $|a|, |b| \ll B$, $|c| \ll B^2$. Every group on it divides $ax + by + c$. This integer is nonzero by the hypothesis, and has size $O(NB)$. Coprimality bounds the number of these groups by $O_\tau(1)$. A line through only one grid point needs no separate point count. Finally, all groups at one point divide its true value $x + i_a = O(N)$; their multiplicity is $O_\tau(1)$. There are at most d components. These observations prove the bound. □

The groups in this lemma can be composite. Only divisibility and coprimality are used, and the auxiliary curve is not enumerated. Its degree is chosen after τ and before N.

Theorem T8.2 (Dense primitive triples). *For fixed $\eta, \varepsilon > 0$, primitive triples of true windows with $\rho_{\mathrm{rel}}(T) \ge \eta B$ have $O_{\eta,C,\varepsilon}(N^\varepsilon)$ hosts. Their homogeneous mass is $O(N^{-2+\varepsilon})$. The total absolute centered activity of all primitive triples is $O(N^{-3/2+\varepsilon})$.*

Proof. Realize the relation space R linearly on complements to the local kernels. Its projection into each of the three blocks is injective, by primitivity. Apply Lemma 6.1 with $d = 2$, choosing a fixed t large enough that the loss is at most $\eta B/4$. On the resulting space R', each nonzero coordinate label has multiplicity at most $K = t - 1$, and a label has at most $2t$ representations by two distinct nonzero labels.

A prime above H occurs oddly at most once per block. On the active coordinates of R', its row has either two equal nonzero labels or three distinct labels summing to zero; it cannot have weight one. Group the primes by their active physical support A, and let G_A be their product. The groups are pairwise coprime and divide all true values at their support positions. Every active coordinate belongs to at most

$$C_t = K + 2tK^2$$

groups. This grouping uses supports after restriction; a former ternary row that has become binary is included as a binary group.

Assign each group to a pair of blocks it meets. For a primitive triple no such pair has two exact units: the corresponding even selections would give equal nonzero square classes in the two

blocks. Their nonzeroness follows from Lemma T1.1 applied to two distinct values. Lemma T8.1 therefore shows that only $o(B)$ groups exceed N^{τ}. Set every position incident to one of these groups equal to zero. This loses at most three times their number of dimensions. The surviving space has dimension at least $\eta B/2$ for large N, and its projections remain injective.

Every still-active position belonged only to original groups of size at most N^{τ}. All its odd primes above H were in such groups; hence $K_H(n) \le N^{C_t\tau}$. Each block has at least two active positions, because its projection contains a nonzero even vector. Its centre lies in $\mathcal{X}_{2C_t\tau}$. Lemma T6.1 bounds the number of triples by $N^{6C_t\tau+3\varepsilon'}$. Choose t after η, and then τ, ε' after the final margin.

For a primitive triple $\rho_{\mathrm{rel}} \le L$, so its homogeneous weight is $O(N^{-2})$. For the total activity split at $\rho_{\mathrm{rel}} = \eta L$. Below that threshold, Corollary 10.2 gives $N^{-3/2+\eta+\varepsilon'}$; above it use the just proved bound. A primitive coefficient has magnitude at most its homogeneous weight, whether compatible or incompatible. Choose η within the final margin. □

Corollary T8.3 (Dense nonaligned pairs)**.** *For fixed $\eta, \varepsilon > 0$, nonaligned pairs with $d_{xy} \ge \eta B$ have $O(N^{\varepsilon})$ hosts.*

Proof. Realize their effective relation space with injective projections. Regularize with Lemma 6.1, now with $d = 1$, losing at most $\eta B/4$. Every active prime row has two equal labels. A coordinate belongs to at most $K = t - 1$ active supports. Confinement bounds the groups above N^{τ} by $o(B)$. Kill both endpoints of every such group. The remaining positive-dimensional space has two active positions in each block, with kernels at most $N^{K\tau}$. The centres lie in $\mathcal{X}_{2K\tau}$. Their count is at most $N^{4K\tau+2\varepsilon'}$; choose τ, ε' sufficiently small. □

T8.1 Proof of Proposition 12.1

Proof. The kernel of each projection of the relation space has dimension at most θB. Intersect three complements to these kernels. The resulting subspace loses at most $3\theta B$ dimensions and has all projections injective.

In a given pair, two exact units impose a reduced rational alignment of height $O(B)$. Two distinct such alignments would bound the centres by $O(B^3)$, which is impossible at the present scale. The alignment is therefore unique. If it has u units, their even-selection space has dimension $u - 1$; its image loses at most one more dimension by Lemma 2.1. Thus $u \le d_{xy} + 2 \le \theta B + 2$. Kill all the positions of these units in the three pairs, at a cost of at most $6\theta B + 12$ dimensions. The remaining space still has macroscopic dimension and injective projections.

Regularize with a further loss at most $\eta B/8$. No pair of surviving offsets lies on an exact alignment, so the confinement argument in Theorem T8.2 applies. After all positions incident to large groups have been killed, the centres again belong to $\mathcal{X}_{2C_t\tau}$. The same count gives N^{ε} hosts. We never enumerate the chosen subspaces or deletion sets.

Finally, $\rho_{\mathrm{rel}}(T) \le d_z + d_{xy} \le L + \theta B$. The homogeneous weight is at most $N^{-2+\theta+o(1)}$, proving the mass estimate. □

T8.2 Proof of Proposition 12.2

Proof. Choose the strong pair $e = \{x, y\}$. Put $\delta_z = \dim(S_z \cap (S_x + S_y)) = \rho_{\mathrm{rel}}(T) - d_{xy}$. Connectedness gives $\delta_z > 0$, and compatibility gives $q(T) = q(e)p_z 2^{\delta_z}$.

For $0 < \delta_z < \eta' L$, the conditional count in Theorem 10.1 gives $N^{1/2+\varepsilon'}$ targets of weight at most $N^{-1+\eta'+o(1)}$. Lemma T1.8, together with Corollary 10.3 for the independent part, bounds the total joint mass of strong nonaligned pairs by $N^{-1+\varepsilon'}$. This branch costs $N^{-3/2+\eta'+O(\varepsilon')}$.

For $\delta_z \ge \eta' L$, the effective-space form of Proposition 6.2, explained after (2.8), gives $N^{\varepsilon'}$ targets after the fixed pair. Corollary T8.3 also gives $N^{\varepsilon'}$ source pairs. Each triple has $q(T) \ll N^{-3+\sigma+o(1)}$. Choose η' and the internal margins after ε. □

T9 The short-basis and common-curve constructions

The notation is that of Section 13. This section supplies the complete local estimates, basis extraction, and algebraic construction used by the main rank argument. The final degree-uniform point count and the nonaligned contradiction are proved in that section of the article.

T9.1 Local square relations and defective positions

For primes larger than the window length, a local square relation can use only positions having no odd valuation at any such prime. The next elementary argument bounds those positions without an admissibility assumption.

Lemma T9.1 (A local square threshold)**.** *Let $B \ge 2$, let $I \subset [-B, B] \cap \mathbb{Z}$ have positive even cardinality m, and let x be a positive integer. If $x > (B+1)^{12m}$, then $\prod_{a\in I}(x+a)$ is not an integer square.*

Proof. Write $m = 2n$ and

$$P(T) = \prod_{a\in I}(T+a), \qquad \sqrt{P(T)} = T^n\Psi(T^{-1}), \qquad \Psi(z) = \prod_{a\in I}(1+az)^{1/2} = \sum_{j\ge 0} q_j z^j.$$

The binomial branches have constant term one. Their coefficients give $q_j \in 4^{-j}\mathbb{Z}$. With $R = 2(B+1)$, Cauchy's estimate on $|z| = R^{-1}$ gives $|q_j| \le 2^m R^j$. The polynomial part $Q(T) = \sum_{j=0}^{n} q_j T^{n-j}$ has denominator dividing $M = 4^n$, and, for $x \ge 2R$,

$$|\sqrt{P(x)} - Q(x)| \le \frac{2^{m+1}R^{n+1}}{x}.$$

The assumed threshold makes this less than $1/M$, since

$$2^{m+1}R^{n+1}M = 2^{5n+2}(B+1)^{n+1} \le (B+1)^{6n+3} \le (B+1)^{9n}.$$

If $\sqrt{P(x)}$ were an integer, it would therefore equal $Q(x)$.

However, $E = M^2(P - Q^2)$ is a nonzero integer polynomial: the distinct linear factors make P squarefree. Its coefficient height satisfies

$$\|P\|_1 \le (B+1)^{2n}, \qquad \|Q\|_1 \le 2^{3n+1}(B+1)^n \le (B+1)^{5n}, \qquad H(E) \le (B+1)^{15n}.$$

Cauchy's root bound for the integer polynomial E would give $x \le 1 + H(E) \le (B+1)^{16n}$, contrary to $x > (B+1)^{24n}$. □

Lemma T9.2 (Few defective positions in every critical window)**.** *The number of $n \in V_x$ for which*

$$\prod_{\substack{p>10B \\ v_p(n)\ \text{odd}}} p = 1$$

is $O_C(B/\log B)$, uniformly over all critical windows.

Proof. We use the elementary bound $\pi(t) \ll t/\log t$. For completeness, the product of the primes in $(u, 2u]$ divides $\binom{2u}{u} \le 4^u$ for integer u; dyadic summation gives $\vartheta(t) = \sum_{p\le t}\log p \ll t$. Splitting at $\sqrt{t}$ gives the stated bound for $\pi(t)$.

Let D be the number of defective positions and $s = \pi(10B)$. Their square classes lie in an s-dimensional space. Set

$$k = \left\lfloor \frac{\log N}{48\log(B+1)} \right\rfloor.$$

For sufficiently large N, $k \asymp_C B/\log B$ and $k \geq 1$. Lemma T9.1 forbids a nonempty even local relation of size at most $2k$, because $24k\log(B+1) \leq \frac{1}{2}\log N < \log x$. Two distinct k-subsets of the defective positions therefore have distinct square classes: their symmetric difference would be such a relation. If $D \geq k$, this gives $\binom{D}{k} \leq 2^s$. When $D \geq 2k$, the inequality $\binom{D}{k} \geq (D/k)^k$ implies $D \leq k2^{s/k} = O_C(k)$; the case $D < 2k$ has the same bound. □

We will also need the elementary prime harmonic estimate

$$\sum_{B^a < p \leq 10B} \frac{1}{p} = \log(1/a) + O_a(1/\log B), \qquad 0 < a < 1 \text{ fixed.} \tag{T9.1}$$

Here is a derivation recording its error term. Chebyshev's bound above implies $\psi(t) = \sum_{p^j \leq t} \log p \ll t$. The identity $\log(n!) = \sum_{d \leq n} \Lambda(d)\lfloor n/d \rfloor$ then gives $\sum_{d \leq t} \Lambda(d)/d = \log t + O(1)$. The contribution of prime powers with exponent at least two is bounded, so $A(t) = \sum_{p \leq t} (\log p)/p = \log t + O(1)$. Partial summation yields

$$\sum_{p \leq t} \frac{1}{p} = \frac{A(t)}{\log t} + \int_2^t \frac{A(u)}{u(\log u)^2}\, du = \log\log t + c_0 + O(1/\log t).$$

The difference at $10B$ and B^a proves (T9.1).

T9.2 Large primes turn intersections into components

Construct a graph on the disjoint formal copies of V_x, V_y. A prime $p > 10B$ has at most one occurrence of odd valuation in either block. If it has two occurrences, join their vertices; if it has only one, mark that vertex as forced to zero. The corresponding binary equation says that the endpoints of an edge have equal coordinates, or that a marked coordinate is zero. A component is free when it contains no marked vertex.

Lemma T9.3 (Free components)**.** *Let c be the number of free components containing an edge. Then*

$$c = d_{xy} + O_C(B/\log B).$$

Each such component meets both blocks and is selected in its entirety or omitted in any solution to the large-prime equations.

Proof. The isolated unmarked vertices are precisely the defective positions. Their total number D_0 is $O_C(B/\log B)$ by Lemma T9.2. The kernel of the large-prime equations has dimension $c + D_0$. Imposing the remaining prime equations and the two block parities costs at most $\pi(10B) + 2$ dimensions. If $\mathcal{R}$ is the space of complete relations and $k_x = \dim\ker\phi_x$, then the map taking a relation to its common class gives an exact sequence with kernel $\ker\phi_x \oplus \ker\phi_y$. Hence

$$c + D_0 - \pi(10B) - 2 \leq \dim\mathcal{R} = k_x + k_y + d_{xy} \leq c + D_0.$$

A local relation can use only defective positions, so $k_x + k_y \leq D_0$. These inequalities prove the assertion. □

T9.3 Proof of Theorem 13.1

Proof. A graph estimate. A multigraph with n vertices and $e/n \to \infty$ has a cycle of length $O(1 + \log(n+1)/\log(e/n))$. Loops and parallel edges give cycles of length one and two. Otherwise delete vertices of degree less than $t = \lfloor e/(2n) \rfloor$. Not all vertices can disappear, since fewer than $nt \leq e/2$ edges would then be deleted. The remaining graph has minimum degree at least t. If

its girth exceeds $2r+1$, a breadth first search contains at least $1+t\sum_{j<r}(t-1)^j$ vertices. This bounds r and proves the estimate.

Consequently, when $n \ll B/\log B$ and $e \le B$, the binary cycle space contains at least $e-O(B/\sqrt{\log B})$ independent cycles of length $O(\log B/\log\log B)$. Indeed, while at least $B/\sqrt{\log B}$ edges remain, record a short cycle and delete just one of its edges. The recorded cycles are independent: the first deleted edge belongs to no later cycle, and the same argument applies successively to all deleted edges.

Medium-prime columns. Start with the free nontrivial components of Lemma T9.3, and discard those with more than $M = \lfloor \log B \rfloor$ vertices. This loses at most $2B/(M+1)$ components. The remaining number is $d_{xy} + O_C(B/\log B)$. Put $z = B^a$. To each retained component attach its parity column on the primes $z < p \le 10B$, and let w_j be that column's Hamming weight. Each nonzero entry consumes a divisible position, whence

$$\sum_j w_j \le \sum_{z<p\le 10B} \#\{n \in V_x \sqcup V_y : p \mid n\} \le 2B \sum_{z<p\le 10B} \frac{1}{p} + 2\pi(10B) \le 2B\log(1/a) + O_a(B/\log B).$$

Discard only columns of weight at least three. Their number is at most one third of this sum, so at least

$$e \ge d_{xy} - \frac{2}{3}B\log(1/a) - O_{a,C}(B/\log B)$$

columns remain. There is no factor M in this count: it bounds the number of discarded columns, not their physical size.

Use the primes $z < p \le 10B$ and one extra vertex ∞ as vertices of a multigraph. A column of weight two joins its two primes; a column of weight one joins its prime to ∞; a zero column is a loop at ∞. Its binary incidence matrix has the same kernel as the parity matrix, because the row at ∞ is the sum of the other rows. The graph has $n \le \pi(10B)+1 \ll B/\log B$ vertices and $e \le B$ edges. The preceding graph argument therefore supplies a space $\mathcal{U}$ of dimension at least $e - O(B/\sqrt{\log B})$ with a basis of short cycles. Each basis vector uses $O(\log^2 B/\log\log B)$ physical positions.

Small primes and parity. All large and medium primes already cancel on $\mathcal{U}$. The primes at most z and the two block parities impose at most $s = \pi(z)+2 \ll_a B^a/\log B + 2$ further equations on its cycle coordinates. Their kernel is $\mathcal{K}_a$; since $s = o(B/\sqrt{\log B})$, it has the dimension in (13.4). A fundamental kernel basis for a matrix with at most s rows uses at most $s+1$ columns per vector: express each nonpivot column in terms of pivot columns. Thus each resulting vector uses at most $s+1$ short cycles, giving (13.5).

Finally, the original free nontrivial components are disjoint and meet both blocks. Every nonzero selection of them contains a component carrying a large prime which occurs once in each block and in no other component. Its common square class is therefore nonzero. This also proves injectivity of both physical projections and of the common-class map. The imposed parity equations complete the proof. □

T9.4 Two relations produce a polynomial of controlled height

The short basis will have support $B^{a+o(1)}$. We therefore need an auxiliary polynomial whose degree and logarithmic height grow linearly with the support, and a simultaneous construction whose threshold is quadratic in that support. The following elementary factor bound will keep the height under control when common polynomial factors are removed.

Lemma T9.4 (Height of a bivariate factor)**.** *Let $P \in \mathbb{Z}[X,Y] \setminus \{0\}$, with partial degrees d_X, d_Y. Every primitive integer factor Q of P, and the corresponding integer quotient, satisfy*

$$H(Q) \le 2^{d_X+d_Y}\|P\|_2 \le 2^{d_X+d_Y}\sqrt{(d_X+1)(d_Y+1)}\,H(P).$$

Proof. The multivariate Mahler measure $M(P)$ is multiplicative, at most $\|P\|_2$, and at least one for a nonzero integer polynomial. Jensen's formula in one variable, followed by the leading coefficient in that variable, proves the last assertion by iteration. The one-variable coefficient bound also iterates: integrating first in X and then in Y gives

$$|[X^iY^j]P| \le \binom{d_X}{i}\binom{d_Y}{j}M(P).$$

Thus $H(Q) \le 2^{\deg_X Q+\deg_Y Q}M(Q)$; multiplicativity and $M(P/Q) \ge 1$ prove the result. The quotient has the same bound, with the integer content handled separately. □

Proposition T9.5 (Two complete witnesses)**.** *There is an absolute constant $c > 0$ with the following property. Let u, v be independent formal words, even in each block, with offsets in $[-B, B]$. For $w = u, v, u + v$, write $|I_w| = 2r_w$, $|J_w| = 2s_w$, and assume $\max(r_w, s_w) \le q$, where $1 \le q \le B$. Suppose that $F_u(x, y)$ and $F_v(x, y)$ are integer squares, with $x, y \in [N, 2N)$. If $N > (B + 1)^{cq}$, there is a nonzero $R \in \mathbb{Z}[X, Y]$ such that*

$$R(x, y) = 0, \qquad \deg_X R, \deg_Y R \le 12q, \qquad \deg R \le 24q, \qquad H(R) \le (B + 1)^{cq}. \tag{T9.2}$$

Moreover, R is the norm of a biquadratic auxiliary function whose four polynomial coefficients have no common nonconstant factor.

Proof. Set $F_1 = F_u$, $F_2 = F_v$, $G = \gcd(F_1, F_2)$, with monic gcd, and $F_3 = F_{u+v} = F_1F_2/G^2$. Formal independence makes

$$K = \mathbb{Q}(X, Y)(W_1, W_2), \qquad W_i^2 = F_i,$$

a degree-four extension. With $W_0 = 1$ and $W_3 = W_1W_2/G$, the four W_χ form a basis. At the given integer point use positive square roots. Then all $W_\chi(x, y)$ are integers: $F_3(x, y)$ is an integer which is a rational square, hence an integer square, and $W_3(x, y) = \sqrt{F_1(x, y)}\sqrt{F_2(x, y)}/G(x, y)$.

The linear system at infinity. Put $r_0 = s_0 = 0$, $k = 3q$, and consider

$$\mathcal{A} = \sum_{\chi=0}^{3} A_\chi(X, Y)W_\chi, \qquad \deg_X A_\chi \le k - r_\chi, \quad \deg_Y A_\chi \le k - s_\chi.$$

There are

$$n = \sum_{\chi=0}^{3}(k - r_\chi + 1)(k - s_\chi + 1) \ge 21q^2 + 18q + 4$$

unknown coefficients, and $n \le 4(3q+1)^2$. Writing $\xi = 1/X$, $\eta = 1/Y$, the positive formal branches are

$$W_\chi = X^{r_\chi}Y^{s_\chi}\prod_{u\in I_\chi}(1 + u\xi)^{1/2}\prod_{v\in J_\chi}(1 + v\eta)^{1/2}.$$

Thus $f(\xi, \eta) = \xi^k\eta^k\mathcal{A}$ is a formal power series. Require its coefficients of total degree at most $2k$ to vanish. This gives at most

$$t = \frac{(2k + 1)(2k + 2)}{2} = 18q^2 + 9q + 1, \qquad n - t \ge 3q^2 + 9q + 3, \qquad \frac{t}{n - t} < 6$$

homogeneous linear equations.

All coefficients required from the binomial series have denominator dividing 4^{2k}. Cauchy's estimate on circles of radius $(2B)^{-1}$ bounds their numerators, after this common denominator is cleared, by $(B + 1)^{c_0q}$ for an absolute c_0. An elementary pigeonhole form of Siegel's lemma yields a nonzero integer solution of height $(B + 1)^{c_1q}$. Indeed, if the integer matrix has height A, the

images of $\{0,\dots,M\}^n$ take at most $(2nAM+1)^t$ values. Choosing $M=\lceil(3nA)^7\rceil$ makes two images equal, because $t/(n-t)<6$. Their difference is a nonzero kernel vector; $\log n = O(\log B)$ is absorbed in the stated height.

Primitive coefficients. Let D be the primitive polynomial gcd of the four A_χ, omitting zero coefficients, and write $A_\chi = DA'_\chi$. The factor bound of Lemma T9.4 keeps $H(A'_\chi)\le(B+1)^{O(q)}$. Write $a=\deg_X D$, $b=\deg_Y D$, $e=\deg D$. Then

$$f=\widetilde{D}f',\qquad \widetilde{D}=\xi^a\eta^b D(\xi^{-1},\eta^{-1}),\qquad f'=\xi^{k-a}\eta^{k-b}\sum_\chi A'_\chi W_\chi.$$

These are power series, and $\operatorname{ord}\widetilde{D}=a+b-e$ for total order. Since order is additive in the power-series domain,

$$\operatorname{ord} f'\ge 2k+1-a-b+e=(k-a)+(k-b)+e+1. \tag{T9.3}$$

Thus removing D preserves vanishing beyond the remaining pole weight. This is a formal division, never a division by the possibly zero value $D(x,y)$. Replace the auxiliary by its primitive version from now on.

Integer separation and the norm. On the bidisc $|\xi|,|\eta|\le(4B)^{-1}$, the analytic numerator is bounded by $(B+1)^{O(q)}$. Cauchy's estimate and the order of vanishing give, in the comparable range $x,y\in[N,2N)$,

$$|\mathcal{A}(x,y)|\le\frac{(B+1)^{c_2 q}}{N}. \tag{T9.4}$$

To see the power of N explicitly before removing D, the tail of the numerator is bounded by its coefficient bound times

$$\sum_{\ell\ge 2k+1}(\ell+1)(4B/N)^\ell\ll(k+1)(4B/N)^{2k+1}.$$

Multiplication by $x^k y^k\le(2N)^{2k}$ gives (T9.4). After removing D, both pole weights decrease, and (T9.3) gives the same estimate with the new height bound.

The specialization $\mathcal{A}(x,y)$ is an integer, so it vanishes for the stated threshold. The auxiliary is nonzero in K by independence of the four characters. Its norm is therefore nonzero. An explicit polynomial expression is

$$\begin{aligned}U&=A_0^2+A_1^2F_1-A_2^2F_2-A_3^2F_3,\\ V&=A_0A_1-A_2A_3(F_2/G),\\ R&=U^2-4F_1V^2.\end{aligned}$$

Here F_2/G is a polynomial. The partial degrees of U are at most $2k$, and those of V are at most $2k-r_1$, $2k-s_1$. For the second term in V, use $r_3=r_1+r_2-\deg_X G$ and its Y analogue. It follows that the partial degrees of R are at most $4k=12q$. The coefficient ℓ^1 norms of all factors in this formula are $(B+1)^{O(q)}$, which proves the height bound and completes (T9.2). □

T9.5 Proof of Theorem 13.2

Proof. For $i<j$, the half-cardinalities of w_i, w_j, w_i+w_j are at most $q=\min(2q_0,B)$, since an even subset of $[-B,B]\cap\mathbb{Z}$ has size at most $2B$. Proposition T9.5 gives a nonzero polynomial R_{ij} of degree at most $48q_0$ and height $(B+1)^{O(q_0)}$, vanishing at (x,y) and arising from an auxiliary with primitive coefficients. Choose an irreducible primitive factor C_0 of R_{12} which vanishes at this point. The factor-height lemma preserves its height bound. If C_0 depended on only one variable, its integer root of size at least N would contradict Cauchy's root bound. Thus it depends on both.

If C_0 failed to divide some R_{ij}, the resultant $\mathrm{Res}_Y(C_0, R_{ij})$ would be a nonzero integer polynomial vanishing at x. Its height is $(B+1)^{O(q_0^2)}$: its Sylvester matrix has size $O(q_0)$, entries of coefficient norm $(B+1)^{O(q_0)}$, and $O(q_0)!$ determinant terms. This again contradicts the root bound under the stated threshold. Hence C_0 divides every R_{ij}.

No factor $X+u$ or $Y+v$ vanishes identically on this curve, since $C_0(x,y)=0$ with both coordinates larger than B. The displayed square classes are therefore defined in K_0. If the classes of F_{w_i}, F_{w_j} were independent there, they would still generate a degree-four field extension. Specializing the polynomial norm formula gives zero, since $C_0 \mid R_{ij}$; in a field this forces the auxiliary to vanish. Independence of its four characters would force all four coefficient polynomials to be divisible by C_0, contrary to their primitivity. Thus the images of every pair of basis vectors are dependent. Over $\mathbb{F}_2$ this means that all nonzero images are equal, so the image has dimension at most one. □

T9.6 Proof of Proposition 13.7

Proof. Fix $\tau < 1$. Theorem 13.1 gives a space $V = \mathcal{K}_\tau$ of dimension

$$t \geq d_{xy} - \frac{2}{3}B\log(1/\tau) - O_{\tau,C}(B/\sqrt{\log B}),$$

with a basis of half-cardinalities at most $q_0 = B^{\tau+o(1)}$. The case $t<2$ is covered by the displayed error term. For $t \geq 2$, apply Proposition T9.5 to each pair of basis vectors. Its threshold holds because $q_0 \log(B+1) = o(B) = o(\log N)$.

Put $\ell(X,Y) = bX - aY - h$. Every resulting norm polynomial R_{ij} is divisible by ℓ. Otherwise, with $d_Y = \deg_Y R_{ij}$, the nonzero integer polynomial

$$a^{d_Y} R_{ij}\left(X, \frac{bX-h}{a}\right)$$

would vanish at x. Its height is $(B+1)^{O(q_0)}$ since $\max(a,b,|h|) \leq 2B^4$, contradicting the integer root bound for $x \geq N$. The primitive-norm argument in Theorem 13.2 now applies to this line and shows that the image of V in $\mathbb{Q}(\ell=0)^\times/\mathbb{Q}(\ell=0)^{\times 2}$ has dimension at most one.

Let V_0 be its kernel, of dimension at least $t-1$. On the line, $X+u$ and $Y+v$ have the same root precisely when

$$h + bu = av. \tag{T9.5}$$

Every word of V_0 must pair all its first-block roots with second-block roots. The integer solutions of (T9.5) advance by (a,b) in two intervals of B consecutive offsets, so their number r satisfies

$$r \leq 1 + \frac{B-1}{\max(a,b)}.$$

Injectivity of the physical projection gives $t-1 \leq \dim V_0 \leq r$, proving (13.10) after absorbing the bounded terms.

For compatible distinct centres with $\max(a,b) \geq 2$, this gives $d_{xy} \leq B/2 + (2/3)B\log(1/\tau) + o_{\tau,C}(B)$. Choose τ sufficiently close to one after fixing a desired error margin, then let N tend to infinity. This order of choices proves the uniform $B/2 + o_C(B)$ bound.

For $a=b=1$, compatibility implies $|x-y| \geq L$: if $x<y<x+L$, the first run makes $f(y-1)=f(y)$, whereas the second requires their signs to be opposite. At distance exactly $L=B-1$, the true windows share one position; at larger distance they are disjoint. Thus (T9.5) has at most one solution. A nonzero word in V_0 would need at least two pairings, because its selections are nonempty and even. Consequently $V_0=0$, $t \leq 1$, and the dimension estimate followed by $\tau \uparrow 1$ gives $d_{xy} = o_C(B)$. □

T10 All triples above finitely many rational pair channels

The canonical overlap inequality and the positive $10 : 12 : 15$ witness are proved in Section 14. The remaining technical input is uniformity of the upper bound across every triple above a fixed finite set of aligned channels, including arithmetically exceptional source pairs. We prove that input here.

T10.1 Proof of Proposition 14.3

Proof. Fix one channel $a : b$ and orient the source pair $e = \{x, y\}$. Choosing $B^{O_{a,b}(1)}$ offsets, write $x = at - i_0$, $y = bt - j_0$ and $X = abt$. Merge repeated values in $bV_x \cup aV_y$ to obtain $U_X \subset X + [-DB, DB]$, where D depends only on the fixed channel. Multiplying an even selection by a constant does not change its square class, so

$$S_x + S_y \subseteq \operatorname{span}\{v(n) : n \in U_X\}. \tag{T10.1}$$

This inclusion holds regardless of injectivity on U_X. The uniform aligned rank bound of Lemma T1.9 gives

$$q(e) \le N^{-\kappa_b + o(1)}, \qquad \kappa_b = 2 - \frac{1}{b} \ge \frac{5}{3}. \tag{T10.2}$$

For fixed height the residual defect is uniform, so the same bound includes admissible local rank defects.

Targets not aligned with the compressed source. For $z \notin e$, put $\delta = \dim(S_z \cap (S_x + S_y))$. Connectedness of $T = e \cup \{z\}$ implies $\delta > 0$. The inclusion (T10.1) bounds δ by the asymmetric rank loss $\delta^*(X, z)$ of Lemma T1.10. That lemma permits an arbitrary true set U_X of this fixed logarithmic width; it does not require injectivity. It gives

$$\sum_{X,z \text{ nonaligned}} (2^{\delta^*(X,z)} - 1) \ll N^{31/24+\varepsilon'}.$$

Since $q(T) = q(e)p_z 2^{\delta}$, $p_z \le 2p$, and $2^{\delta} \le 2(2^{\delta^*} - 1)$, these targets have total mass at most

$$N^{-\kappa_b - 1 + 31/24 + \varepsilon' + o(1)} \ll N^{-11/8+\varepsilon}. \tag{T10.3}$$

The number of source-offset patterns is polynomial in B and is absorbed into the exponent margin.

The residual kernel for an aligned target. Write an aligned target as $z = (ct + h)/d$, with $(c, d) = 1$ and $c, d, |h| \le B^{O_{a,b}(1)}$, as in the proof of Lemma T2.6. Before this final normalization, U_X and V_z each have fixed logarithmic width. Choose $H > C_D B$ larger than both diameters and remove the prime rows at most H and those dividing the normalization multipliers. Each remaining prime row meets at most one distinct old value and at most one new value. This uses the fact that repeated old values were merged before the target was added.

After the common normalization, merge equal values once more. There are at most $3B$ distinct integers, all at height at least a constant multiple of N and in an interval of diameter $B^{O_{a,b}(1)}$. The remaining prime rows have weight at most two, and leave one variable per unpinned component. We give the uniform residual estimate, since the coefficients c, d need not be fixed.

For a fixed integer K, restrict to components of at most K vertices. The removed prime rows and a total-parity row form a code on these components of redundancy

$$s \le \pi(H) + O_{a,b}(\log B) + 1 = O_{a,b}(B/\log B).$$

A nonzero word in its kernel gives an even square product of distinct normalized integers. Lemma T1.1, applied with the polynomial-in-B diameter just specified, forces it to contain

$\gg_{a,b} \log N/\log B$ physical vertices and hence $\gg_{a,b,K} \log N/\log B$ components. If c_K is the number of these components, the Hamming inequality with $t \asymp_{a,b,K} \log N/\log B$ gives

$$\sum_{j=0}^{t} \binom{c_K}{j} \le 2^s, \qquad c_K = O_{a,b,K}(B/\log B).$$

Here t is chosen below half the component minimum distance. Components larger than K number at most $3B/K$. Choosing K arbitrarily large before N therefore bounds the entire value kernel on the distinct normalized set by $o_{a,b}(L)$, uniformly over c, d, h.

The relevant map factors as

$$\bigoplus_{v\in T} E_v \longrightarrow \{\text{merged normalized coordinates}\} \longrightarrow \{\text{square classes}\}.$$

Its nullity is at most the nullity of merging plus the residual $o(L)$ just proved. Internal defects of the old windows are included in this composition and do not require any assumption of good normalization.

Counting the exact coincidences. In the common parameter the physical coefficients are ad, bd, c. Suppose first that they are distinct. Divide by their common gcd and order them as $u < v < w$. Unless the parameter belongs to a $B^{O_{a,b}(1)}$ endpoint population, dyadic geometry gives $w < 2u$. Indeed a rational ratio of polynomial height cannot cross two by $B^{O(1)}/N$, and equality forces the extreme windows to the two dyadic endpoints. The lattice calculation (14.3) and Lemma 14.2 show that merging loses at most $19L/30+O(1)$ dimensions. The residual estimate therefore gives

$$\dim(S_x + S_y + S_z) \ge \frac{71}{30}L - o(L), \qquad q(T) \le N^{-71/30+o(1)}.$$

There are $NB^{O_{a,b}(1)}$ aligned triples of this form, including the choices of offsets and rational coefficients. Their total mass is $O(N^{-41/30+\varepsilon})$.

If two physical coefficients are equal, compatibility makes the interiors of the corresponding parameter intervals disjoint, since distinct compatible starts have separation at least L. Their combined coincidences with the third lattice are bounded by the length of that third interval: at most $L/a + O(1)$ or $L/b + O(1)$ units. As $a \ge 2$, the total merging nullity is at most $L/2 + O(1)$. Including the residual, this gives mass $O(N^{-3/2+\varepsilon})$. Endpoint populations have only $B^{O_{a,b}(1)}$ source and target choices; the bound $q(T) \le q(e)$ from (T10.2) gives $O(N^{-5/3+\varepsilon})$. These estimates include exact contacts. Strict contacts are incompatible and do not enter the sum.

The least of the exponents obtained is $41/30$. Sum over the fixed finite family $\mathcal{F}$ to conclude. □

T11 Conditional dense targets and small-kernel counts

These are the two geometric proofs used in Section 15. Their source counts, widths, and rank thresholds are fixed before N tends to infinity. They do not use the finite-transfer theorem or the law approximation.

T11.1 Proof of Lemma 15.2

Proof. We first count target pairs, keeping the number of binary components in *each* target explicit. All component counts below include isolated active coordinates.

The two injective projections. Put

$$R = \ker\bigl(W \oplus S_y \oplus S_z \longrightarrow W + S_y + S_z\bigr), \qquad \dim R = d.$$

If an element of R has zero y-projection, its other two entries belong to $W \cap S_z = 0$. Hence the projection $R \to S_y$ is injective; the other projection is injective by $W \cap S_y = 0$. In particular $d \le \min(\dim S_y, \dim S_z) \le L$. Represent W by a linear section in the physical source coordinates, and each target space by a section in its formal even-selection space. These sections preserve both injective target projections and all actual prime-valuation equations, including on an irregular window.

Fix $0 < \delta < 1/2$, to be chosen inside the final exponent margin, and choose fixed positive ξ, θ with

$$\xi \le \tfrac{1}{4}\min(\eta, \delta), \qquad \theta \le \tfrac{1}{8}\min(\eta, \delta).$$

Apply Lemma 6.1 to obtain $R' \subseteq R$ with $d' := \dim R' \ge d - \xi L$. Choose its recovery parameter large enough to give this loss and to control circuits in the $s+2$ physical blocks. Every nonzero coordinate label on R' then has a fixed bounded multiplicity K.

For each prime $\ell > H \asymp B$, pair equal active labels in its row. All edges of the resulting binary graph join equal nonzero labels, so every component lies within one label class and has at most K vertices. The coordinate labels in either target span $(R')^*$, since its projection is injective. A component contributes only one label to this span, even if it meets that target several times. Consequently each target meets at least d' binary components. This is a separate lower bound for each target, not just for their union.

Enumerating coefficients from the fixed sources. After the equal labels have been paired, the remaining distinct labels in a prime row split into minimal circuits. Each such circuit uses at least three physical blocks: $\ell > H$ has at most one occurrence in any block. Since only two blocks are free, every circuit contains a source occurrence. The product of primes assigned to a circuit support therefore divides a fixed source value. The recovery bound gives a fixed bound on the number of circuit supports incident to any selected coordinate. For a fixed number of selected binary components, enumerate their positions, these circuit supports, one source occurrence on each support, and divisors of the corresponding source values. The cost is $B^{O(1)}N^{o(1)} = N^{o(1)}$. Small-prime parities cost $2^{O(\pi(H))} = N^{o(1)}$ as well. Thus the squarefree coefficients of their component equations have $N^{o(1)}$ choices and height N^{A_0} for a fixed A_0. The height bound per component is fixed once s, the recovery parameters and the width multiplier are fixed; it does not increase when more disjoint component equations are selected. The number of equations required by Lemma T1.7 is chosen afterwards. No subspace or full deletion pattern is enumerated.

Removing the small aligned curves. Two exact units between the targets determine a reduced alignment of height $O(B)$. Two distinct alignments of this height would force $y, z = O(B^3)$, impossible in the critical range. If there are fewer than two exact units, no removal is needed. Otherwise let $u \ge 2$ be the number of units on the unique alignment. Their even selections have dimension $u-1$ before passage to square classes, and a target has at most one local rank defect. Their common image therefore gives

$$u - 2 \le \dim(S_y \cap S_z) \le \theta L$$

in the weak aligned case. Keep at most one of these units and exclude from selection every binary component meeting an endpoint of another unit. At most $2(u-1) \le 2\theta L + 2$ components are excluded. This deletes components from the selectable collection; it does not restrict R' again. The surviving positions contain no two exact units on an alignment, which excludes the small-curve alternatives in the proof of Lemma T1.7. No removal is needed for a nonaligned pair.

For $t = y, z$, write m_t for the number of selectable components meeting target t. The separate projection bounds now give

$$m_t \geq d' - 2\theta L - 2 \geq d - (\xi + 2\theta)L - 2.$$

Thus $m_t \geq \eta L/3$ for large N whenever $d \geq \eta L$; if $d > (1/2 + \delta)L$, then $m_t > (1/2 + \delta/3)L$ for both targets.

The first stratum: a positive rank proportion. A component type records its number of coordinates in each physical block. There are finitely many types, because the component size is at most K. If a type meeting both targets occurs sufficiently often, select the fixed number of its components required by the two-variable geometric count. With the small curves excluded above and coefficients of height N^{A_0}, the proof of Lemma T1.7 gives $N^{1/2+\varepsilon'}$ target pairs, including the $N^{o(1)}$ pattern and coefficient choices.

Otherwise only a bounded number of components meet both targets. Each target still has a positive proportion of L components avoiding the other target. In each target choose a univariate type occurring with positive frequency, and then the fixed number of its components needed by the one-variable part of Lemma T1.7, taking at least two. Their component equations form a one-variable system with fixed coefficients and at least two distinct selected offsets; different components have disjoint vertices. In the degree-one case, these are precisely the two linear factors for Lemma T1.2. The one-variable count gives $N^{\varepsilon'}$ target pairs after allocating the margin between the two centers and the coefficient choices. Every selected set has fixed size, so its even-selection square-class map is injective for large N by Lemma T1.1; irregularity of the entire window is irrelevant to this local use. We have proved

$$\#\{(y, z) : d \geq \eta L\} \ll N^{1/2+\varepsilon'}.$$

Both counts here and below impose the pair restriction in the lemma.

The second stratum: rank greater than half a window. Suppose $d > (1/2 + \delta)L$. Let $n_{1,t}$ be the number of selectable components using exactly one coordinate in target t. Every other component meeting that target uses at least two of its $B = L + 1$ coordinates. Hence, separately for $t = y, z$,

$$B \geq n_{1,t} + 2(m_t - n_{1,t}), \qquad n_{1,t} \geq 2m_t - B > \frac{2\delta}{3}L - 1.$$

If a mixed type with one coordinate in either target occurs sufficiently often, the degree-one case of the geometric count gives $N^{\varepsilon'}$ pairs directly. Otherwise only boundedly many of the $n_{1,t}$ components are mixed. The displayed inequality then provides two components for each target, each using one coordinate of that target and no coordinate of the other. The resulting two distinct linear factors give the uniform two-value Pell count for each center. Thus

$$\#\{(y, z) : d > (1/2 + \delta)L\} \ll N^{\varepsilon'}.$$

This explains the threshold $L/2$: above it, the coordinate budget forces singleton occurrences in each target.

Finally $p_y, p_z \ll N^{-1}$ and $2^L \asymp_C N$. The stratum $\eta L \leq d \leq (1/2 + \delta)L$ has weighted cost $N^{1/2+\varepsilon'}N^{-2+1/2+\delta}$, and the stratum above it has cost $N^{\varepsilon'}N^{-2+1}$. Choose $\delta > 0$ and then $\varepsilon' > 0$ so that $\delta + \varepsilon' < \varepsilon$; all preceding choices are fixed before N. This proves (15.4). In a primitive family, each target is direct from the prescribed $r - 2$ sources separately, as well as from the other target; the full nullity is at most L. Multiplication by the $r - 2$ source marginals and summation over the prescribed source collection gives (15.5), after dividing the final exponent margin between the source count and the conditional estimate. □

T11.2 Proof of Lemma 15.4

Proof. For the first assertion fix the m offsets, at a polylogarithmic cost, and write $d_i = n_i/K_Y(n_i) = s_i a_i^2$. Since $d_i \gg N^{1-b}$ and $\gcd(d_i, d_j) \ll B$,

$$\operatorname{lcm}(d_1, \dots, d_m) \geq \frac{\prod_i d_i}{\prod_{i<j} \gcd(d_i, d_j)} \geq N^{m(1-b)-o(1)} > N.$$

The a_i are pairwise coprime. If $\prod_i a_i^2 > N$, multiply the square factors in a fixed order until their product first exceeds N. The resulting square D lies between N and N^{1+u}, so it has at most $N^{(1+u)/2}$ possible roots. Assign its complete prime powers to the m offsets; the number of assignments is $m^{\omega(D)} = N^{o(1)}$. The CRT leaves at most one center modulo $D > N$ in the interval.

If $\prod_i a_i^2 \leq N$, start with this product and add the small prime factors from the least common multiple until the first crossing of N. Now $N < D \leq NY$ and $D = sa^2$ with s Y-smooth. For fixed $\sigma > \max(1/2, 1 - 1/A)$, the number of such modules is at most

$$(NY)^\sigma \sum_{a \geq 1} a^{-2\sigma} \prod_{\ell \leq Y} (1 - \ell^{-\sigma})^{-1} = N^{\sigma + o(1)}.$$

Choose σ just above $1 - 1/A$. Assigning the complete prime powers again costs $N^{o(1)}$ and the CRT gives at most one center. This proves (15.6).

For the second assertion, write $n_1 = c_1 d^2 v$ and $n_2 = c_2 e^2 w$, where c_1, c_2 are squarefree and H-smooth and v, w are the odd kernels above H. The kernels are coprime because their common prime would divide a nonzero difference of size less than H. The product $vw \leq N^q$, its division into v, w, the two smooth parts, and the offsets have $N^{q+o(1)}$ choices. For these choices, $c_2 we^2 - c_1 vd^2 = n_2 - n_1 \neq 0$ has $N^{o(1)}$ solutions by Lemma T1.2, uniformly in the effective polynomial heights. □

T12 The first retained class on the regular set

This section specializes the general stopping identity of Section 5 to the two leading pair channels. It supplies the boundary estimates used by the first reconstruction and the positional comparison. The optimal scalar reconstruction uses the larger class $\mathscr{D}_2$ in Section 15. Here and in the regular-set reconstruction below, write

$$G = G_N, \quad n = |G|, \quad Z_G = Z_N^\circ, \quad \mu = np, \quad E_N = \mathcal{E}_N, \qquad a_N = \mathbb{E}(Z_N)_2 - \lambda_N^2.$$

The pair calculation and mean estimate give $a_N - \alpha_N = O_{C,\varepsilon}(N^{-1+\varepsilon})$ and $a_N \asymp_C LN^{-2/3} > 0$ for large N. Every space on G has dimension L, and every marginal is p.

T12.1 Retaining a direct family and one compressed pair

We now use the regular set G of arithmetic windows. For $I \subseteq G$, recall

$$\rho(I) = |I|L - \dim \sum_{x \in I} S_x.$$

Every S_x has dimension L, and a family is called direct when $\rho(I) = 0$. Let E_N be the family of good pairs from Corollary 3.4. Its pairs are compatible and dependent, and

$$\rho(e) \leq L/3 + O(1) \qquad (e \in E_N). \tag{T12.1}$$

For $e = \{x, y\}$ put $V_e = S_x + S_y$ and $q_e = q(e)$. Define

$$\Gamma_e(h) = \{U \subseteq G \setminus e : |U| = h, \quad V_e, \ (S_z)_{z \in U} \text{ are in direct sum}\}. \tag{T12.2}$$

The direct-sum condition concerns the single compressed block V_e and the other blocks. It does not assert independence of the two spaces inside e. The affine probability formula gives

$$q(e \cup U) = q_e p^{|U|} \qquad (U \in \Gamma_e(|U|)). \tag{T12.3}$$

Let $\mathscr{D}$ consist of all direct families together with all families $e \cup U$ occurring in (T12.2). Thus a family in $\mathscr{D}$ is allowed one good pair, treated as a single block, but no further dependence.

Lemma T12.1 (Heredity and boundary nullity)**.** *The family $\mathscr{D}$ is hereditary. Each non-direct member contains a unique dependent pair $e \in E_N$, and its nullity equals $\rho(e)$. Hence, for all sufficiently large N,*

$$\partial\mathscr{D} \subseteq \{I \notin \mathscr{D} : 0 < \rho(I) \le 4L/3 + O(1) < 3L/2\}. \tag{T12.4}$$

Proof. Every subfamily of a direct family is direct. Consider $e \cup U$ in the second part of the definition. A subfamily which retains both endpoints of e still consists of V_e and some of the independent singleton blocks. A subfamily which removes an endpoint is direct, since the remaining endpoint space is contained in V_e. This proves heredity.

The direct-sum condition excludes a dependent pair inside U and excludes a dependent pair joining U to either endpoint of e. Thus e is the unique dependent pair. Additivity of the ranks of the compressed blocks then gives $\rho(e \cup U) = \rho(e)$.

Adding one L-dimensional space increases nullity by at most L. The proper prefix of a boundary member has nullity at most $L/3 + O(1)$, so the member itself has nullity at most $4L/3 + O(1)$. Finally, nullity zero would make it a direct family and therefore a member of $\mathscr{D}$. □

The retained dependence has the size of the principal correction. The strict upper bound on the boundary nullity is what makes the stopped remainder easier to estimate than the unrestricted absolute expansion.

T12.2 The factorial tail of the boundary

Write $n = |G|$ and $\mu = np$, and choose a fixed upper bound Λ for μ throughout the critical band. We need a cutoff large enough to absorb the loss $2^{3L/2}$ in (T12.4), yet small enough for the critical-size component estimate. The following explicit choice meets both requirements:

$$K_N = \min\left\{K \in \mathbb{N} : K \ge \max(4, \lceil 4e\Lambda \rceil), \quad K \log \frac{K}{2e\Lambda} \ge 3 \log N\right\}. \tag{T12.5}$$

It satisfies

$$K_N = (3 + o(1))\frac{\log N}{\log\log N} = o(L); \qquad K_N \le L/4 \quad \text{for all sufficiently large } N. \tag{T12.6}$$

The coefficient 3 is a convenient sufficient choice.

Lemma T12.2 (Large boundary families)**.** *Uniformly in the fixed critical band,*

$$\sum_{\substack{I \in \partial\mathscr{D} \\ |I| > K_N}} 2^{|I|} q(I) = O_C(N^{-3/2}). \tag{T12.7}$$

Proof. The affine law and (T12.4) imply

$$q(I) \le 2^{3L/2} p^{|I|} \ll_C N^{3/2} p^{|I|}.$$

This remains valid for incompatible families, whose probability is zero. Therefore

$$\sum_{\substack{I\in\partial\mathscr{D}\\ |I|>K_N}} 2^{|I|}q(I) \ll_C N^{3/2}\sum_{r>K_N}\binom{n}{r}(2p)^r \le C_C N^{3/2}\sum_{r>K_N}\frac{(2\mu)^r}{r!}.$$

By $K!\ge (K/e)^K$ and (T12.5), $(2\mu)^{K_N}/K_N!\le N^{-3}$. The ratios of subsequent terms are at most $2\Lambda/(K_N+1)<1/2$, so the last tail is at most $2N^{-3}$. This proves the assertion. □

Only families of size at most K_N remain. They have nullity at most $3L/2$, a bound fixed relative to L. The component estimate in Theorem T16.3 supplies the arithmetic input for this range. Section T17 will convert that centered estimate into the positive boundary mass required by (5.5).

T13 Proofs of the uniform certificate tools

All definitions and parameter conditions are those of Section 7. The counts are geometric, and all constants are uniform in the size of the family being summed.

T13.1 Proof of Lemma 7.1

Proof. Write uniquely $v=ksa^2$, where $k=K_Y(v)$ and s is the squarefree product of the primes at most Y occurring to odd order. The inequality $1\le (C_1N/(ksa^2))^\sigma$ on the range being counted gives an upper bound

$$(C_1N)^\sigma\left(\sum_{k\le T}k^{-\sigma}\right)\left(\sum_{a\ge1}a^{-2\sigma}\right)\prod_{\ell\le Y}(1+\ell^{-\sigma}).$$

Enlarging the sum over k to all positive integers only increases it. The three factors are respectively $O_\sigma(T^{1-\sigma})$, $\zeta(2\sigma)$, and at most $\exp(\sum_{j\le Y}j^{-\sigma})$. This proves the bound. □

T13.2 Proof of Lemma 7.2

Proof. There are at most rNB^2 choices for the anchor and its two offsets. The product Q has at most m prime factors, and each has at most rB destinations. Once these assignments are fixed, the Chinese remainder theorem leaves at most $1+N/D_j$ possibilities for centre j. At most two of the D_j exceed N, since $Q\le 9N^2$. For those two modules, replacing $1+N/D_j$ by $2N/D_j$ costs at most $(6B)^2$. Therefore

$$\prod_{j\ne\text{anchor}}(1+N/D_j)\le 2^{r-1}(6B)^2N^{r-1}T^{-2}.$$

This proves (7.4); allowing collisions and incompatible assignments only enlarges the count.

After multiplication by $(up)^r/r!$, the remaining series is, up to a fixed factor,

$$N^{-2\beta}B^{m+4}\sum_{r\ge0}\frac{z^r r^{m+1}}{r!},\qquad z=2uNp=O_u(1).$$

Since $r^{m+1}\le (m+1)!e^r$, this series is at most $N^{-2\beta}B^{m+4}(m+1)!\exp(ez)$. Finally

$$m=\left(\frac{2}{A_0}+o(1)\right)\frac{H}{\log H},\qquad \log(B^m(m+1)!)=\left(\frac{4}{A_0}+o(1)\right)H.$$

This yields (7.5). The factorial here divides the orderings of a set of distinct windows; it is not an assumption on the growth of a fixed-order arithmetic constant. □

T13.3 Proof of Lemma 7.3

Proof. For an outgoing edge, choose two anchor offsets in B^2 ways. Each of the at most m primes is either unused or assigned to one of B offsets in the receiver. A qualifying code has modulus $D > N$ and therefore determines at most one receiving centre in I_N. This gives $B^2(B+1)^m = N^{2/A_0+o(1)}$ possibilities.

For the incoming degree, fix the receiving window y and let Ω_y be the primes greater than Y dividing any of its values. Writing $R = \lceil B \log(3N)/\log Y \rceil$, we have $|\Omega_y| \le R$. A code chooses two anchor offsets, at most m primes from Ω_y, and one of the two offsets for each chosen prime. For a qualifying code their product exceeds N, so its congruences determine at most one anchor centre. Thus the number of possible incoming neighbours is at most

$$B^2 \sum_{j=0}^{m} \binom{R}{j} 2^j \le B^2(m+1)(2eR/m)^m = N^{2/A_0+o(1)}.$$

The displayed binomial bound holds for large N, since $m \le R/2$. Nonqualifying codes may be included in the number of codes; the one-centre conclusion is applied only to qualifying ones. No receiving offset needs to be chosen in this incoming count: the receiver has already been fixed. Symmetrisation adds the two degree bounds. □

T13.4 Proof of Lemma 7.4

Proof. There are at most $n(4\Delta)^{h-1}$ connected vertex sets of size $h \ge 2$: encode a canonical rooted spanning tree by its plane-tree shape, its root and the choices of neighbours along its edges. The number of plane-tree shapes is at most 4^{h-1}. Allowing repeated vertices only enlarges this count.

Decompose each admissible set into its induced connected components. Singleton components must be marked. Dropping disjointness and the absence of edges between the other components bounds the generating function by $\exp(Mz + \sum_{h\ge2} n(4\Delta)^{h-1} z^h)$. In the exponential, the orderings of distinct components cancel the factorial denominator; repeated components merely contribute additional nonnegative terms. Summing the geometric series proves the claim. □

T14 From fixed orders to all primitive families

The estimates at each fixed order do not control the growing order needed to reconstruct a distribution. We now obtain that uniformity without estimating the growth of the fixed-order constants. The argument has two parts. A large squarefree kernel produces congruences which count the possible centres. When too much of this kernel is received by one other window, it produces an edge in a graph of small degree. A generating function then controls all large families at once. Only a finite set of small cardinalities is left to the fixed-order estimates.

Throughout this section, a family of distinct windows is *primitive* if its spaces are not in direct sum but every proper subfamily is in direct sum. Connectedness refers to the canonical decomposition into direct-sum components in Lemma 4.4. All families in this section consist of windows in the regular set G_N. The quantities $q(I)$, $c(I)$, $\rho(I)$ and $\rho_*(I)$ retain their definitions from Section 4. All weights and all auxiliary parameters in this section are fixed before N tends to infinity.

T14.1 The centred estimate at each fixed order

The small-cardinality part of the argument needs an estimate for absolute centred coefficients, including incompatible affine systems. Positive probabilities alone do not give this estimate: a joint event can be impossible while its centred coefficient is nonzero. For separated windows we

remove this difficulty by using a homogeneous majorant. Contacts require a different argument, since their homogeneous probabilities can be large.

For $x \in G_N$, let H_x be the indicator of the homogeneous event that every character in S_x takes the value 1. Equivalently, the random multiplicative function is constant on the $L+1$ values in window x. Set

$$W(I) = \mathbb{E} \prod_{x \in I} H_x = p^{|I|} 2^{\rho(I)}. \tag{T14.1}$$

All homogeneous systems are compatible. Moreover, the full-support Fourier formula of Lemma 4.2, after taking absolute values, gives

$$|c(I)| \le W(I). \tag{T14.2}$$

Here the geometric good-pair class includes pairs whose prescribed affine run-start events are incompatible; denote this enlarged class by $\mathcal{A}_{\mathrm{raw}}$.

Lemma T14.1 (Absolute centred mass at a fixed order)**.** *For every fixed integer $r \ge 3$ and $\varepsilon > 0$,*

$$\sum_{\substack{C \subseteq G_N,\ |C|=r \\ C \text{ connected}}} |c(C)| \ll_{r,\varepsilon} N^{-4/5+\varepsilon}. \tag{T14.3}$$

The sum over all r-element families containing a contact, with no connectedness restriction, is $O_{r,\varepsilon}(N^{-1+\varepsilon})$.

Proof. First suppose that all windows are separated. Call a family residual if its relation space is neither zero nor exactly the relation space of one geometric good pair, extended by zero on the other blocks. A connected family of at least three windows is residual. We prove the stronger bound obtained by summing $W(C)$ over all residual separated families of size r.

Proposition 6.3 applies to the homogeneous targets as well as to the prescribed affine targets. Its rank increment $\dim(S_x \cap \sum_{y \in I} S_y)$ is geometric, and the conditional probability is p times the corresponding power of 2 in either case. Consequently a seed may be completed to any fixed total cardinality at a cost N^{ε}, after reducing the error exponent used for the seed. The following four cases exhaust residual families.

If a family contains a dependent pair outside $\mathcal{A}_{\mathrm{raw}}$, the first seed estimate of Proposition T1.12 gives total weight $O_\varepsilon(N^{-4/5+\varepsilon})$. Propagation adds the other $r-2$ windows at the stated cost. If it contains an original primitive subfamily of size $h \ge 3$, with $h \le r$, Proposition T4.3 gives the same bound for its homogeneous weight. There are only finitely many such values of h, so propagation again gives the required estimate.

Next suppose that the family contains two distinct geometric good pairs. Their union has three or four windows. By Proposition T1.12, the total homogeneous weight of these unions is

$$O_\varepsilon(N^{-5/4+\varepsilon} + L^2 N^{-4/3}) = O_\varepsilon(N^{-5/4+\varepsilon}).$$

Completing the union to r windows costs only N^{ε}. This case includes a pair sharing one endpoint with the other pair.

The remaining family has a unique geometric good pair e and an additional relation. Compress e exactly, and choose a minimal dependent subfamily of the resulting blocks. If it avoids e, it is one of the original primitive seeds already considered. Otherwise it consists of the compressed pair and $h-1$ original windows. For $h=2$, the pair-and-target seed estimate of Proposition T1.12 gives $O_\varepsilon(N^{-4/3+\varepsilon})$ for channel $2:3$ and $O_\varepsilon(N^{-5/4+\varepsilon})$ for channel $3:4$. For $h \ge 3$, Proposition T4.4 gives

$$O_{h,\varepsilon}(N^{-\kappa_b+1/5+\varepsilon}), \qquad \kappa_b = 2 - 1/b, \quad b \in \{3,4\}.$$

The largest of these bounds is $O(N^{-22/15+\varepsilon})$. Propagation is applied after replacing the compressed seed by its original windows. The number of seeds and placements in a family of fixed size is

bounded in terms of r. These cases prove the separated estimate by (T14.2). No compatibility restriction has been introduced in this argument.

For contacts, return to the actual affine events. Expanding the centred product gives

$$|c(C)| \le \sum_{U \subseteq C} p^{r-|U|} q(U). \tag{T14.4}$$

Fix $h = |U|$ and write $u = r - h$. If U already contains a contact, the positive contact seed followed by Proposition 6.3 gives

$$\sum_{\substack{|U|=h \\ U \text{ contains a contact}}} q(U) \ll_{h,\varepsilon} N^{-1+\varepsilon}.$$

The remaining u centres and their factors p^u cost at most $(Np)^u = O_r(1)$. If U is separated but C contains a contact, at least one endpoint of a contact is among the u new centres. For fixed U there are at most $O_r(BN^{u-1})$ such completions: either a new centre is within $O(B)$ of U, or two new centres are within $O(B)$ of each other. The cases where no completion is possible contribute zero. Since

$$\sum_{|U|=h} q(U) = \mathbb{E}\binom{Z_N^\circ}{h} = O_h(1)$$

by the fixed moments in Theorem 6.4, these completions contribute $O_r(B/N)$. Summing over the finitely many values of h proves the contact assertion. The moments used here are proved from the positive affine classification before this lemma; they do not depend on any all-cardinality centred estimate. □

T14.2 Primitive families at all cardinalities

Theorem T14.2 (Primitive families at all cardinalities)**.** *For every fixed $w > 0$ and $\varepsilon > 0$,*

$$\sum_{\substack{C \subseteq G_N \text{ primitive} \\ |C| \ge 3}} w^{|C|}\big(q(C) + |c(C)|\big) \ll_{w,\varepsilon} N^{-4/5+\varepsilon}. \tag{T14.5}$$

The sum includes overlapping windows and incompatible affine systems.

T14.3 One rank loss for a primitive family

If C is primitive, the projection of its relation space onto any one block is injective. Indeed, an element in the kernel of that projection would be a relation supported on a proper subfamily. Since a regular window has rank L, a primitive family of $r \ge 3$ windows satisfies

$$1 \le \rho(C) \le L, \qquad q(C) \le p^{r-1}, \qquad c(C) = q(C) - p^r. \tag{T14.6}$$

The last identity follows by expanding the centred product: every proper subfamily has joint probability equal to the product of its marginals. Consequently $|c(C)| \le p^{r-1}$, including when $q(C) = 0$.

A nonzero relation of a primitive family uses every block. Its projection on a regular window corresponds to a nonempty even selection of the $B = L + 1$ values in that window. It therefore selects at least two distinct values there. This elementary fact is what permits a two-value counting argument, rather than a one-value friability estimate.

The kernel, moderate-certificate, saturation graph and generating-function estimates used next are stated in Section 7 and proved in Section T13. Their p-weighted forms are used here on the regular set.

T14.4 The finite-small-order, uniform-large-order split

Take

$$A_0 = 40, \quad \beta = \frac{24}{25}, \quad \sigma = \frac{49}{50}, \quad \delta = \frac{1}{1250}, \quad \theta = \frac{2}{A_0} = \frac{1}{20}. \tag{T14.7}$$

These values leave a strict margin in all subsequent inequalities. For a fixed $w > 0$, substitute $z = wpN^{\delta/2}$ into Lemma 7.4. The three quantities in its exponent or denominator satisfy

$$Mz \le N^{-\delta/2+o(1)}, \qquad \Delta z \le N^{-1+\theta+\delta/2+o(1)}, \qquad n\Delta z^2 \le N^{-1+\theta+\delta+o(1)}.$$

They tend to zero. Consequently, for every fixed integer $s \ge 3$,

$$\sum_{\substack{C \text{ primitive, } |C|\ge s \\ C \text{ without a moderate certificate}}} w^{|C|}(q(C) + |c(C)|) \ll_w N^{1-s\delta/2}. \tag{T14.8}$$

Indeed, removing the tilt contributes $N^{-s\delta/2}$ and the primitive rank bound contributes at most $2p^{-1} = O(N)$. For the moderate families, Lemma 7.2 and the same rank bound give $N^{1-2\beta+4/A_0+o(1)} = N^{-41/50+o(1)}$.

Proof of Theorem T14.2. Use $s = 5000$, so the nonmoderate tail in (T14.8) is $O_w(N^{-1})$. For each of the finitely many sizes $3 \le r < s$, the fixed-order primitive estimate in Proposition T4.3 gives $O_{r,\varepsilon}(N^{-4/5+\varepsilon})$ for the positive mass of separated families. Compatible contacts and their fixed-order extensions contribute $O_{r,\varepsilon}(N^{-1+\varepsilon})$. Lemma T14.1 controls $|c(C)|$, including contacts and incompatibility. Alternatively, for a primitive one can use $|c(C)| \le q(C) + p^r$ and the fixed-order support count. Replacing its fixed weight by w^r costs only a constant on this finite set of sizes. Add these estimates to the two large-size bounds. No fixed-order estimate has been applied with $r = r(N)$. □

T15 One distinguished pair after compression

The approximation will keep the contribution of one good pair. To control additional relations above it, the pair must first be treated as a single affine block. Centring this new block is different from centring its two original indicators; we retain the distinction until the final identity of this section.

For an affine-compatible good pair $e = \{x, y\}$ of channel $(a, b) = (2, 3)$ or $(3, 4)$, let $J_e = J_x J_y$, $q_e = \mathbb{E}J_e$ and $V_e = S_x + S_y$. Its exact compression has centre $X_e = bx$, offset $h_e = ay - bx = O(B)$, and values

$$U_e = b\mathcal{V}_x \cup a\mathcal{V}_y, \qquad \mathcal{V}_x = \{x - 1, x, \dots, x + L - 1\}.$$

Thus U_e is a set of integer values, while V_e is a vector space. Lemma 3.5 gives an injective even-selection representation on the distinct values of U_e, width $O(B)$, at most $2B$ values, and

$$r_e = \dim V_e \ge \kappa_b L - O(1), \qquad q_e = 2^{-r_e} \ll N^{-\kappa_b}, \qquad \kappa_b = 2 - 1/b. \tag{T15.1}$$

There are $B^{O(1)}$ possible motifs; for a fixed motif its centre determines the pair subject to the integral congruences. The affine event itself is preserved, without an additional random sign from the multipliers.

For a set A of original windows disjoint from e, put

$$q(e, A) = \mathbb{E}\left[J_e \prod_{z\in A} J_z\right], \qquad \widehat{c}(e, A) = \mathbb{E}\left[(J_e - q_e) \prod_{z\in A} (J_z - p)\right].$$

The configuration is primitive after compression when the spaces $V_e, (S_z)_{z\in A}$ are primitive. Its number of compressed blocks is $r = |A| + 1$ and its number of original windows is $r + 1$.

Theorem T15.1 (A distinguished compressed pair at all cardinalities)**.** *For fixed $w > 0$, let*

$$\mathcal{P}_b(w) = \sum_{\substack{(e,A) \text{ primitive after compression} \\ e \text{ in channel } a{:}b,\ |A|\geq 1}} w^{|A|+2}\{q(e,A) + |\widehat{c}(e,A)| + |c(e \cup A)|\}.$$

Then, for each $\varepsilon > 0$,

$$\mathcal{P}_3^{\mathrm{sep}}(w) \ll_{w,\varepsilon} N^{-4/3+\varepsilon}, \qquad \mathcal{P}_4^{\mathrm{sep}}(w) \ll_{w,\varepsilon} N^{-5/4+\varepsilon}, \tag{T15.2}$$

$$\mathcal{P}_b(w) \ll_{w,\varepsilon} N^{-1+\varepsilon}, \qquad b = 3, 4. \tag{T15.3}$$

The superscript restricts all original windows to be separated. The pair remains distinguished in each sum.

Proof. Projection onto any original singleton block is injective on the primitive relation space. Thus its nullity is at most L, and

$$\widehat{c}(e,A) = q(e,A) - q_e p^{r-1}, \qquad q(e,A),\ |\widehat{c}(e,A)| \leq q_e p^{r-2} \ll N^{2-\kappa_b} p^r. \tag{T15.4}$$

For an incompatible configuration, the centred value is exactly $-q_e p^{r-1}$. These estimates retain the saving from the pair rank.

Use the parameters (T14.7). The two-anchor certificate works on the compressed block as well: all its distinct values lie in an interval of width $O(B)$ and are $O(N)$, so their large kernels are coprime. Replace the moderate threshold by $C_2 NB$ for a sufficiently large fixed constant, and put $m_* = \lceil 2\log(C_5 N)/\log Y \rceil$, where C_5 bounds all compressed values divided by N. With $P_N = B^{O(1)}$ possible pair motifs, the ordered moderate host count is at most

$$P_N C_3^r r^2 B^4 N^r T^{-2} (C_4 r B)^{m_*}.$$

There is only one factor P_N, since only one compressed block is present. Dividing by $r!$, using (T15.4) and summing the exponential series as before gives

$$N^{2-\kappa_b-2\beta+4/A_0+o(1)} = N^{-\kappa_b+9/50+o(1)} \tag{T15.5}$$

for each of the first two masses. The factors P_N and $C_4^{m_*}$ are $N^{o(1)}$.

For the nonmoderate sector use a typed graph whose vertices are all original windows and all possible compressed good-pair blocks of the fixed channel. It has $n_* = N^{1+o(1)}$ vertices, $M_* = N^{1-\delta+o(1)}$ marked vertices, and maximum saturation degree $\Delta_* = N^{\theta+o(1)}$. In the incoming and outgoing degree counts a fixed motif adds only $B^{O(1)}$ choices. Every unmarked vertex of a primitive configuration of at least three blocks has a saturated neighbour. Lemma 7.4, with $z = wpN^{\delta/2}$, is still bounded because $M_* z$, $\Delta_* z$, and $n_* \Delta_* z^2$ tend to zero with strict power margins. We may enlarge the generating function to all typed families, then retain (T15.4) for the configurations with exactly one distinguished pair. For each fixed $s \geq 3$, this bounds their tail by

$$O_w(N^{2-\kappa_b-s\delta/2}).$$

At $s = 5000$ this is $O_w(N^{-\kappa_b})$.

It remains to treat finitely many $2 \leq r < 5000$. For separated configurations with fixed $r \geq 3$, Proposition T4.4 bounds their homogeneous mass by $O_{r,\varepsilon}(N^{-\kappa_b+1/5+\varepsilon})$. This bounds both q and $|\widehat{c}|$, since the latter is at most the homogeneous mass even when the full affine configuration is incompatible. For $r = 2$, the pair-and-target estimate of Proposition T1.12 gives respectively $O_\varepsilon(N^{-4/3+\varepsilon})$ and $O_\varepsilon(N^{-5/4+\varepsilon})$ for the same homogeneous majorant. Thus no phase restriction is needed in the separated estimate.

For the contact sector, Proposition 6.3 bounds the mass of the $r+1$ original windows by $O_{r,\varepsilon}(N^{-1+\varepsilon})$. The number of possible distinguished pairs in such a fixed-size family is bounded

in terms of r. The deterministic host count for these contacts is $N^{r-1+o(1)}$: the good pair costs $NB^{O(1)}$, and a contact removes one remaining free centre. Multiplying by the centering term $q_e p^{r-1}$ gives $N^{-\kappa_b+o(1)}$. The identity in (T15.4) therefore bounds $|\widehat{c}|$ on contacts, including incompatible configurations. Together with the two tails, these bounds prove (T15.2)–(T15.3) for q and $|\widehat{c}|$.

Finally, A is a nonempty direct family, and expansion of $(J_x-p)(J_y-p)$ gives the exact identity

$$c(e\cup A)=\widehat{c}(e,A)-p\,c(\{x\}\cup A)-p\,c(\{y\}\cup A). \tag{T15.6}$$

Each endpoint family is direct or primitive: a proper subset containing the endpoint is contained in a proper direct family of compressed blocks. A fixed original primitive family of size r can be extended by a distinguished companion in at most $O(rB)$ ways, since the deterministic good-pair degree is $O(B)$. Hence the sum of the two corrections is bounded by a fixed multiple of

$$wpB\sum_{r\ge 2} rw^r \sum_{\substack{D\ \text{primitive}\\ |D|=r}} |c(D)|.$$

For $r\ge 3$, absorb r into a slightly larger fixed weight and use Theorem T14.2; for $r=2$, use the total pair activity $O(LN^{-2/3})$. The result is $O_{w,\varepsilon}(pB(LN^{-2/3}+N^{-4/5+\varepsilon}))=O_{w,\varepsilon}(N^{-5/3+\varepsilon})$. This is smaller than each claimed bound and proves the assertion for the original centred coefficient as well. □

T15.1 The active structure outside a small event

Lemma T15.2 (A single pair is the only remaining active component)**.** *There exists an event E_* with $\mathbb{P}(E_*)\ll_\varepsilon N^{-4/5+\varepsilon}$ such that, outside E_*, all active spaces are a direct sum of singleton spaces and at most one compressed good-pair space. In particular every active subfamily has nullity at most $L/3+O(1)$.*

Proof. Exclude an active contact, an active nongood dependent pair, an active primitive family of at least three original windows, two distinct active good pairs, and an active primitive family after compression of a good pair. The pair and contact estimates, the two-pair seed bound in Proposition T1.12, and Theorems T14.2 and T15.1 bound the union by the asserted probability. All sums here are positive.

If no good pair is active, a minimal dependent active subfamily would be one of the excluded families. If exactly one good pair is active, compress it. A minimal dependent subfamily of the resulting blocks either avoids the compressed block, giving an excluded original primitive, or contains it, giving an excluded compressed primitive. Thus the remaining compressed blocks are direct. The only rank loss is that of the good pair, at most $L/3+O(1)$. □

T16 Connected coefficients with bounded compatible nullity

The preceding theorems concern minimal dependent families. The boundary of the stopped expansion can contain larger connected families. We first control all cardinalities when the compatible nullity is less than two window ranks. We then remove the restriction on that fixed nullity level over the growing range needed by the expansion.

T16.1 A two-anchor estimate at every cardinality

Theorem T16.1 (All cardinalities below two ranks)**.** *For fixed $0\le\eta<2$, $w>0$ and $\varepsilon>0$,*

$$\sum_{\substack{C\subseteq G_N\ \text{connected},\ |C|\ge 3\\ \rho_*(C)\le\eta L}} w^{|C|}|c(C)| \ll_{\eta,w,\varepsilon} N^{-\min\{4/5,\,2-\eta\}+\varepsilon}. \tag{T16.1}$$

Proof. Expansion of the centred product gives

$$|c(C)| \le 2^{|C|}p^{|C|}2^{\rho_*(C)} \ll_\eta N^\eta(2p)^{|C|}. \tag{T16.2}$$

The Fourier expansion also shows that $c(C) \neq 0$ implies a relation meeting every block: centring removes the zero Fourier coordinate in each factor. Such a relation again selects at least two values per block. Therefore a family with nonzero coefficient and no moderate certificate has only marked isolated vertices in the saturation graph. No independence assertion about the edges of that graph is used.

With the parameters in (7.2), the moderate sector, by Lemma 7.2, costs $N^{\eta-2\beta+4/A_0+o(1)}$. The graph generator bounds its nonmoderate tail beyond any fixed size s by $O_{\eta,w}(N^{\eta-s\delta/2})$. For $0 < \varepsilon < \min\{1, 2-\eta\}$ choose

$$A_0 > \max\{40, 32/\varepsilon\}, \qquad \beta = 1-\varepsilon/16, \qquad \sigma = 1-1/(2A_0).$$

Then $\delta = \varepsilon/(32A_0)$, $2/A_0 + \delta < 1$, and $2-2\beta+4/A_0 < \varepsilon/4$. Choose a fixed integer s with $s\delta/2 \ge \eta+1$. The moderate part has the required $2-\eta$ exponent, the nonmoderate tail is $O(N^{-1})$, and the finitely many connected sizes $3 \le r < s$ are handled by Lemma T14.1. Shrinking the margins there gives (T16.1). Larger values of ε follow from smaller ones. □

At $\eta = 1$, this already gives the $4/5$ exponent at all sizes. At $\eta = 3/2$, however, the two-anchor estimate would give only the exponent $1/2$. We need more anchors where a block carries many independent relation coordinates.

T16.2 Projection dimensions and positive cores

For a family D, let $\mathcal{K}(D)$ be the kernel of the sum map $\bigoplus_{x\in D} S_x \to \sum_{x\in D} S_x$. The dimension of its projection onto S_x is

$$d_x(D) = \rho(D) - \rho(D\setminus\{x\}) = \dim\left(S_x \cap \sum_{y\in D\setminus\{x\}} S_y\right). \tag{T16.3}$$

The kernel of that projection is exactly $\mathcal{K}(D\setminus\{x\})$, which proves the identity.

If t is even, a subspace of dimension at least $t-1$ in the even-selection space contains a vector of weight at least t. Indeed, choose $t-1$ pivot coordinates and set all of them equal to one. The resulting vector has weight at least $t-1$, and its weight is even. Call D a *t-core* when $d_x(D) \ge t-1$ for every $x \in D$. At each of its blocks a relation therefore selects at least t distinct values. The selected relation may depend on the block.

Proposition T16.2 (Positive mass of high-nullity cores)**.** *Fix $\eta > 1$, an even integer $t \ge \eta+2$, and $v > 0$. Then*

$$\sum_{\substack{D \text{ compatible, } D\ t-\text{core}\\ L/2<\rho(D)\le\eta L}} v^{|D|}q(D) \ll_{\eta,t,v,\varepsilon} N^{-4/5+\varepsilon} \tag{T16.4}$$

for every $\varepsilon > 0$, with no restriction on the cardinality.

Proof. We give the multiple-anchor count explicitly. Set

$$A_0 = 8t, \qquad Y = (\log N)^{A_0}, \qquad \beta = 1-\frac{1}{4t}, \qquad T = N^\beta,$$
$$\sigma = 1-\frac{1}{16t}, \qquad \delta = \frac{1}{64t^2}, \qquad m_t = \left\lceil \frac{t\log(3N)}{\log Y}\right\rceil.$$

The number of marked windows remains $N^{1-\delta+o(1)}$. At an unmarked anchor choose t selected values. Their kernel product Q is squarefree, with $T^t < Q \le (3N)^t$ and at most m_t prime factors.

A moderate allocation has $D_j \le 6NB$ for each receiver. At most t received modules exceed N, so its CRT cost is

$$\prod_{j \ne \text{anchor}} (1 + N/D_j) \le 2^{r-1}(6B)^t N^{r-1} T^{-t}.$$

The corresponding ordered host count is at most $6^t r B^{2t} 2^{r-1} N^r T^{-t} (rB)^{m_t}$. Multiplication by $(up)^r/r!$ and the same exponential-series estimate as before give, for fixed $u > 0$,

$$\sum_r \frac{(up)^r}{r!} |\mathcal{U}_{r,t}| \le N^{-t\beta + 2t/A_0 + o(1)}.$$

Here $\log(B^{m_t+2t}(m_t+1)!) = (2t/A_0 + o(1)) \log N$. Since $q(D) \ll_\eta N^\eta p^{|D|}$, the positive moderate mass is

$$N^{\eta - t\beta + 2t/A_0 + o(1)} = N^{\eta - t + 1/2 + o(1)} \le N^{-3/2 + o(1)}. \tag{T16.5}$$

For the remaining cores, define saturation using t anchor values. The outgoing degree is bounded by $B^t(B+1)^{m_t}$. For the incoming degree, the fixed receiver has at most $R = \lceil B \log(3N)/\log Y \rceil$ possible primes; the codes are bounded by $B^t \sum_{j \le m_t} \binom{R}{j} t^j$. As in Lemma 7.3, a qualifying code has modulus greater than N and determines at most one centre. Both degrees are $N^{t/A_0 + o(1)} = N^{1/8 + o(1)}$. Every unmarked vertex of a nonmoderate t-core has a saturated neighbour. Lemma 7.4, tilted by $N^{\delta/2}$, therefore bounds the positive mass beyond fixed size s by $O_{\eta,v}(N^{\eta - s\delta/2})$. Choose $s \ge 128t^2(\eta+1)$, so this is $O(N^{-1})$.

For a fixed size $h < s$, compatibility and $\rho(D) > L/2$ imply that activation of D occurs inside the event E_* of Lemma T15.2. Indeed, its complement permits nullity at most $L/3 + O(1)$. For every fixed integer $k > 1$, Hölder and the bounded fixed moments of Theorem 6.4 give

$$\sum_{\substack{|D|=h,\ D \text{ compatible} \\ \rho(D) > L/2}} q(D) \le \mathbb{E}\left[\binom{Z_N^\circ}{h} ; E_* \right] \le C_{h,k} \mathbb{P}(E_*)^{1-1/k}.$$

Choose k large enough that $4/(5k) < \varepsilon/3$, then use a sufficiently small fixed margin in the bound for $\mathbb{P}(E_*)$. The result is $O_{h,\varepsilon}(N^{-4/5+\varepsilon})$. Only finitely many sizes h are involved. Adding these to (T16.5) and the saturated tail proves (T16.4). The distributional approximation being proved in this paper is not used in this argument. □

T16.3 Removing blocks with small relation projections

Theorem T16.3 (Bounded compatible nullity on the growing range)**.** *For every fixed $\eta \ge 0$, $w > 0$, $\varepsilon > 0$ and every $K_N = o(L)$,*

$$\sum_{\substack{C \subseteq G_N \text{ connected} \\ 3 \le |C| \le K_N,\ \rho_*(C) \le \eta L}} w^{|C|} |c(C)| \ll_{\eta,w,\varepsilon} N^{-4/5+\varepsilon}. \tag{T16.6}$$

For $\eta > 1$, choose a fixed even $t \ge \eta + 2$ and set $d = t - 2$. The estimate holds uniformly whenever $K_N \le L/(2d)$. In particular, for $\eta = 3/2$ one may take $t = 4$, $d = 2$ and $K_N \le L/4$.

Proof. Theorem T16.1 handles $\rho_*(C) \le L$. For $L < \rho_*(C) \le \eta L$, choose a compatible subfamily $A \subseteq C$ attaining $\rho(A) = \rho_*(C)$, using a fixed lexicographic rule if needed. Remove from A any block whose relation projection has dimension at most d, until none remains. Let D be the resulting family. Equation (T16.3) shows that

$$0 \le \rho(A) - \rho(D) \le d(|A| - |D|) \le dK_N \le L/2.$$

Thus D is a nonempty compatible t-core with $L/2 < \rho(D) \le \eta L$. If $r = |C|$ and $h = |D|$, the centred expansion and the rank bound give

$$|c(C)| \le 2^r p^r 2^{\rho_*(C)} \le 2^r (2^d p)^{r-h} q(D). \tag{T16.7}$$

The elements of $C \setminus A$ are included in p^{r-h}; they need not have been removed by the algorithm. This also covers incompatible families C.

Sum (T16.7) over all admissible cores and all their supersets. Dropping restrictions on the latter only increases the sum. With $n = |G_N|$ and $np \le Np = \lambda$, this gives

$$\begin{aligned}\sum_{\substack{|C|\le K_N \\ L<\rho_*(C)\le \eta L}} w^{|C|}|c(C)| &\le \sum_{D \text{ admissible}} (2w)^{|D|} q(D)(1+2^{d+1}wp)^{n-|D|} \\ &\le e^{2^{d+1}w\lambda} \sum_{D \text{ admissible}} (2w)^{|D|} q(D).\end{aligned}$$

The last sum is bounded by Proposition T16.2. There is no factor for the deletion order: a deterministic choice assigns one core to each family, and summing over all supersets already overcounts the original family. The exponential factor is constant because d, w are fixed and λ is bounded. Adding the sector $\rho_*(C) \le L$ proves the theorem. □

For the required value $\eta = 3/2$, the choices in the positive-core proof are $A_0 = 32$, $\beta = 15/16$, $\sigma = 63/64$, $\delta = 1/1024$ and $s = 5120$. Its moderate mass is $N^{-2+o(1)}$, its saturated tail is $O(N^{-1})$, and only fixed sizes below 5120 use fixed moments. This supplies the whole range containing $K_N \asymp \log N / \log\log N$ without promoting a fixed-order estimate to a growing-order one.

The same deletion argument gives the positive corollary

$$\sum_{\substack{D \text{ compatible, } |D|\le K_N \\ L<\rho(D)\le \eta L}} w^{|D|} q(D) \ll_{\eta,w,\varepsilon} N^{-4/5+\varepsilon}.$$

The centred form (T16.6), rather than this additional corollary, is the estimate used by the stopped expansion below.

T17 Recovering the law from the retained pair data

The stopped expansion requires a positive sum of joint probabilities. The component theorem controls centered coefficients instead. The two quantities cannot be interchanged by taking absolute values of the entire expansion: a single good pair has mass of the same order as the correction we wish to retain. We separate that pair explicitly, estimate its failed extensions after compression, and use the component bound only for the remaining terms.

Throughout this section, $\mathscr{D}$ is the class from Section T12, $n = |G|$, $\mu = np$, and

$$\beta_N = \sum_{e \in E_N} q_e.$$

All asymptotic constants may depend on the fixed critical band.

T17.1 Two deficits under independent counting weights

For $r \ge 0$ define

$$F_r = p^r \#\{I \subseteq G : |I| = r,\ I \text{ is not direct}\}, \tag{T17.1}$$

$$G_r = \sum_{e \in E_N} q_e p^{r-2} \left\{ \binom{n-2}{r-2} - \#\Gamma_e(r-2) \right\} \qquad (r \ge 2), \tag{T17.2}$$

and put $G_0 = G_1 = 0$. We use zero for binomial coefficients whose lower index lies outside the usual range. In particular, both sequences vanish for $r > n$, and $F_0 = F_1 = 0$. The weighted deficits are

$$F = \sum_{r\ge 0} 2^r F_r, \qquad \mathfrak{G} = \sum_{r\ge 0} 2^r G_r. \tag{T17.3}$$

These are deterministic counting deficits. The added centers carry independent weights p, even when the corresponding arithmetic events are dependent.

Lemma T17.1 (Bounds for the independent deficits)**.** *For every fixed $\varepsilon > 0$,*

$$F \ll_\varepsilon N^{-4/5+\varepsilon} + B/N, \qquad \mathfrak{G} \ll_\varepsilon N^{-1+\varepsilon}. \tag{T17.4}$$

Proof. Let $\mathcal{P}$ denote the minimal non-direct families, including pairs, and set

$$S_0 = \sum_{C\in\mathcal{P}} (2p)^{|C|}.$$

Every non-direct family contains some $C \in \mathcal{P}$. After choosing it, sum all added centers with weight $2p$. The resulting union bound gives

$$F \le (1+2p)^n S_0 \le e^{2\mu} S_0. \tag{T17.5}$$

For a minimal non-direct family of size at least three, every proper subfamily is direct. Expanding the centered product therefore gives $c(C) = q(C) - p^{|C|}$, and hence

$$p^{|C|} \le q(C) + |c(C)|.$$

Theorem T14.2, with fixed weight 2, bounds the contribution of these families by $O_\varepsilon(N^{-4/5+\varepsilon})$. For pairs, the deterministic host bounds of Theorem 3.3, including the $O(NB)$ contacts and good-pair hosts, give

$$\sum_{\substack{C\in\mathcal{P}\\ |C|=2}} (2p)^2 \ll_\varepsilon N^{-4/5+\varepsilon} + B/N.$$

These host counts include incompatible pairs; their independent weights are still p^2. This proves the bound for S_0 and F.

Fix now $e \in E_N$ and a family of singleton blocks which is not direct after adjoining V_e. A minimal non-direct subfamily either lies entirely among the singletons or contains V_e. The first possibility contains an original member of $\mathcal{P}$. Write (e, C) for a minimal non-direct family of the second kind, where $\varnothing \ne C \subseteq G \setminus e$. Summing additional centers with their independent weights gives

$$\mathfrak{G} \le 4e^{2\mu} \left\{ \beta_N S_0 + \sum_{(e,C)\ \text{minimal}} 2^{|C|} q_e p^{|C|} \right\}. \tag{T17.6}$$

The factor 4 accounts for the two original endpoints of e.

To use the compressed estimate, regard $J_e = J_x J_y$ as one indicator of mean q_e and set

$$\widehat{c}(e, C) = \mathbb{E}\left[(J_e - q_e) \prod_{z\in C} (J_z - p) \right].$$

For a minimal non-direct family of these blocks, every proper block subfamily is direct. The same centered expansion yields

$$\widehat{c}(e, C) = q(e \cup C) - q_e p^{|C|}, \qquad q_e p^{|C|} \le q(e \cup C) + |\widehat{c}(e, C)|.$$

Theorem T15.1, with the pair distinguished, bounds the second sum in (T17.6) by $O_\varepsilon(N^{-1+\varepsilon})$. The factor $4\,2^{|C|} = 2^{|C|+2}$ is exactly the weight of the original windows; it introduces no loss depending on the size.

Finally, $\beta_N = O(a_N)$ and $a_N \asymp LN^{-2/3}$. The already obtained bound for S_0 therefore gives $\beta_N S_0 = O_\varepsilon(N^{-1+\varepsilon})$, after choosing a smaller fixed error margin in the input bounds if necessary. Equation (T17.6) proves the second assertion. □

The extensions in this proof are summed under the independent weights. In particular, no identity $q(C \cup U) = q(C)p^{|U|}$ has been used for a family which is not in direct sum.

T17.2 The exact measure retained by the hereditary class

Let $b = \delta_0 + p\Delta$, the Bernoulli law of mean p, and define

$$\mathsf{B}_\beta = b^{*n} + \beta_N \Delta^{*2} * b^{*(n-2)}. \tag{T17.7}$$

At this stage it is convenient to work with a signed correction to the binomial law.

Proposition T17.2 (Evaluation of the retained polynomial)**.** *For the class $\mathscr{D}$,*

$$T_{\mathscr{D}} = \mathsf{B}_\beta - \sum_{r\geq 0}(F_r + G_r)\Delta^{*r}, \qquad \|T_{\mathscr{D}} - \mathsf{B}_\beta\|_1 \leq F + \mathfrak{G}. \tag{T17.8}$$

Proof. The total joint probability of direct families of size r is $\binom{n}{r}p^r - F_r$. By Lemma T12.1, every non-direct member of $\mathscr{D}$ contains a unique good pair. Equations (T12.2) and (T12.3) show that their total contribution is

$$\sum_{e\in E_N} q_e p^{r-2} \#\Gamma_e(r-2) = \beta_N \binom{n-2}{r-2} p^{r-2} - G_r.$$

This term is zero for $r < 2$. Summing against Δ^{*r} and applying the binomial formula gives the identity. The norm bound follows from $F_r, G_r \geq 0$ and $\|\Delta^{*r}\|_1 = 2^r$. □

The coefficient β_N is the sum of the joint probabilities of good pairs. It will become the covariance correction after the independent contribution of those pairs and the binomial diagonal have been accounted for.

T17.3 From centered components to positive boundary mass

For $K \leq L/4$ put

$$\mathscr{Q} = \{I \subseteq G : \rho(I) \leq 3L/2\}, \qquad R_K = \sum_{\substack{I\in\mathscr{Q}\setminus\mathscr{D}\\ |I|\leq K}} 2^{|I|} q(I).$$

The boundary of size at most K is contained in this sum. Define the component activities

$$\begin{aligned} A_{\mathrm{good}} &= 4\sum_{e\in E_N} c(e),\\ A_{\mathrm{bad}} &= 4\sum_{\substack{I\subseteq G,\ |I|=2\\ I\notin E_N}} |c(I)|,\\ B_K &= \sum_{\substack{C \text{ connected},\ 3\leq|C|\leq K\\ \rho_*(C)\leq 3L/2}} 2^{|C|}|c(C)|. \end{aligned} \tag{T17.9}$$

For a good pair, compatibility and positive nullity give $c(e) = q_e - p^2 > 0$. In particular, $0 \leq A_{\mathrm{good}} \leq 4\beta_N = O(a_N)$.

Proposition T17.3 (Positive mass outside the retained class)**.** *For every $K \le L/4$,*

$$R_K \le F + \mathfrak{G} + e^{2\mu}\{\exp(A_{\text{good}} + A_{\text{bad}} + B_K) - 1 - A_{\text{good}}\}. \tag{T17.10}$$

Consequently, uniformly in that range,

$$R_K \ll_\varepsilon N^{-4/5+\varepsilon} + B/N + a_N^2. \tag{T17.11}$$

Proof. Expand each indicator about its mean:

$$q(I) = \sum_{J \subseteq I} p^{|I|-|J|} c(J), \qquad c(\varnothing) = 1. \tag{T17.12}$$

If $I \in \mathscr{Q}$, then every $J \subseteq I$ and every canonical component C of J satisfy $\rho_*(C) \le \rho(C) \le \rho(I) \le 3L/2$. The first inequality follows from the definition of compatible nullity, and the second from monotonicity of nullity.

The term $J = \varnothing$, summed over $I \in \mathscr{Q} \setminus \mathscr{D}$, contributes at most F: all such I are non-direct.

For a term $J = e \in E_N$, write $I = e \cup U$. Since $I \notin \mathscr{D}$, the family U does not belong to $\Gamma_e(|U|)$. Using $0 < c(e) \le q_e$ and then dropping the restrictions on I gives the upper bound

$$4 \sum_{e \in E_N} q_e \sum_{\substack{U \subseteq G \setminus e \\ U \notin \Gamma_e(|U|)}} (2p)^{|U|} = \mathfrak{G}.$$

This is the only part of the argument where a term of order a_N has to be removed before taking an absolute majorant.

It remains to sum nonempty J which are not a single good pair. The coefficient $c(J)$ factors over its canonical direct-sum components. A singleton component has zero coefficient. Every remaining nonzero family is therefore a collection of components of size at least two, containing either a bad pair, a component of size at least three, or at least two good pairs. Its component activities are included in (T17.9).

The sum over these collections is bounded by

$$\sum_{\text{remaining } J} 2^{|J|} |c(J)| \le \exp(A_{\text{good}} + A_{\text{bad}} + B_K) - 1 - A_{\text{good}}. \tag{T17.13}$$

Indeed, a collection of m distinct components has $m!$ orders, cancelling the factorial in the exponential series. Allowing repeated, intersecting or non-independent components only adds nonnegative terms to the majorant. The empty collection and a single good-pair component are precisely the two terms subtracted on the right.

For each fixed J, summing the additional members of $I \setminus J$ in (T17.12) costs at most $(1 + 2p)^{n-|J|} \le e^{2\mu}$. This proves (T17.10).

Theorem 3.3 and Theorem T16.3, with fixed $\eta = 3/2$ and $w = 2$, give

$$A_{\text{bad}} + B_K \ll_\varepsilon N^{-4/5+\varepsilon} + B/N.$$

Use a small fixed error margin so that this quantity tends to zero. For $u, v \ge 0$ tending to zero, $e^{u+v} - 1 - u = O(u^2 + v)$. The bounds for A_{good}, F and $\mathfrak{G}$ now give (T17.11). An estimate for a smaller error margin implies the displayed estimate for every prescribed $\varepsilon > 0$. □

T17.4 Converting the binomial reference to Poisson

Set

$$C_2 = \sum_{\substack{I \subseteq G \\ |I|=2}} c(I), \qquad \alpha_G = \operatorname{Var}(Z_G) - \mu = 2C_2 - np^2. \tag{T17.14}$$

The second identity is the variance formula for a sum of indicators of common mean p. Moreover,

$$\beta_N - C_2 = |E_N|p^2 - \sum_{\substack{|I|=2\\ I\notin E_N}} c(I), \qquad |\beta_N - C_2| \le |E_N|p^2 + \frac{1}{4}A_{\mathrm{bad}}. \tag{T17.15}$$

Thus this difference is $O_\varepsilon(N^{-4/5+\varepsilon} + B/N)$.

Lemma T17.4 (Poisson–Charlier conversion)**.** *With the parameters of the critical band,*

$$\|\mathsf{B}_\beta - Q_{\mu,\alpha_G}\|_1 \le 4|\beta_N - C_2| + O_C(N^{-2} + a_N/N). \tag{T17.16}$$

Proof. Work in the Banach algebra of absolutely summable signed measures. Since $2p < 1$ for large N, the convolution logarithm converges absolutely and gives

$$\log_*(\delta_0 + p\Delta) = p\Delta - \frac{p^2}{2}\Delta^{*2} + O_{\ell^1}(p^3).$$

The convolution exponential of $\mu\Delta$ is the Poisson probability Π_μ, of norm one. Multiplying the preceding identity by n and exponentiating therefore yields

$$b^{*n} = \Pi_\mu - \frac{np^2}{2}\Pi_\mu * \Delta^{*2} + O_{\ell^1}(np^3 + (np^2)^2). \tag{T17.17}$$

The same calculation, retaining only the leading term, gives

$$b^{*(n-2)} = \Pi_\mu + O_{\ell^1}(p + np^2).$$

First replace β_N by C_2 in (T17.7). This costs at most $4|\beta_N - C_2|$, because $b^{*(n-2)}$ is a probability and $\|\Delta^{*2}\|_1 = 4$. The two expansions then give coefficient $C_2 - np^2/2 = \alpha_G/2$ in front of $\Pi_\mu * \Delta^{*2}$. Their remaining error is bounded by

$$O_C(np^3 + (np^2)^2 + |C_2|(p + np^2)).$$

Now $n = O(N)$, $p = O_C(N^{-1})$ and $C_2 = O(a_N)$. This is the error stated in (T17.16). □

T17.5 The first approximation and its uniform boundary

Proposition T17.5 (The first quadratic approximation)**.** *Uniformly in the critical band, for every $\varepsilon > 0$,*

$$\|\mathcal{L}Z_N - Q_{\mu_N,\alpha_N}\|_1 \ll_{C,\varepsilon} N^{-4/5+\varepsilon}.$$

The same bound holds with μ_N replaced by λ_N.

This estimate identifies the leading correction and proves the boundary estimates used below. The optimal scalar reconstruction in Section 15 keeps exact marginals on the whole set of admissible windows.

Proof. Choose $\mathscr{D}$ and K_N as in Section T12. For all sufficiently large N, $K_N \le L/4$. The stopped identity, the boundary-nullity bound, Lemma T12.2 and Proposition T17.3 give

$$\|\mathcal{L}Z_G - T_{\mathscr{D}}\|_1 \le R_{K_N} + O_C(N^{-3/2}) \ll_\varepsilon N^{-4/5+\varepsilon} + B/N + a_N^2 + N^{-3/2}.$$

Proposition T17.2 adds the two deficits $F + \mathfrak{G}$. Lemma T17.4, together with (T17.15), then gives

$$\begin{aligned}\|\mathcal{L}Z_G - Q_{\mu,\alpha_G}\|_1 &\ll_\varepsilon N^{-4/5+\varepsilon} + B/N + a_N^2 + N^{-3/2} + N^{-2} + a_N/N\\ &\ll_\varepsilon N^{-4/5+\varepsilon},\end{aligned} \tag{T17.18}$$

since $B \asymp \log N$ and $a_N \asymp LN^{-2/3}$.

It remains to restore the excluded windows and the desired parameters. The independent reference bounds (8.1) and (8.2) will transfer the parameters.

Proposition 3.1 gives

$$\|\mathcal{L}Z_N - \mathcal{L}Z_G\|_1 + |\mu_N - \mu| + |\alpha_N - \alpha_G| + |\mu_N - \lambda_N| \ll_\varepsilon N^{-1+\varepsilon}.$$

Apply (8.2) first to transfer (μ, α_G) to (μ_N, α_N), and then to replace μ_N by λ_N while retaining α_N. Both costs are absorbed in (T17.18). This proves the two asserted forms of the approximation, uniformly in the fixed critical band.

Finally, if $0 < \varepsilon < 2/15$, then

$$\frac{N^{-4/5+\varepsilon}}{a_N} = O_C\left(\frac{N^{-2/15+\varepsilon}}{\log N}\right) \longrightarrow 0.$$

Thus the corrected error is smaller than the retained pair term. □

T17.6 The total-variation coefficient

Proof of Corollary 1.2. Theorem 1.1 and the reverse triangle inequality first give

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\mu_N}) = \alpha_N \mathcal{C}(\mu_N) + O_{C,\varepsilon}(N^{-4/3+\varepsilon}).$$

Use $|\mu_N - \lambda_N| \ll N^{-1+\varepsilon}$ and (8.2) to obtain

$$\mathcal{L}Z_N - \Pi_{\lambda_N} = \frac{\alpha_N}{2}\Pi_{\lambda_N} * \Delta^{*2} + O_{\ell^1,C,\varepsilon}(N^{-1+\varepsilon}).$$

The reverse triangle inequality, followed by the factor $1/2$ converting the ℓ^1 norm to total variation, yields

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\lambda_N}) = \alpha_N \mathcal{C}(\lambda_N) + O_{C,\varepsilon}(N^{-1+\varepsilon}). \tag{T17.19}$$

The pair asymptotic from Theorem 3.3 now gives

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_{\lambda_N}) \sim \mathcal{C}(\lambda_N)\frac{r_N + 2}{18\, 2^{r_N/3}}\, \lambda_N^{5/3} LN^{-2/3}, \qquad r_N = L \bmod 3. \tag{T17.20}$$

The lower bound in (8.6) is uniform on the fixed critical band, and the remainder is $o(LN^{-2/3})$. This proves the sharp order asserted in the corollary. The equivalence is relative to the displayed oscillating expression; no convergence of its phase or intensity factors is required. □

Replacing α_N by its leading equivalent proves (T17.20). It does not by itself preserve the error $O(N^{-1+\varepsilon})$ in (T17.19), which uses the exact second factorial cumulant.

For the representative case $\lambda = 1$, the only negative masses of $\Pi_1 * \Delta^{*2}$ occur at $k = 1$ and $k = 2$. They are $-e^{-1}$ and $-e^{-1}/2$, respectively. The total mass is zero, so

$$\|\Pi_1 * \Delta^{*2}\|_1 = \frac{3}{e}, \qquad \mathcal{C}(1) = \frac{3}{4e}.$$

Along $N = 2^L$ with L divisible by three,

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \Pi_1) \sim \frac{1}{12e}\, LN^{-2/3}. \tag{T17.21}$$

This computes the coefficient in the example announced in the introduction.

The total-variation asymptotic uses the exact-marginal reconstruction and the explicit pair coefficient. The independent positivity and reference-continuity estimates do not use the reconstruction.

T18 Norms of Poisson differences

The following calculation proves Proposition 19.2. It fixes both the signed-difference convention and the factor of two between ℓ^1 distance and total variation.

Proof of Proposition 19.2. Write $v_k = (\Pi_\mu * \Delta^{*2})\{k\}$. Direct calculation gives

$$v_k = \pi_\mu(k) - 2\pi_\mu(k-1) + \pi_\mu(k-2) = \frac{\pi_\mu(k)}{\mu^2}\big((k-\mu)^2 - k\big).$$

The quadratic has roots u_-, u_+ and is strictly negative exactly at the integers $a, \dots, b$. These form a nonempty interval; integer roots contribute zero and are excluded by the definitions of a, b. Since $\sum_k v_k = 0$,

$$\begin{aligned}\sum_k |v_k| &= -2\sum_{k=a}^{b} v_k \\ &= 2\{\pi_\mu(a-1) - \pi_\mu(a-2) + \pi_\mu(b-1) - \pi_\mu(b)\},\end{aligned}$$

by telescoping. This proves (19.1). For $\mu = 2$, the quadratic is $(k-1)(k-4)$: the negative masses are $v_2 = -e^{-2}$ and $v_3 = -2e^{-2}/3$. Their sum, multiplied by -2, gives (19.3). The same calculation at $\mu = 1$ gives the value $3/e$ used in (T17.21).

For the general bound, define the normalized Charlier polynomials

$$C_r(k;\mu) = \sum_{j=0}^{r} \binom{r}{j} (-1)^{r-j} \frac{(k)_j}{\mu^j}.$$

Then $(\Pi_\mu * \Delta^{*r})\{k\} = \pi_\mu(k) C_r(k;\mu)$. Their exponential generating function is

$$\sum_{r\ge 0} C_r(k;\mu)\frac{t^r}{r!} = e^{-t}(1+t/\mu)^k.$$

For $P \sim \mathrm{Pois}(\mu)$, the expectation of the product of the generating functions with variables s, t is

$$\mathbb{E}\Big[e^{-s-t}(1+s/\mu)^P (1+t/\mu)^P\Big] = e^{st/\mu}.$$

Coefficient comparison, justified in a neighborhood of $(s,t) = (0,0)$ by the Poisson exponential moments, yields

$$\mathbb{E}[C_r(P;\mu)C_\ell(P;\mu)] = \mathbf{1}_{\{r=\ell\}}\frac{r!}{\mu^r}.$$

Cauchy–Schwarz now gives the second bound in (19.2). The first follows from $\|\Pi_\mu\|_1 = 1$, $\|\Delta\|_1 = 2$, and submultiplicativity of the convolution norm. □

T19 Tails, deletion, and limits of positive comparison methods

The stopped expansion controls a distribution without imposing an exponential moment bound on the original count. This distinction is necessary in the present model. We first record the exact support and local structure, then give a nearby count with factorial tails. A real arithmetic cylinder shows why these tails cannot simply be transferred back. The final parts examine deletion of active pairs and the positive cost of forcing private prime signs.

T19.1 Exact support and the local rank-one structure

Write $Z = Z_N$, $p = 2^{-L}$ and $R_N = \lceil N/L \rceil$. For $j \geq 1$, $(z)_j = z(z-1)\cdots(z-j+1)$.

Proposition T19.1 (Support and factorial lattice)**.** *For every $N \geq 2$ and $L \geq 1$,*

$$0 \leq Z \leq R_N, \qquad \mathbb{E}(Z)_j \in j!p^j\mathbb{Z}_{\geq 0} \quad (j \geq 1). \tag{T19.1}$$

Proof. Two starts at distance less than L cannot both be active: the predecessor of the later start lies inside the earlier constant run. Thus the active starts in an interval of N integers are L-separated, which gives the support bound. For j distinct starts, their jL relative sign prescriptions form an affine binary system. Its joint probability is either zero or 2^{-r} for an integer $r \leq jL$. It is therefore an integer multiple of p^j. Finally, $(Z)_j = j!\sum_{|I|=j}\prod_{x\in I} J_x$. □

The local exceptional windows have an exact description. We use the rank-one conclusion of Lemma 2.1; a bound $o(L)$ for the local nullity would not suffice for what follows.

The proof of Lemma 2.1 also gives raw-value nullity at most two at critical length, because the even relation code has codimension at most one in the full relation code. It also gives exact k-wise independence of the star coordinates $Y_i = f(U)f(U+i)$, $1 \leq i < B$, whenever $(k+1)\log(64B) < \log U$: every nonempty product of at most k relative signs selects at most $k+1$ values, and Runge excludes a square product. These are sufficient bounds. They do not establish that critical windows with a nonzero relation actually exist.

Let $\mathcal{R}$ be the collection of nonempty even sets S of integers whose product is a square and which are contained in at least one V_x, $x \in I_N$. Write

$$a_S = \min S, \qquad b_S = \max S, \qquad d_S = b_S - a_S,$$
$$B_S = \{b_S - L + 1, \ldots, a_S + 1\} \cap I_N, \qquad e_S = \mathbf{1}_{\{a_S+1 \in I_N\}}.$$

Proposition T19.2 (Exact local blocks)**.** *For sufficiently large N in a fixed critical band, the intervals B_S, $S \in \mathcal{R}$, are nonempty and pairwise disjoint. They partition the locally deficient starts. On such a block,*

$$\mathbb{P}(J_x = 1) = \begin{cases} 0, & x = a_S + 1, \\ 2p, & x \in B_S \setminus \{a_S + 1\}, \end{cases} \qquad \sum_{x\in B_S} J_x \sim \mathrm{Bernoulli}\big(2p(|B_S| - e_S)\big). \tag{T19.2}$$

Outside these blocks the marginal is p. In particular,

$$\mathbb{E}Z - Np = p\sum_{S\in\mathcal{R}}(|B_S| - 2e_S), \tag{T19.3}$$

$$\sum_{x\in I_N} |\mathbb{P}(J_x = 1) - p| = p\sum_{S\in\mathcal{R}} |B_S|. \tag{T19.4}$$

A full block contributes $p(L - d_S - 1)$ to the mean discrepancy. There are at most two blocks truncated by the endpoints of I_N, whose total boundary correction is at most $2Lp$ in absolute value.

Proof. The condition $S \subseteq V_x$ is exactly $b_S - L + 1 \leq x \leq a_S + 1$. If two distinct such sets belonged to the same window, they would give two independent nonzero relative relations, contrary to Lemma 2.1. Thus their intervals are disjoint and every deficient window lies in exactly one. Its unique nonzero relation has affine phase $(-1)^{\mathbf{1}_{\{x-1\in S\}}}$, so Fourier inversion gives $p(1+(-1)^{\mathbf{1}_{\{x-1\in S\}}})$. Since $x - 1 \leq a_S$, the predecessor belongs to S exactly when $x = a_S + 1$. This proves the marginal formulas. The diameter of B_S is at most $L - 1$; consequently at most one of its starts is active. Its count is Bernoulli with mean the sum of the displayed marginals. Summation proves (T19.3) and (T19.4). A full interval has $L - d_S + 1$ starts. Disjoint intervals allow at most one truncation at each endpoint, and each full interval has at most L starts. □

No independence between the Bernoulli block counts is asserted. For the complete relative sign vector $Y_x \in \{\pm 1\}^L$ one also has the exact identity

$$d_{\mathrm{TV}}(\mathcal{L}Y_x, \mathrm{Unif}\{\pm 1\}^L) = 1 - 2^{-r_x}, \qquad r_x = L - \dim S_x.$$

Indeed Y_x is uniform on a binary image of size 2^{L-r_x}. Thus this local distance is either zero or $1/2$ at critical length. A nonzero S must have diameter at least $(1 - o(1)) \log N$, by the subcritical part of Lemma 2.1; its full block has length at most $(1/\log 2 - 1 + o(1)) \log N$.

T19.2 A nearby count with factorial tails at every order

Let $G = G_N$, $n = |G|$, $\mu = np$, and let E'_N be the exceptional event of Lemma T15.2. It contains the occurrence of a primitive family of size at least three, an active contact, an active bad dependent pair, two good active pairs, and an additional dependence after compressing the unique good pair. Its probability δ_N is $O_\varepsilon(N^{-4/5+\varepsilon})$. Outside it, the active original spaces are a direct sum of single windows and at most one good pair. Deleting one endpoint of that pair, if present, therefore gives a direct family.

Theorem T19.3 (A modified count with all factorial tails)**.** *On the original probability space define* $\widetilde{Z}_N = Z_N^\circ \mathbf{1}_{(E'_N)^c}$. *Then*

$$d_{\mathrm{TV}}(\mathcal{L}Z_N, \mathcal{L}\widetilde{Z}_N) \ll_\varepsilon N^{-4/5+\varepsilon}. \tag{T19.5}$$

Uniformly for every integer $k \geq 1$, *every real* $s \geq 1$, *and every integer* $r \geq 1$,

$$\mathbb{P}(Z_N^\circ \geq k) \leq \frac{\mu^{k-1}}{(k-1)!} + \delta_N, \tag{T19.6}$$

$$\mathbb{P}(\widetilde{Z}_N \geq k) \leq \frac{\mu^{k-1}}{(k-1)!}, \tag{T19.7}$$

$$\mathbb{E}s^{\widetilde{Z}_N} \leq s(1 + p(s-1))^n \leq se^{(s-1)\mu}, \tag{T19.8}$$

$$\mathbb{E}(\widetilde{Z}_N)_r \leq \mu^r + r\mu^{r-1}. \tag{T19.9}$$

Proof. Let R be the maximum cardinality of a direct active subfamily of G. Directness makes the prescribed events exactly independent. A union bound gives

$$\mathbb{P}(R \geq k) \leq \sum_{\substack{|I|=k \\ I \text{ direct}}} q(I) \leq \binom{n}{k} p^k \leq \frac{\mu^k}{k!}.$$

Choose one maximizing direct family in each realization. Its $\binom{R}{r}$ subsets of size r are all direct and active, whence

$$\mathbb{E}\binom{R}{r} \leq \binom{n}{r} p^r \quad (r \geq 0).$$

Outside E'_N we have $Z_N^\circ \leq R + 1$, and everywhere $\widetilde{Z}_N \leq R + 1$. The two tail bounds follow. Expanding $s^R = (1 + (s-1))^R$ with nonnegative coefficients gives $\mathbb{E}s^R \leq (1 + p(s-1))^n$. Finally $\binom{R+1}{r} = \binom{R}{r} + \binom{R}{r-1}$ proves the factorial moment bound. The coupling error is at most δ_N on G; Proposition 3.1 restores the deleted starts. □

These inequalities concern the modified variable. The event defining it depends on the whole active configuration, so its indicators no longer have the original affine marginals or centred factorization. In particular, (T19.5) does not bound $\mathbb{E}[s^{Z_N}; E'_N]$.

T19.3 Effective tails of the original count

A different argument gives a quantitative tail bound for Z_N itself, at a larger threshold. It tracks the moment order explicitly. Put $H = \log N$ and $h = \log H$.

Lemma T19.4 (Branching with an explicit order dependence)**.** *For any j old starts, let $\mathcal{F}_{\mathrm{old}}$ contain all their raw sign values. There is a deterministic exceptional set of new starts of size at most*

$$D_{j,N} = j(2L+1) + B\sqrt{3N}\exp\left(C\sqrt{m_{j,N}/\log(m_{j,N}+2)}\right), \qquad m_{j,N} = \left\lceil \pi(B) + \frac{jB\log(3N)}{\log B}\right\rceil,$$

outside which the L new relative signs are uniform and independent of $\mathcal{F}_{\mathrm{old}}$. In particular, uniformly for $1 \le j \le H$,

$$\log(\lambda_N + D_{j,N}) \le \frac{H}{2} + C_C\left(h + \frac{\sqrt{j}\,H}{h}\right). \tag{T19.10}$$

If $F_k = \mathbb{E}(Z_N)_k$, then

$$F_k \le C_C \exp\left\{\frac{k-2}{2}H + C_C\left(kh + \frac{k^{3/2}H}{h}\right)\right\} \quad (2 \le k \le H). \tag{T19.11}$$

Proof. Let $\mathcal{P}$ consist of the primes at most B and the primes larger than B dividing one of the old values. It has at most $m_{j,N}$ members. The number of integers at most $3N$ whose squarefree part is supported on $\mathcal{P}$ is at most

$$\sqrt{3N}\prod_{\ell\in\mathcal{P}}(1+\ell^{-1/2}).$$

For a set of m primes, the estimate $\ell_i \gg i\log(i+1)$ for its ith member implies

$$\log\prod_{\ell\in\mathcal{P}}(1+\ell^{-1/2}) \le C\sqrt{m/\log(m+2)}.$$

Delete starts whose new window overlaps an old one or contains at least two integers with squarefree part supported on $\mathcal{P}$. Counting each such integer in at most B windows gives the displayed bound for $D_{j,N}$.

If a nontrivial combination of the new relative rows belonged to the old raw-value span, it would select an even nonempty set of new values. A prime larger than B in such a value is private in the new window, so must also divide an old value. Every selected value would consequently have squarefree part supported on $\mathcal{P}$. At least two are selected, contradicting the deletion rule. Thus the rank increases by exactly L even after conditioning on all old raw values, which proves independence.

For $j \le H$, $m_{j,N} = O_C(jH^2/h)$ and $\log(m_{j,N}+2) \gg h$. This proves (T19.10). Summing the conditional probabilities of the next start, using the bound one on the exceptional set, yields the finite recurrence $F_{j+1} \le (\lambda_N + D_{j,N})F_j$. The pair estimate gives $F_2 = O_C(1)$. Multiplying from $j = 2$ to $k-1$ and summing $\sqrt{j}$ proves (T19.11). □

Proposition T19.5 (A quantified effective support)**.** *For each fixed $D > 0$ and integer $r \ge 0$, there is a constant $K = K(C, D, r)$ such that*

$$\mathbb{E}[Z_N^r; Z_N > A_N] \le N^{-D}, \qquad A_N = \left\lceil \sqrt{N}\exp\left(\frac{K\log N}{(\log\log N)^{2/3}}\right)\right\rceil \tag{T19.12}$$

for sufficiently large N. Hence, for every fixed $\eta, D > 0$ and integer $r \ge 0$, $\mathbb{E}[Z_N^r; Z_N > N^{1/2+\eta}] = O(N^{-D})$.

Proof. Take $k = \lceil h^{2/3} \rceil$. Eventually $k > r$ and $A_N \geq 2k$. On $Z_N > A_N$, $Z_N^r \leq 2^k A_N^{r-k} (Z_N)_k$. The logarithm of the resulting bound from (T19.11) is at most

$$\frac{r-2}{2} H + C_C \left(kh + \frac{k^{3/2} H}{h} \right) - (k-r) \frac{KH}{h^{2/3}} + k \log 2 + O_C(1).$$

Here $kh = o(H)$, $k^{3/2}/h = 1 + o(1)$ and $(k-r)/h^{2/3} = 1 + o(1)$. A sufficiently large fixed K makes this at most $-DH$. The last assertion follows by comparison of thresholds. □

The all-fixed-order bounds of Theorem 6.4 also imply $\mathbb{E}[Z_N^r ; Z_N > N^\eta] = O(N^{-D})$ for fixed η, D, r, by choosing one sufficiently large fixed moment. The value of (T19.12) is its explicit control of a growing order, without extrapolating fixed-order constants.

A larger threshold admits a much smaller exceptional probability. The distinction is useful when the moment order itself increases as fast as $\log N$.

Proposition T19.6 (A very small tail beyond a near-linear threshold)**.** *Let $S = \sqrt{\log N \log\log N}$ and fix $0 < c < \sqrt{13}/4$. For $A_{N,c} = \lceil R_N e^{-cS} \rceil$, one has, for large N,*

$$\mathbb{P}(Z_N > A_{N,c}) \leq e^{-(\log N)^3}, \qquad \mathbb{E}[Z_N^r ; Z_N > A_{N,c}] \leq e^{-(\log N)^3/2} \quad (1 \leq r \leq \log N). \tag{T19.13}$$

Using only the published estimate of de la Bretèche–Drappeau, the same conclusion holds for $c < 2/\sqrt{5}$.

Proof. Write $\kappa(n) = P^+(\mathrm{sf}(n))$, with $P^+(1) = 1$. We first give the analytic input and its transfer to these odd prime pivots. Pascadi's bound [20, Corollary 7.2], with the forms $n, n+h$, implies, for fixed $\eta > 0$ and $\gamma = 13/8 - \eta$,

$$\sum_{1 \leq h \leq B} \#\{n \in [N-B, 2N+B] : \kappa(n), \kappa(n+h) \leq e^{aS}\} \leq N \left\{ e^{-(\gamma/(2a)+o(1))S} + e^{-(a+1/(2a)+o(1))S} \right\}, \tag{T19.14}$$

uniformly for a in a fixed compact subinterval of $(0, \infty)$. Here the parameter in the cited result is fixed before N: its smoothness range contains $y = e^{aS}$ and its coefficient range contains every $h \leq B = O(\log N)$. For ordinary y-smooth pairs the estimate follows from $\Psi(X, y) = X \varrho(\log X / \log y) e^{O(\log X / \log y)}$ and $\log \varrho(u) = -(1 + o(1)) u \log u$.

The difference between ordinary smoothness and $\kappa(n) \leq y$ must also be counted. Such a difference requires $n = \ell^2 m$ for a prime $\ell > y$ and $\kappa(m) \leq y$. For $s > 1/2$, positive summation and the exact Dirichlet series give

$$\#\{n \leq 4N : \kappa(n) \leq y < P^+(n)\} \leq (4N)^s \zeta(2s) \prod_{\ell \leq y} (1 + \ell^{-s}) \sum_{\ell > y} \ell^{-2s}.$$

Put $w = \log y = aS$, $u = \log N / w$ and $s = 1 - (\log u)/w$. Chebyshev's prime bound and partial summation show that the logarithm of the Euler product is $O(\log w + u/\log u) = o(S)$. Extending the last sum to all integers gives $O(y^{1-2s}/(2s-1))$. The resulting upper bound is $N e^{-(a+1/(2a)+o(1))S}$. Each exceptional integer lies in at most $2B$ relevant pairs, proving (T19.14). Theorem 4.1 of [8] gives the same argument with $\gamma = 8/5 - \eta$.

Choose $\eta > 0$ so small that $a = \sqrt{\gamma/2} > c$, and set $Y = e^{aS}$. Delete windows with at least two pivots at most Y. By (T19.14), their number is $N e^{-(a+o(1))S}$; the second term there is strictly smaller. Conditional on all prime signs at most Y, each remaining window has marginal p: retain its possible defective value as anchor and use a private prime greater than Y for every other value. Join two remaining windows when they share a prime greater than Y with odd valuation. This is an exact conditional dependency graph. A greedy coloring uses at most

$$\chi = \left\lceil 1 + \frac{B^2 \log(3N)}{\log Y} (N/Y + 2) \right\rceil$$

colors. Conditional independence in each color class and Hölder's inequality imply

$$\mathbb{E}[e^{tZ_G} \mid \mathcal{F}_Y] \le \exp\{\lambda_N(e^{\chi t}-1)/\chi\}, \qquad t \ge 0.$$

Chernoff's bound, with $\chi t = \log(z/\lambda_N)$, is at most $\exp\{-(z/\chi)(\log(z/\lambda_N)-1)\}$. Take $z = 2\chi(\log N)^2$. Since $\log\chi = \log N - aS + O(\log\log N)$, this probability is at most $e^{-(\log N)^3}$. The deleted count plus z is $o(R_N e^{-cS})$, proving the first assertion. Finally $Z_N \le R_N \le N$ proves the moment assertion uniformly for $r \le \log N$. □

T19.4 An arithmetic cylinder with large exponential moments

Proposition T19.7 (Divergence of the real exponential moment)**.** *Along odd primes $q \to \infty$, put $N_q = 2^{q-1}$ and $L_q = q-1$. For every fixed $t > 2$,*

$$\liminf_{q\to\infty} \frac{\log N_q}{N_q} \log \mathbb{E}e^{tZ_{N_q,L_q}} \ge (t-2)\log 2 > 0. \tag{T19.15}$$

The same bound holds for the count of positive runs alone.

Proof. Let $M = 2N_q + q$ and prescribe the prime-sign cylinder

$$E_q = \{f(q) = -1,\ f(\ell) = 1 \text{ for every prime } \ell \le M,\ \ell \ne q\}.$$

It has probability $2^{-\pi(M)}$. On this event, $f(n) = (-1)^{v_q(n)}$ for all integers used by the windows. Every start $x = uq+1 \in I_{N_q}$ with $q \nmid u(u+1)$ has a negative predecessor, then exactly $q-1$ positive values, and a negative next value. Their number is

$$K_q = (1-2/q)N_q/q + O(1) = (1-o(1))N_q/q.$$

Therefore $\log \mathbb{E}e^{tZ} \ge tK_q - \pi(2N_q+q)\log 2$. Since $q = \log N_q/\log 2 + 1$, the prime number theorem gives (T19.15). □

The number 2 is a sufficient threshold in this construction; its optimality is not asserted. For every fixed r, the same cylinder contributes at most $R_{N_q}^r 2^{-\pi(M)} = o(1)$ to the rth moment. It is thus compatible with all fixed-moment estimates and with the total-variation approximation.

The cylinder also produces connected families that lie far beyond the bounded-nullity boundary used in the proof. The next elementary rank argument extracts one such family with a positive centred coefficient.

Lemma T19.8 (Deleting large private ranks)**.** *Let A be a compatible family of m affine events, each with constraint-space dimension L, and put $R = \dim\sum_{x\in A} S_x$ and $p = 2^{-L}$. For an integer $1 \le t \le L$, repeatedly remove a member whose current private rank is at least t. The final family A' satisfies*

$$|A'| \ge m - R/t, \qquad \dim\sum_{x\in A'} S_x \le R, \qquad r_x(A') < t \quad (x \in A'),$$

where $r_x(I) = \dim\sum_{y\in I} S_y - \dim\sum_{y\in I\setminus\{x\}} S_y$. Each direct component C of A' satisfies

$$c(C) \ge (1-|C|p2^t)q(C) \ge (1-mp2^t)q(C). \tag{T19.16}$$

If $m - R/t > 0$ and $D = R/(m-R/t)$, some such component obeys

$$\frac{\dim\sum_{x\in C} S_x}{|C|} \le D, \qquad |C| \ge L/D, \qquad \rho(C) \ge L^2/D - L. \tag{T19.17}$$

Proof. Each deletion reduces the current rank by at least t, so at most R/t deletions occur. For a compatible family, all Fourier phases of its relations are positive. The centred Fourier formula is

$$c(C) = p^{|C|}\#\{(s_x) \in \ker(\bigoplus_{x\in C} S_x \longrightarrow \sum_{x\in C} S_x) : s_x \neq 0 \ (x \in C)\}.$$

The image of the projection of this kernel on S_x has dimension $L - r_x(C) \geq L - t + 1$. In the uniform kernel, the probability that $s_x = 0$ is consequently at most $p2^t$. A union bound and $q(C) = p^{|C|}2^{\rho(C)}$ prove (T19.16). The size-weighted average of the rank-to-size ratios of the components is at most D. Choose a component no larger than that average. Its rank is at least L, proving $|C| \geq L/D$. If $D < L$, use $\rho(C) \geq |C|(L - D)$; if $D \geq L$, the stated lower bound for nullity is nonpositive. □

Proposition T19.9 (A real component of quadratic nullity)**.** *Along the subsequence of Proposition T19.7, there are compatible connected families $C_q \subset G_{N_q}$ containing no good pair and satisfying*

$$\begin{gathered} |C_q| \geq (\kappa - o(1))L_q, \qquad \rho_*(C_q) = \rho(C_q) \geq (\kappa - o(1))L_q^2, \\ c(C_q) \geq (1 - O(L_q^{-1/2}))q(C_q) > 0, \qquad \kappa = \frac{6\log 2}{17}. \end{gathered} \tag{T19.18}$$

Proof. Restrict the starts in the cylinder to

$$T_q = \{x = uq + 1 : 3N_q/2 + 4q \leq x < 2N_q, \quad q \nmid u(u+1)\}, \qquad A_q = T_q \cap G_{N_q}.$$

The regular-set estimate gives $m = |A_q| \sim N_q/(2q)$; the cylinder witnesses compatibility. For a good edge $x < y$ in channel $a : b$, its exact unit gives $y = (b/a)x + (b/a)i - j$, with $-1 \leq i, j \leq L_q - 1$ and $b/a \in \{4/3, 3/2\}$. The lower bound $y \geq (b/a)x - (1 + b/a)q$ exceeds $2N_q$ in this interval. Thus A_q has no good edge.

All prime coordinates used by these windows belong to the union

$$[1, 2N_q/3 + O(q)], \quad [3N_q/4 - O(q), N_q + O(q)], \quad [3N_q/2 - O(q), 2N_q + O(q)].$$

Indeed the multiplier of a prime in a used value is respectively at least three, two, or one. The prime number theorem bounds their number by $(17/12 + o(1))N_q/\log N_q$. Hence the total rank R satisfies $R/m \leq 17/(6\log 2) + o(1)$. Apply Lemma T19.8 with $t = \lfloor \frac{1}{2}\log_2 L_q \rfloor$. Then $R/t = o(m)$, $D \leq 17/(6\log 2) + o(1)$ and $mp2^t = O(L_q^{-1/2})$. Equations (T19.16)–(T19.17) give the required component. Every subfamily is compatible, and nullity is nondecreasing under inclusion, so its compatible nullity equals its full nullity. □

This is a statement about which arithmetic terms exist. The probability of the witnessing cylinder is not identified with $q(C_q)$, and the proposition gives no lower bound for the total variation error or for a sum of absolute component activities.

T19.5 Keeping positions while justifying pair deletion

Deleting an active pair is a transformation of the whole field. The scalar approximation alone does not justify this operation. We retain coordinates in the same finite expansion, using the boundary and deficit estimates already proved. Lemma 20.1 compares the unsigned field $(J_x)_{x\in I_N}$ with independent Poisson single arrivals and arrivals along good arithmetic pairs, with an $O_{C,j,\varepsilon}(N^{-4/5+\varepsilon})$ error for every fixed polynomial weight of degree j in the total count. The field retains integer positions, without run signs or exact excess lengths; the target is on the same grid, not a diffuse spatial process. We first work on the regular set $G = G_N$ and then restore the excluded starts.

On $\mathbb{N}^G$, let $\boldsymbol{e}_x$ denote the coordinate unit vector and set

$$\Delta_x = \delta_{\boldsymbol{e}_x} - \delta_{\mathbf{0}}, \qquad b_x = \delta_{\mathbf{0}} + p\Delta_x, \qquad \Delta_I = \mathop{*}_{x\in I} \Delta_x.$$

For a hereditary class $\mathscr{D}$, the same partition by first excluded prefix gives the exact identity

$$\mathcal{L}((J_x)_{x\in G}) - \sum_{I\in\mathscr{D}} q(I)\Delta_I = \sum_{I\in\partial\mathscr{D}} \Delta_I * \boldsymbol{\nu}_I, \quad \boldsymbol{\nu}_I = \mathbb{E}\left[J_I \delta_{\sum_{y>\max I} J_y \boldsymbol{e}_y}\right]. \tag{T19.19}$$

One can obtain it directly by expanding $\prod_x(1+u_x)$ with $u_x = J_x\Delta_x$ in the convolution algebra. The proof of Lemma 5.1 permits a different algebra element at every site. Each $\boldsymbol{\nu}_I$ is positive of mass $q(I)$, and $\|\Delta_I\|_1 = 2^{|I|}$. Thus the boundary estimate is unchanged.

Use the field laws H_N and U_N and the intensities q_e, ι_x, ν_x defined in Section 20. In particular, $\beta = \sum_e q_e$, $\theta = 2\beta$, $\lambda = Np$, and $\iota_x = 0$ outside G. The total-count laws are given by (20.2).

Proof of Lemma 20.1. First work on G in the unweighted norm. The retained reference is now

$$\mathsf{B}_\beta = \mathop{*}_{x\in G} b_x + \sum_{e=\{x,y\}\in E_N} q_e \Delta_x * \Delta_y * \mathop{*}_{z\in G\setminus e} b_z.$$

The exact retained identity subtracts the missing direct families, with coefficient $p^{|I|}$, and the missing compressed extensions, with coefficient $q_e p^{|U|}$. Taking norms of these subtractions costs at most $F + \mathfrak{G}$, with exactly the deficits of (T17.3). There is no scalar cancellation in that estimate. The vector version of the first-boundary remainder is bounded by the same positive sum as (5.5). Hence

$$\|\mathcal{L}((J_x)_{x\in G}) - \mathsf{B}_\beta\|_1 \ll_\varepsilon N^{-4/5+\varepsilon}.$$

Put $\mathsf{P} = \exp_*\{p\sum_{x\in G}\Delta_x\}$. Replacing the product of Bernoulli laws by P costs $O(np^2)$. In each edge term, replacing the product outside e by P costs $O(np^2+p)$ before multiplication by $4q_e$. Thus

$$\mathsf{B}_\beta = \mathsf{P} * \left(\delta_{\mathbf{0}} + \sum_{e=\{x,y\}} q_e \Delta_x * \Delta_y\right) + O_{\ell^1}(np^2 + \beta np^2 + \beta p).$$

The exponential remainder has norm at most $e^{4\beta} - 1 - 4\beta = O(\beta^2)$. Moreover

$$p\sum_x \Delta_x + \sum_{e=\{x,y\}} q_e \Delta_x * \Delta_y = \sum_x (p-\iota_x)\Delta_x + \sum_{e=\{x,y\}} q_e(\delta_{\boldsymbol{e}_x+\boldsymbol{e}_y} - \delta_{\mathbf{0}}),$$

so this convolution exponential is exactly the positive law H_N on G. The errors np^2 and β^2 are $O_\varepsilon(N^{-1+\varepsilon})$. Restoring the deleted starts in both laws costs $N^{-1+o(1)}$.

It remains to justify the weights; total variation alone is insufficient. Both total counts have every fixed moment uniformly bounded: for the real count this is Theorem 6.4, and for the compound target it follows from (20.2) with bounded parameters. For their signed difference σ and any fixed integer $s > j$, Hölder's inequality gives

$$\sum (1+|\boldsymbol{z}|)^j |\sigma\{\boldsymbol{z}\}| \le \|\sigma\|_1^{1-j/s} \left(\sum (1+|\boldsymbol{z}|)^s |\sigma\{\boldsymbol{z}\}|\right)^{j/s}.$$

Choose s sufficiently large after ε, and use the unweighted estimate with a smaller error margin. This proves (20.3) for each fixed j. □

The weights in this lemma have fixed polynomial degree. They are not exponential weights and do not contradict Proposition T19.7.

Let $\Xi = \sum_{e\in E_N} J_e$, $C_N = \{\Xi \ge 1\}$ and $r_N = \mathbb{P}(C_N)$. On C_N, choose one active good edge uniformly and delete its two endpoints, obtaining a binary field $J^\flat$. Off C_N keep the field

unchanged. Also let $\widehat{J}$ be the field obtained by deleting every vertex incident to an active good edge. Their counts are

$$W = \sum_x J^\flat_x = Z - 2\mathbf{1}_{C_N}, \qquad \widehat{W} = \sum_x \widehat{J}_x.$$

The auxiliary random choice affects the field $J^\flat$ but not W.

Proposition T19.10 (Deletion, isolation, and remaining covariances)**.** *Uniformly in the critical band, for every $\varepsilon > 0$,*

$$d_{\mathrm{TV}}(\mathcal{L}J^\flat, \mathsf{U}_N) + d_{\mathrm{TV}}(\mathcal{L}\widehat{J}, \mathsf{U}_N) \ll_\varepsilon N^{-4/5+\varepsilon}, \tag{T19.20}$$

$$\sum_x |\mathbb{E}J^\flat_x - (p - \iota_x)| + \sum_x |\mathbb{E}\widehat{J}_x - (p - \iota_x)| \ll_\varepsilon N^{-1+\varepsilon}. \tag{T19.21}$$

For a counting field X put $K_{xy}(X) = \mathrm{Cov}(X_x, X_y) - \mathbf{1}_{\{x=y\}}\mathbb{E}X_x$. Then

$$\sum_{x,y} |K_{xy}(J^\flat)| + \sum_{x,y} |K_{xy}(\widehat{J})| \ll_\varepsilon N^{-4/5+\varepsilon}. \tag{T19.22}$$

In particular, the absolute second factorial cumulants of W and $\widehat{W}$ have this bound. On the natural coupling,

$$0 \le \beta - r_N \le \tfrac{1}{2}\mathbb{E}(\Xi)_2 = O(a_N^2), \tag{T19.23}$$

$$d_{\mathrm{TV}}(\mathcal{L}W, \mathcal{L}\widehat{W}) \le \tfrac{1}{2}\mathbb{E}(\Xi)_2 = O(a_N^2), \tag{T19.24}$$

$$W_1(\mathcal{L}W, \mathcal{L}\widehat{W}) \le \mathbb{E}(\Xi)_2 = O(a_N^2). \tag{T19.25}$$

Here $a_N \asymp LN^{-2/3}$ as in the main theorem.

Proof. Extend the one-edge deletion kernel to $\mathbb{N}^{I_N}$ by choosing uniformly an edge whose two endpoint counts are positive and subtracting one at each endpoint; if no such edge exists, leave the configuration unchanged. It agrees with the binary deletion kernel. Apply it to the real and compound fields. The unweighted comparison error cannot increase. In the target representation, $T = \sum_e V_e \sim \mathrm{Pois}(\beta)$. If $T = 0$, deletion leaves U unchanged unless two occupied single sites form an edge. This event has probability at most $|E_N|p^2 = O(B/N)$. If $T = 1$, let e be its edge. Deleting e recovers U exactly. A different edge can be active only if its endpoints are occupied singles, or if it shares one endpoint with e and its other endpoint is a single. The corresponding bounds are $|E_N|p^2$ and $O(Bp)$, uniformly in e. Finally $\mathbb{P}(T \ge 2) = O(\beta^2)$. This couples the once-deleted target to U with error $O(B/N + \beta^2)$. For isolation, set all coordinates incident to an active edge to zero; the same argument also excludes singles at the two endpoints of e, at cost $2p$ when $T = 1$. This proves (T19.20).

Every deleted count is bounded by its original total count. The weighted argument at the end of Lemma 20.1 therefore applies also to the two deleted laws and U_N. With weight $(1 + |\boldsymbol{z}|)^2$ it controls, in the sum of absolute entries, their first and second factorial moment measures. The total intensities are bounded, so it controls the tensor squares of their means as well. The independent Poisson field has zero factorial covariance matrix. This proves (T19.22).

For the finer intensity assertion, let $B = J - J^\flat$. Nonnegativity gives the exact identity

$$\sum_x (\iota_x - \mathbb{E}B_x) = 2(\beta - r_N), \qquad 0 \le \mathbb{E}B_x \le \iota_x.$$

For isolation, put $D_x = \sum_{y:\{x,y\}\in E_N} J_y$. Then

$$\mathbb{E}\widehat{J}_x = \mathbb{E}J_x - \iota_x + \mathbb{E}[J_x(D_x - 1)_+], \qquad \sum_x \mathbb{E}[J_x(D_x - 1)_+] \le 2(\beta - r_N).$$

The last inequality holds realization by realization: if there are active edges, at least two vertices are incident to them, whereas the sum of their active degrees is 2Ξ. Now use $\sum_x |\mathbb{E}J_x - p| = N^{-1+o(1)}$ and the positive two-edge bound in Lemma T1.17.

Finally $0 \le k - \mathbf{1}_{\{k\ge 1\}} \le (k)_2/2$ for integers $k \ge 0$. Also $W = \widehat{W}$ when $\Xi \le 1$, and

$$0 \le W - \widehat{W} \le 2(\Xi - \mathbf{1}_{C_N}) \le (\Xi)_2.$$

These inequalities prove (T19.23)–(T19.25). □

The same covariance conclusion applies to the positive cubic count used to approximate isolation. With the active degrees D_x from the proof, put

$$Y_x^{[3]} = J_x\left(1 - D_x + \binom{D_x}{2}\right), \qquad Y^{[3]} = \sum_x Y_x^{[3]}, \qquad \Sigma = \sum_x J_x\binom{D_x}{2}.$$

For an integer $d \ge 1$, $1 - d + \binom{d}{2} = \binom{d-1}{2} \ge 0$, whereas its value at zero is one. Thus $0 \le \widehat{J}_x \le Y_x^{[3]}$, and the fields agree when $\Xi \le 1$. More precisely, if $O = \sum_x J_x(D_x - 1)_+$, then

$$Y^{[3]} - \widehat{W} = \Sigma - O \ge 0, \qquad \Sigma \le \tfrac{1}{2}(\Xi)_2, \qquad \mathbb{E}Y^{[3]} = m - 2\beta + \mathbb{E}\Sigma.$$

The inequality for Σ counts pairs of distinct good edges sharing a distinguished endpoint; in a simple graph each unordered pair can share at most one endpoint. It follows that both the total-variation and the W_1 distances from $Y^{[3]}$ to $\widehat{W}$ are $O(a_N^2)$. Since the deterministic degree is $O(B)$, $Y^{[3]} \ll B^2 Z$. The fixed moments of Z, Hölder's inequality on $\{\Xi \ge 2\}$, and Lemma T1.17 give, for every fixed polynomial weight, a difference $O_\varepsilon(N^{-4/3+\varepsilon})$ between the two laws. In particular,

$$|\operatorname{Var}(Y^{[3]}) - \mathbb{E}Y^{[3]}| \ll_\varepsilon N^{-4/5+\varepsilon}. \tag{T19.26}$$

This deduction needs no asymptotic for the rarer configurations where the cubic and isolated counts differ.

Let $P_0 = \mathcal{L}(Z \mid C_N^c)$, $\mu_0 = \mathbb{E}[Z \mid C_N^c]$, and $m = \mathbb{E}Z$. The preceding proof also gives, for every fixed polynomial weight, a comparison of P_0 with $\operatorname{Pois}(\lambda - \theta)$ at order $N^{-4/5+\varepsilon}$. Indeed absence of an observed edge in the compound field forces $T = 0$; the remaining conditional law is U conditioned on having no accidental edge. Removing this conditioning costs $O(B/N)$, and $\mathbb{P}(C_N^c) = 1 - o(1)$ permits conditioning the comparison in Lemma 20.1. Consequently

$$2r_N = \theta + O(a_N^2), \qquad \mathbb{E}W = m - 2r_N = \lambda - \theta + O_\varepsilon(N^{-1+\varepsilon}), \qquad \mu_0 = \lambda - \theta + O_\varepsilon(N^{-4/5+\varepsilon}). \tag{T19.27}$$

The identity for the mean of W is exact before its last estimate. One may replace θ by the two-channel expression of Theorem 3.3, or by the actual second factorial cumulant, at cost $O_\varepsilon(N^{-4/5+\varepsilon})$.

For clarity, the corresponding localization statements are

$$\big|d_{\mathrm{TV}}(\mathcal{L}Z, H_{\lambda,\theta}) - (1 - r_N)d_{\mathrm{TV}}(P_0, \operatorname{Pois}(\mu_0))\big| \ll_\varepsilon N^{-4/5+\varepsilon}, \tag{T19.28}$$

$$\big|d_{\mathrm{TV}}(\mathcal{L}Z, H_{\lambda,\theta}) - d_{\mathrm{TV}}(\mathcal{L}W, \operatorname{Pois}(\mathbb{E}W))\big| \ll_\varepsilon N^{-4/5+\varepsilon}, \tag{T19.29}$$

$$\big|d_{\mathrm{TV}}(\mathcal{L}((J_x)_x), \mathsf{H}_N) - d_{\mathrm{TV}}(\mathcal{L}J^\flat, \mathsf{U}_N)\big| \ll_\varepsilon N^{-4/5+\varepsilon}. \tag{T19.30}$$

They follow by the triangle inequality from the comparisons just proved; the changes of Poisson means cost at most their absolute differences. Thus the full stopped expansion supplies, in particular, the first-order reduction to the background. Small second cumulants alone would not supply any of these distributional comparisons.

T19.6 Signed Palm identities and weighted recurrence errors

It is useful to state precisely which information a scalar Stein comparison retains. For a probability P on $\mathbb{N}$ with finite first moment, put $P_k = P\{k\}$ and $P_k = 0$ for $k < 0$, and define

$$r_k(P;\mu,\theta) = kP_k - (\mu-\theta)P_{k-1} - \theta P_{k-2}, \qquad \mathcal{R}(P;\mu,\theta) = \sum_{k\ge 0} |r_k(P;\mu,\theta)|. \tag{T19.31}$$

Lemma T19.11 (A scalar recurrence criterion)**.** *If $0 \le \theta < \min(\mu, 1/2)$, then*

$$d_{\mathrm{TV}}(P, H_{\mu,\theta}) \le \frac{\mathcal{R}(P;\mu,\theta)}{1-2\theta}. \tag{T19.32}$$

If $m = \mathbb{E}X > 0$, $\kappa = \operatorname{Var} X - \mathbb{E}X$ and $0 \le \kappa < \min(m, 1/2)$, the size-biased variable X^s satisfies the exact identity

$$\mathcal{R}(\mathcal{L}X; m, \kappa) = 2m d_{\mathrm{TV}}\left(\mathcal{L}X^s, (1-\kappa/m)\mathcal{L}(X+1) + (\kappa/m)\mathcal{L}(X+2)\right). \tag{T19.33}$$

Proof. For an indicator test h, a Poisson Stein solution satisfies $\mu f(k+1) - kf(k) = h(k) - \Pi_\mu h$ and may be chosen with $\|f\|_\infty \le 1$, $\|\Delta f\|_\infty \le 1$. One proof uses the immigration–death process at time t, whose law from k is $\mathrm{Bin}(k, e^{-t}) + \mathrm{Pois}(\mu(1-e^{-t}))$. One extra initial individual survives with probability e^{-t}, and two survive with probability e^{-2t}. Integrating these couplings in the Poisson equation gives respectively the bounds $\int_0^\infty e^{-t}dt = 1$ and $\int_0^\infty 2e^{-2t}dt = 1$; set $f(0) = f(1)$.

The operator $A_\theta f(k) = \mu f(k+1) + \theta\Delta f(k+1) - kf(k)$ has mean zero under $H_{\mu,\theta}$, by applying the elementary Poisson identity separately to its single and double counts. Changing indices gives $PA_\theta f = -\sum_k r_k f(k)$. The difference of the expectations of $\Delta f(k+1)$ under P and $H_{\mu,\theta}$ is at most $2d_{\mathrm{TV}}(P, H_{\mu,\theta})$. Taking the supremum over h and absorbing this perturbation proves (T19.32). The two measures in (T19.33) differ at k by $r_k(\mathcal{L}X; m, \kappa)/m$; total variation is half their ℓ^1 distance. □

The constant in (T19.32) is not claimed optimal. The criterion is sufficient, and the raw norm is not bounded by total variation alone. Parameter changes satisfy the elementary bound

$$|\mathcal{R}(P;\mu,\theta) - \mathcal{R}(P;\mu',\theta')| \le |\mu-\mu'| + 2|\theta-\theta'|. \tag{T19.34}$$

When the first three moments exist and $\mu = m$, $\theta = \kappa$, the signed residual has $\sum_k r_k = \sum_k k r_k = 0$ and $\sum_k k^2 r_k = \kappa_3^{(F)}(X)$. The last identity tests it by an unbounded function and is not an estimate for its norm.

Proposition T19.12 (The signed Palm measure)**.** *For any finite family of indicators, let $Z = \sum_x J_x$, $p_x = \mathbb{E}J_x$, $m = \mathbb{E}Z$, $P = \mathcal{L}Z$, and $T_x = Z - J_x$. Omit sources with $p_x = 0$, and define*

$$\mathcal{D} = \sum_x p_x\{\mathcal{L}(T_x \mid J_x = 1) - \mathcal{L}(T_x)\}.$$

Then

$$\left|\mathcal{R}(P; m, \theta) - \|\mathcal{D} - \theta\Delta * P\|_1\right| \le 2\sum_x p_x^2. \tag{T19.35}$$

Writing $q_A = \mathbb{E}\prod_{x\in A} J_x$, the coefficient of $(z-1)^j$ in its generating polynomial is exactly

$$d_j = \sum_{|A|=j}\sum_{x\notin A}(q_{A\cup\{x\}} - p_x q_A). \tag{T19.36}$$

If all n marginals equal p and $B_j = \mathbb{E}\binom{Z}{j}$, this becomes $d_j = (j+1)B_{j+1} - p(n-j)B_j$.

Proof. Let SP translate a measure by one and let $(KP)_k = kP_k$. The size-bias identity is $KP = S\sum_x p_x \mathcal{L}(T_x \mid J_x = 1)$. Consequently the residual in (T19.31) is the translate of

$$\mathcal{D} + \mathcal{B} - \theta\Delta * P, \qquad \mathcal{B} = \sum_x p_x\{\mathcal{L}(T_x) - P\}.$$

In the natural coupling, T_x and Z differ only when $J_x = 1$, so $\|\mathcal{B}\|_1 \le 2\sum_x p_x^2$. Translation preserves ℓ^1 norm, and the reverse triangle inequality proves (T19.35). Expanding $\prod_{y\neq x}(1 + (z-1)J_y)$ before summing over x gives (T19.36). □

The order of summation matters: both the sum over intersections of a fixed size and the sum over sources occur before the norm. In this model $p_x \le 2p$ for large N, so the diagonal error is $O(N^{-1})$. Taking absolute values separately for each forced coordinate would remove cancellations present in $\mathcal{D}$.

For reference, two sufficient uncorrected criteria follow without any affine assumption. If

$$v_x = d_{\mathrm{TV}}(\mathcal{L}T_x, \mathcal{L}(T_x \mid J_x = 1)), \qquad w_x = W_1(\mathcal{L}T_x, \mathcal{L}(T_x \mid J_x = 1)),$$

then

$$\mathcal{R}(P; m, 0) \le 2\sum_x p_x(p_x + v_x), \qquad d_{\mathrm{TV}}(P, \mathrm{Pois}(m)) \le \sum_x p_x(p_x + w_x). \tag{T19.37}$$

The first follows from the preceding proposition. For the second, couple the two scalar T_x laws optimally and lift that coupling to the original joint law using conditional distributions given T_x. The extra difference from Z to T_x has mean p_x. The Poisson Stein equation with $\|\Delta f\|_\infty \le 1$ gives the displayed bound.

The polynomially weighted comparison already proved also makes the scalar recurrence comparisons legitimate after deletion. In particular, with $\mathcal{R}_0(P;\mu) = \mathcal{R}(P;\mu,0)$,

$$\left|\mathcal{R}(\mathcal{L}Z;\lambda,\theta) - (1 - r_N)\mathcal{R}_0(P_0;\mu_0)\right| \ll_\varepsilon N^{-4/5+\varepsilon}, \tag{T19.38}$$

$$\left|\mathcal{R}(\mathcal{L}Z;\lambda,\theta) - \mathcal{R}_0(\mathcal{L}W;\mathbb{E}W)\right| \ll_\varepsilon N^{-4/5+\varepsilon}. \tag{T19.39}$$

Indeed the operator in (T19.31) is bounded from the norm $\sum_k (1+k)|\sigma_k|$ to ℓ^1, uniformly for the present parameters; its value on the compound target is zero. The analogous statement holds for the Poisson background. Lemma 20.1, the deletion comparison, and (T19.34) prove both formulas. This argument uses fixed moments as well as the distributional approximation.

T19.7 Removing the remaining active pair dependencies

The recurrence norm weighted by $1/k$ has a useful stability property under small changes of a law. For any probability P on $\mathbb{N}$ and $\mu > 0$ define

$$\mathcal{V}(P;\mu) = \sum_{k\ge 1} \frac{|kP_k - \mu P_{k-1}|}{k}.$$

It is always finite, even without a first-moment assumption.

Lemma T19.13 (A recurrence norm equivalent to total variation)**.** *For every P and $\mu > 0$,*

$$e^{-\mu} d_{\mathrm{TV}}(P, \Pi_\mu) \le \mathcal{V}(P;\mu) \le 2(1+\mu) d_{\mathrm{TV}}(P, \Pi_\mu). \tag{T19.40}$$

Moreover, if $0 < \mu, \nu \le M$, then

$$|\mathcal{V}(P;\mu) - \mathcal{V}(Q;\nu)| \le 2(1+M) d_{\mathrm{TV}}(P,Q) + |\mu - \nu|. \tag{T19.41}$$

Proof. Write $a_k = P_k - \Pi_\mu\{k\}$ and $b_k = a_k - (\mu/k)a_{k-1}$ for $k \geq 1$. The upper bound follows by summing absolute values. The forward recurrence gives

$$a_k = \frac{\mu^k}{k!}a_0 + \sum_{j=1}^{k} \frac{\mu^{k-j} j!}{k!} b_j.$$

The sum over $k \geq j$ of the coefficient of b_j is at most e^μ. Since $\sum_k a_k = 0$, it follows first that $|a_0| \leq \sum_j |b_j|$ and then that $\sum_k |a_k| \leq 2e^\mu \sum_j |b_j|$. All series are absolutely convergent by the same coefficient bound. This proves the lower estimate. Applying the upper operator bound to $P - Q$ and bounding $\sum_{k\geq 1} Q_{k-1}/k \leq 1$ proves stability. □

Let E_N^{all} be the graph on G containing contacts at distance exactly L and all separated pairs with nonzero joint relative kernel. It contains E_N. Define

$$\Xi^{\mathrm{bad}} = \sum_{e \in E_N^{\mathrm{all}} \setminus E_N} J_e, \qquad \Lambda^{\mathrm{loc}} = \sum_{x \notin G} J_x, \qquad \widetilde{J}_x = J_x \mathbf{1}_{\{x\in G\}} \mathbf{1}_{\{\sum_{y:\{x,y\}\in E_N^{\mathrm{all}}} J_y = 0\}}, \qquad \widetilde{W} = \sum_x \widetilde{J}_x.$$

There is no degree bound imposed on E_N^{all}.

Proposition T19.14 (All active pair dependencies can be removed)**.** *For every $\varepsilon > 0$,*

$$\tau_N := \mathbb{E}\Xi^{\mathrm{bad}} + \mathbb{E}\Lambda^{\mathrm{loc}} \ll_\varepsilon N^{-4/5+\varepsilon}, \tag{T19.42}$$

$$d_{\mathrm{TV}}(\mathcal{L}\widetilde{W}, \mathcal{L}\widehat{W}) \leq \tau_N, \qquad 0 \leq \mathbb{E}\widehat{W} - \mathbb{E}\widetilde{W} \leq 2\tau_N. \tag{T19.43}$$

In every realization, distinct retained original spaces are pairwise direct and their starts are strictly separated. The values of $\mathcal{V}$ at the respective actual means differ by $O_\varepsilon(N^{-4/5+\varepsilon})$. Thus, for a fixed $0 < \varepsilon < 2/15$, the condition

$$\mathcal{V}(\mathcal{L}\widetilde{W}; \mathbb{E}\widetilde{W}) = o(a_N)$$

is equivalent to the Poisson approximation of the erased background, and hence to $d_{\mathrm{TV}}(\mathcal{L}Z, H_{\lambda,\theta}) = o(a_N)$.

Proof. A compatible separated dependent pair in G has $q_e = p^2 2^{\rho(e)} \geq 2p^2$, whence $q_e \leq 2|q_e - p^2|$. The absolute bad-pair estimate therefore bounds their positive event mass. Contacts have mass $N^{-1+o(1)}$ and locally deficient windows have total marginal mass $N^{-1+o(1)}$. This proves (T19.42). Pointwise,

$$0 \leq \widehat{W} - \widetilde{W} \leq 2\Xi^{\mathrm{bad}} + \Lambda^{\mathrm{loc}}, \qquad \{\widehat{W} \neq \widetilde{W}\} \subseteq \{\Xi^{\mathrm{bad}} + \Lambda^{\mathrm{loc}} > 0\}.$$

Markov's inequality gives (T19.43). Two retained starts cannot be closer than L by incompatibility, cannot be at distance L by the filter, and cannot have a nonzero pair kernel by its definition. The means stay in a fixed compact subset of $(0, \infty)$, so Lemma T19.13 transfers the criterion. Use (T19.24) and (T19.29) for the final equivalence. □

Pairwise directness here is a property of the retained original constraint spaces in a given realization. The filtered indicators themselves depend on the absence of all neighboring events. It does not make these indicators independent, or exclude primitive dependencies among three or more original spaces. The stopped expansion supplies the further information needed for their distribution.

T19.8 The cost of forcing private prime signs

A small distance between distributions need not be visible in the positive cost of a particular conditional coupling. Here this distinction can be quantified within the arithmetic model. We use $\kappa(n) = P^+(\mathrm{sf}(n))$, $P^+(1) = 1$.

To state its precise logarithmic scale, let $H_X = \log X$ and use the following definitions from [22, Proposition 4.2 and Appendix F.3]. For $v \geq e$, let $u(v) \geq 1$ solve $e^{u(v)}/u(v) = v$ and put

$$D(v) = vu(v) - \mathrm{Ei}(u(v)), \qquad V_X = D(H_X/V_X), \qquad \nu_X = H_X/V_X.$$

The indicated positive solution is unique for large X. At the precision used below,

$$2V_X = \sqrt{2H_X(\log H_X + \log\log H_X - \log 2 - 2)} + o(\nu_X), \qquad \log\log X = o(\nu_X). \tag{T19.44}$$

Lemma T19.15 (Reciprocal odd-prime pivots)**.** *As $X \to \infty$,*

$$\sum_{n \leq X} \frac{1}{\kappa(n)} = X \exp\{-2V_X + o(\nu_X)\}. \tag{T19.45}$$

The sum over $N \leq n < 2N$ is $N \exp\{-2V_N + o(\nu_N)\}$.

Proof. The asymptotic of Erdős–Ivić–Pomerance [11, Theorem 1 and equation (1.3)] gives this logarithmic formula for $S^+(X) = \sum_{n \leq X} 1/P^+(n)$. Its first correction is the explicit square root in (T19.44); the later terms are $O(\sqrt{H_X \log H_X}(\log\log H_X)^2/(\log H_X)^2) = o(\nu_X)$. Since $\kappa(n) \leq P^+(n)$, this proves the lower bound.

For the upper bound, write uniquely $n = a^2 v$ with v squarefree and use positivity to obtain

$$\sum_{n \leq X} \frac{1}{\kappa(n)} \leq \sum_{a \leq \sqrt{X}} S^+(X/a^2).$$

The terms $a > e^{3V_X}$ are $O(Xe^{-3V_X})$. On the remaining range $\log(X/a^2) = H_X - O(V_X)$ and $\nu_{X/a^2} \sim \nu_X$. The explicit function $G(H) = \sqrt{2H(\log H + \log\log H - \log 2 - 2)}$ has $G'(H) = o(1)$ uniformly there. The stated classical remainder, without differentiation, is uniformly $o(\nu_X)$. Thus

$$S^+(X/a^2) \leq X \exp\{-2V_X + o(\nu_X)\} a^{-2+o(1)}.$$

These weights have bounded sum. This also explains the square-part reduction following Proposition F.5 in [22].

For a dyadic lower bound, send each $n \leq N$ with $\kappa(n) > 2$ to the unique $2^j n \in [N, 2N)$. The pivot is unchanged, and each image has $O(\log N)$ preimages. The omitted integers have squarefree part 1 or 2, so number $O(\sqrt{N})$. The factor $\log N$ is absorbed in $\exp\{o(\nu_N)\}$. The upper bound follows from the sum up to $2N$, whose scales agree at this precision. □

Consider a specific forcing for the signed exact-run field. Let $E = \lceil 3V_N/\log 2 \rceil$, $B_* = L + E + 2$, and $Y = 2B_*$. Retain starts whose maximal support $\{x-1, \ldots, x+L+E\}$ has every pivot larger than Y. Let $W_{x,e,\sigma}$ be the indicator of the exact word

$$(-\sigma, \underbrace{\sigma, \ldots, \sigma}_{L+e}, -\sigma), \qquad 0 \leq e \leq E, \quad \sigma \in \{-1, 1\}.$$

Its mean is $\lambda_{x,e,\sigma} = p2^{-e-2}$. For a source $a = (x, e, \sigma)$, fix every nonpivot prime sign and set each private pivot of its actual word to make that word occur. The resulting full field W^a has exactly $\mathcal{L}(W \mid W_a = 1)$: conditional on the nonpivots, its raw word bits were independent and fair and its prescribed probability was constant. This is the private-pivot conditional coupling of [22, Proposition F.7]. Its nonnegative coordinate cost is

$$\mathcal{E}_N = \sum_a \lambda_a \mathbb{E}\|W - (W^a - \boldsymbol{e}_a)\|_1.$$

Proposition T19.16 (Exact logarithmic cost of this forcing)**.** *For this specified signed forcing,*

$$\mathcal{E}_N = \exp\{-2V_N + o(\nu_N)\}. \tag{T19.46}$$

Consequently this positive coordinate-by-coordinate cost is not $O(N^{-\alpha})$ for any fixed $\alpha > 0$.

Proof. A target can change only if it contains a multiple of a source pivot. Summing multiples, then reciprocals, bounds the number of directed source–target possibilities by $O(NB_*^2 \sum_{n\le 3N} 1/\kappa(n) + NB_*^2)$. The finite Palm ledger [22, Lemma F.2 and equation (F.6)] therefore gives

$$\mathcal{E}_N \ll Np^2 + p^2 \left\{ NB_* + NB_*^2 \sum_{n\le 3N} \frac{1}{\kappa(n)} + R_2^{\mathrm{val}}(N, B_*) \right\}.$$

Here R_2^{val} is the sum of $2^{\rho^{\mathrm{val}}} - 1$ over separated maximal supports. The full-value estimate [22, Proposition 3.24] gives $R_2^{\mathrm{val}}(N, B_*) \le N^{5/3+o(1)}$, since $B_* = \log_2 N + o(\log N)$. The type weights sum to at most one after the common factor p has been removed. Lemma T19.15 proves the upper bound.

For a lower bound use only the type $e = 0, \sigma = 1$, of mean $p_0 = p/4$, and restrict both maximal supports to pivots larger than $Y_0 = (\log N)^5$. Rankin's inequality with $s = 9/10$ gives

$$\#\{n \le 3N : \kappa(n) \le Y_0\} \le (3N)^{9/10} \zeta(9/5) \prod_{\ell \le Y_0} (1 + \ell^{-9/10}) = N^{9/10+o(1)}.$$

Thus only $N^{9/10+o(1)}$ starts are excluded. For each source choose $q = \kappa(x-1) \le e^{3V_N}$ and count targets with $q \mid y - 1$. The dyadic reciprocal estimate shows that these pairs number $N^2 \exp\{-2V_N + o(\nu_N)\}$.

Deleting excluded sources or targets costs at most $N^{19/10+o(1)}$ pairs; the tail $q > e^{3V_N}$ costs $O(N^2 e^{-3V_N})$, and nearby pairs cost $O(NB_*)$. The condition $q^2 \mid y-1$ removes at most a proportion Y_0^{-1} of the main count plus $O(N)$. For any other source pivot q' and target offset, the Chinese remainder theorem counts at most $N/(qq') + 1$ targets. All these extra intersections cost at most $O(B_*^2/Y_0)$ times the main count plus $O(NB_*^2)$. Finally the full-value host estimate [22, Proposition 3.7] excludes at most $N^{3/2+o(1)}$ pairs with a nonzero raw relation. Every removed term is negligible.

For each remaining pair the two raw word vectors U, V are independent and uniform, while the forcing changes only the first target bit, by the fair bit $U_0 + b_0$. The prescribed target word and its first-bit translate are disjoint, each of probability p_0. Therefore $\mathbb{E}|W_{y,0,+} - W^a_{y,0,+}| = p_0$ exactly. Multiplication by the source weight p_0 and summation over the surviving pairs prove the lower bound in (T19.46). □

The two original words in each lower-bound pair are independent. Their positive contribution is caused by the chosen forcing. It is therefore not a lower bound for their dependence, for total variation, or for the scalar cost after summing all changed coordinates with their signs.

There is also an obstruction that does not depend on the choice of an arithmetic anchor. Let the maximal support have length $B_* = O_C(\log N)$, and retain starts with at most one pivot at most $Y = 2B_*$. For $\iota \in \{0, 1\}$, use the unsigned word of length $B_\iota = L + 1 + \iota$: the start event when $\iota = 0$, and exact excess zero when $\iota = 1$. Its mean on this retained set is $q_\iota = p2^{-\iota}$. Choose an anchor as any deterministic function of the integer arguments and their factorizations; if there is a pivot at most Y in the actual word, it must be the anchor. Preserve its value and force the other private pivots to impose the word up to a global sign. All other prime signs are left fixed. Let

$$\mathcal{E}_{\iota,N} = \sum_x q_\iota \sum_{y \ne x} \mathbb{E}|W_{y,\iota} - W^{x,\iota}_{y,\iota}|$$

be the same-type, off-site part of its coordinate cost.

Proposition T19.17 (An obstruction for every such anchor)**.** *For every fixed $0 < a < 1/10$, uniformly over these arithmetic anchor rules,*

$$\mathcal{E}_{\iota,N} \geq (1 - o_{C,a}(1))\, 2^{-2\iota} \lambda_N^2 N^{-a}. \tag{T19.47}$$

The same assertion with an additional factor $1/2$ holds if, instead, the orientation is any fair randomized function of the source word alone and only its private pivots are changed.

Proof. Apply Matomäki–Radziwiłł [16, Corollary 6] with the fixed parameter $u = 2/a$ and interval length $\psi(t) = \lfloor \log t/(2 \log 2) \rfloor$. For all but $o_a(N)$ sources, the actual word contains at least two ordinary N^a-smooth integers. Indeed their number is asymptotic to $\varrho(2/a)\psi(t) \to \infty$ in almost every such interval, and $t^{a/2} \leq N^a$ for large N. The passage from natural density to $[N, 2N)$ leaves $o_a(N)$ exceptions.

Restrict again to maximal supports whose every pivot exceeds $Y_0 = (\log N)^5$. The preceding Rankin bound loses only $N^{9/10+o(1)}$ starts. At most one of the two small pivots can be the anchor, so at each of the remaining $(1 - o_a(1))N$ sources choose a forced pivot $Y_0 < q_x \leq N^a$. There are at least $(1 - o_a(1))N^{2-a}$ pairs with $q_x \mid y - 1$. The clean-pair deletions from the preceding proof apply: excluded targets cost $N^{19/10+o(1)} = o(N^{2-a})$; nearby targets, valuations at least two, and other forced pivots cost $o(N^{2-a})$ as well. Full raw dependence has at most $N^{3/2+o(1)}$ hosts.

For a surviving pair, U, V are independent uniform vectors in $\mathbb{F}_2^{B_\iota}$. If the selected source index is i and the anchor is $j \neq i$, the flip is $U_i + U_j + b_i + b_j$, a fair bit independent of V. The allowed relative word is the two-point coset $\{b, b + \mathbf{1}\}$, disjoint from its translate by the first coordinate vector. The expected target indicator change is therefore exactly q_ι. Weighting by the source mean gives $q_\iota^2(1 - o(1))N^{2-a}$, proving (T19.47).

For the final assertion, write the prescribed source as $b + g_x(U, \xi)\mathbf{1}$, where ξ is independent auxiliary randomness and g_x is fair for uniform U. This guarantees the exact conditional source law even after all nonpivot bits are fixed. For two distinct small-pivot indices i, j, the flips satisfy

$$\delta_i + \delta_j = U_i + U_j + b_i + b_j.$$

The right side is fair, so $\mathbb{P}(\delta_i = 1) + \mathbb{P}(\delta_j = 1) \geq 1/2$. Choose, before drawing signs, an index with flip probability at least $1/4$. Repeat the clean-pair count, now excluding every other source pivot from the target. Since V is independent of (U, ξ), each target change has expectation at least $2q_\iota/4$. This gives the additional factor $1/2$. □

For any proposed power $\alpha > 0$, choose a fixed $a < \min(\alpha, 1/10)$ in (T19.47); then $N^\alpha \mathcal{E}_{\iota,N} \to \infty$. The quantifier fixes a before N and invokes no uniform short-interval smoothness estimate with a growing parameter. The result concerns only forcings that change the indicated private pivots. An orientation using the outside environment, a different full conditional coupling, or a signed scalar comparison is not covered.

T20 Offset geometry and residual relation spaces

The group budget used in the main proof keeps the product of primes sharing an offset. Two further geometric consequences explain what such products can force. A line of offsets has a bounded total prime product; a large product on three offsets gives a plane involving all three windows. These statements also give a more detailed description of the relation space after a covering set of coordinates has been removed.

Throughout this section $B = L + 1$, $I_N = [N, 2N) \cap \mathbb{Z}$, and $H = C_0 B$ for a fixed $C_0 > 1$. We consider three distinct regular windows with starts $X = (x_1, x_2, x_3) \in I_N^3$ and pairwise direct relative spaces. Their offsets lie in $\mathcal{O} = \{-1, 0, \ldots, L - 1\}$, whose diameter is less than B. Every prime $\ell > H$ having odd valuation in all three windows has a unique offset $u_\ell \in \mathcal{O}^3$. Such primes

are called shared in the statements below. No independence of the three-window system is assumed. Whenever a primitive mass is mentioned, it is the sum of the original affine joint probabilities over ordered primitive triples. Each of these probabilities is at most p^2.

T20.1 Whole prime products on lines and planes

Lemma T20.1 (The prime product on an offset line)**.** *Let $\mathcal{L}$ be a line in offset space, and let $\mathcal{Q}_{\mathcal{L}}$ be the shared primes whose offsets lie on it. If their distinct offsets consist of one point, then*

$$\prod_{\ell\in\mathcal{Q}_{\mathcal{L}}} \ell \le 3N.$$

If there are at least two distinct offsets, the line has a primitive integer direction d and

$$\prod_{\ell\in\mathcal{Q}_{\mathcal{L}}} \ell \le 6N\|d\|_\infty \le 6NB. \tag{T20.1}$$

For $Y \ge H$, the number of distinct offsets on this line supplied by primes greater than Y is at most

$$M_Y = 1 + \frac{\log(6NB)}{\log Y} = O_{C_0,C}(B/\log B) = o(B).$$

Proof. At a single offset the product of the distinct primes divides each coordinate of $X+u$, all of which are at most $3N$ for large N. Otherwise fix one occurring offset u_0. Every integer point on the line has the form u_0+td, with $t\in\mathbb{Z}$. Each of its shared primes divides the same integer vector

$$d\times(X+u_\ell) = d\times(X+u_0).$$

This vector is nonzero. If it were zero, positivity of $X+u_0$ would make all coordinates of d nonzero. Two distinct offsets would give two positive proportional value vectors, with two distinct values in each block. Their projections on any two blocks would give equal nonzero square classes of even selections. Injectivity and the direct sum of those two spaces exclude this relation.

A nonzero coordinate of the displayed cross product has absolute value at most $6N\|d\|_\infty$ and is divisible by every prime in the product. Two offsets in $\mathcal{O}^3$ give $\|d\|_\infty < B$. This proves (T20.1). Choose one prime per distinct offset and take logarithms to obtain the last assertion; the extra one covers a line carrying only one offset. □

Lemma T20.2 (An integer product tail)**.** *For fixed $s>1$ and $T\ge 1$,*

$$\sum_{\substack{a,b\ge 1\\ ab>T}} (ab)^{-s} \ll_s T^{1-s}\log(2T).$$

Proof. For $a\le T$, the inner sum is at most $C_s(T/a)^{1-s}$, giving $C_sT^{1-s}\sum_{a\le T}a^{-1}$ after multiplication by a^{-s}. For $a>T$, use the full sum $\zeta(s)$ and $\sum_{a>T}a^{-s}=O_s(T^{1-s})$. □

Let $\mathscr{P}_B$ be the family of distinct affine planes

$$A\cdot X = h, \qquad A=(u_2-u_1)\times(u_3-u_1)\neq 0, \qquad h=-A\cdot u_1, \qquad u_1,u_2,u_3\in\mathcal{O}^3.$$

There are at most B^9 such planes, and these representations satisfy $\|A\|_\infty\le 2B^2$ and $|h|\le 6B^3$.

Proposition T20.3 (A forced plane uses all three coordinates)**.** *Two distinct shared primes satisfy $\ell\ell'\le 6NB$. If three distinct shared primes have product greater than $18NB^2$, then X belongs to a plane of $\mathscr{P}_B$ whose normal satisfies $A_1A_2A_3\neq 0$. Every further shared prime $\ell>12B^3$ has*

$$A\cdot u_\ell = -h. \tag{T20.2}$$

The primitive mass of triples belonging to two distinct planes of $\mathscr{P}_B$ is $O_C(N^{-1+o(1)})$.

Proof. For two primes, either they share a value in one of two selected blocks, so their product is at most $3N$, or their occurrences are distinct in both. In the latter case their product divides $nm' - n'm$. Its vanishing gives the forbidden pair relation $nn'mm' = (nm')^2$. Since $n, m \le 3N$ and both within-block differences have absolute value less than B, its nonzero absolute value is at most $6NB$.

For three primes, their product divides

$$\det(X + u_1, X + u_2, X + u_3) = (X + u_1) \cdot ((u_2 - u_1) \times (u_3 - u_1)),$$

whose absolute value is at most $18NB^2$. Hence it vanishes. The offsets cannot lie on a line, by Lemma T20.1, including the case of repeated offsets. Thus $A \neq 0$ and $A \cdot X = -A \cdot u_1$.

Suppose, for example, that $A_1 = 0$. The other two coefficients must both be nonzero, since $A \cdot (X + u_i) = 0$ cannot have just one nonzero coefficient at a positive vector. The projections of all three value vectors onto coordinates two and three are proportional. If two projections differ, both their coordinates differ, giving a forbidden pair relation. If none differ, the two offset differences are parallel to the first axis, contradicting $A \neq 0$. The other coordinates are handled in the same way.

A further prime divides $A \cdot (X + u_\ell) = h + A \cdot u_\ell$, an integer of absolute value at most $12B^3$. A larger prime forces it to vanish. Finally, two distinct planes are disjoint or meet in a line, which has at most N integer points in I_N^3. There are at most B^{18} pairs of planes. Multiplication by the upper probability p^2 gives the mass assertion. □

Lemma T20.4 (Many planar supports force a small normal)**.** *Let $A \cdot u = -h$ be a plane of offsets, with $A \in \mathbb{Z}^3$ primitive and nonzero, and $h \in \mathbb{Z}$. Let $m_{A,Y}$ count its distinct offsets supplied by shared primes greater than $Y \ge H$. Then*

$$m_{A,Y} \le M_Y + \frac{6B^2}{\|A\|_\infty}. \tag{T20.3}$$

If the hyperedges given by these offsets require at least δL vertices to cover them, where $\delta > 0$ is fixed, then for large N,

$$\|A\|_\infty \le 24B/\delta, \qquad |h| \le 72B^2/\delta.$$

There are consequently $O_\delta(B^5)$ possible such integer planes.

Proof. The lattice $\Lambda = A^\perp \cap \mathbb{Z}^3$ has covolume $\|A\|_2$. Indeed, a lattice basis d, e can be completed by an integer vector w with $A \cdot w = 1$ to a basis of $\mathbb{Z}^3$, so $d \times e = \pm A$. Choose a shortest nonzero lattice vector d, of length ℓ. It is primitive and extends to a lattice basis. Unless the cube has at most one integer point on the plane, $\ell \le \sqrt{3}B$. Parallel lattice lines in direction d have spacing $\|A\|_2/\ell$. At most $1 + \sqrt{3}B\ell/\|A\|_2$ meet the cube, and each contains at most $1 + \sqrt{3}B/\ell$ integer points. By Lemma T20.1, each carries at most M_Y supports. It follows that

$$m_{A,Y} \le \left(1 + \frac{\sqrt{3}B\ell}{\|A\|_2}\right) \min\left(M_Y, 1 + \frac{\sqrt{3}B}{\ell}\right) \le M_Y + \frac{6B^2}{\|A\|_\infty}.$$

The omitted one-point case satisfies the same bound. The covering number is no larger than the number of supports. Since $M_Y = o(B)$ and $L = B - 1$, a covering number at least δL implies the asserted bound for A, with room in the constant. The existence of an offset gives $|h| \le 3B\|A\|_\infty$. Counting the normals and the constants gives $O_\delta(B^5)$ planes. □

T20.2 Root separation and counting on a plane

Lemma T20.5 (At most one common root)**.** *Let $y = \alpha x + \beta$, with rational $\alpha \neq 0$, pass through a positive pair of regular windows with direct relative spaces. Choose offsets distinct within each*

block. The two root lists $-i$ and $-(\beta+j)/\alpha$ have at most one common element. If these offsets are partitioned among equations with disjoint supports, discarding at most two equations leaves squarefree polynomials with pairwise disjoint root sets after substitution of the line.

Proof. A common root says $\beta+j=\alpha i$, and hence $y+j=\alpha(x+i)$. Positivity makes $\alpha>0$. Two common roots would give two distinct values on each side in the same ratio. Their even square classes are nonzero by injectivity and equal, contradicting directness. The sole possible common root belongs to at most two equations. Removing those equations removes both repeated roots within an equation and roots shared between equations. □

Proposition T20.6 (Many transverse square products on prescribed planes)**.** *Fix $\eta,\varepsilon>0$ and $K>0$. Let a family of at most B^K planes have integer equations $A_1x+A_2y+A_3z=h$, with nonzero normal and all coefficients of absolute value at most B^K. The number of primitive triples on these planes admitting at least ηB disjoint sets of values, each meeting all three windows and having square product modulo primes at most H, is $O_{C,C_0,\eta,K,\varepsilon}(N^{1+\varepsilon})$. Their primitive mass is $O(N^{-1+\varepsilon})$.*

Proof. A plane with only one nonzero coordinate coefficient cannot meet the dyadic region for large N. Otherwise permute coordinates so that $A_1A_2\neq0$. The budget of $3B$ values leaves at least $\eta B/2$ sets of size at most $D_0=\lceil 6/\eta\rceil$. Select $k+2$ of them, where k is fixed below. Their equations are

$$w_\nu^2=d_\nu P_\nu(x)Q_\nu(y)R_\nu(z),$$

with nonempty split products, disjoint root lists in each window, total degree at most D_0, and squarefree H-smooth d_ν. Their offsets and coefficients have $N^{o(1)}$ possibilities for fixed k,D_0, and $d_\nu\le\prod_{\ell\le H}\ell=N^{O_{C,C_0}(1)}$.

Fix z. The plane gives $y=\alpha x+\beta$, where $\alpha=-A_1/A_2$ and $\beta=(h-A_3z)/A_2$. Lemma T20.5 leaves k equations whose polynomials $P_\nu(x)Q_\nu(\alpha x+\beta)$ have simple disjoint roots. Their square classes are geometrically independent, since valuation at a root belonging to one equation detects that equation. They define a geometrically integral curve of degree at least 2^k in the coordinates $(x,w_1,\dots,w_k)$: its generic projection to the x-line has that degree. Its degree is also bounded above in terms of k,D_0 only.

Clearing powers of A_2 from the equations preserves the original integer coordinates w_ν. Their heights are at most N^{q_0} for fixed $q_0=q_0(C,C_0,D_0,K)$ independent of k,z and the motifs; this follows from the original three-variable products. Pila's uniform curve estimate [21, Theorem A] therefore gives $O(N^{q_0/2^k+q_0\zeta})$ points for each motif and fixed z. Choose k sufficiently large, then ζ sufficiently small, to make this exponent less than $\varepsilon/3$. Sum the $O(N)$ values of z, the $N^{o(1)}$ motifs and the B^K planes. Finally use $q(\{x,y,z\})\le p^2$ to obtain the mass bound. □

T20.3 Deleting a cover of the three-block constraints

For a separated primitive triple, let $\mathcal{H}_3$ have the $3B$ formal window positions as vertices and the three occurrences of each shared prime as a hyperedge. Repeated supports may be identified. Let D be any vertex cover of these edges, with $t=|D|$. For the relative relation space R, define $R_D=\{v\in R:v|_D=0\}$.

Lemma T20.7 (Covering cost and free components)**.** *One has*

$$\dim R\le t+\dim R_D.$$

After coordinates in D are set to zero, the equations from primes greater than H form a graph with possibly pinned vertices. If $c_3(D)$ counts its unpinned components meeting all three blocks, then

$$\dim R_D\le c_3(D)+O_{C,C_0}(B/\log B),$$

uniformly in D. The product of the values in every free component is a square modulo primes at most H.

Proof. The first statement is rank–nullity for the coordinate restriction $R \to \mathbb{F}_2^D$. Every row previously of weight three meets D, so its remaining weight is at most two. A row of weight two equates its coordinates, and a row of weight one pins its coordinate to zero; pinning propagates through the component.

A free isolated vertex was already H-defective. Otherwise a large prime incident to it would remain either an edge or a pin. There are $O(B/\log B)$ such vertices by Lemma T1.3. A nonsingleton component cannot be contained in one block because edges join distinct blocks. For a fixed pair of blocks, there are at most $\pi(H)+2$ free components meeting exactly that pair: otherwise their small-prime valuation vectors, augmented by the two block parities, have a nonzero zero-sum combination. Disjointness of the components would then give a forbidden relative pair relation. These components also number $O(B/\log B)$. All remaining free components meet the three blocks and each has one binary coordinate before the additional equations are imposed. This proves the dimension estimate. Every large-prime row meets a free component zero or two times, proving its product assertion. □

Proposition T20.8 (A six-vertex budget for the residual space)**.** *For every fixed $\varepsilon > 0$, there is a set of primitive triples of total mass $O_{C,C_0,\varepsilon}(N^{-1+\varepsilon})$ outside which every cover D simultaneously satisfies*

$$\dim R_D \le \frac{3B - |D|}{6} + O_{C,C_0}(B/\log B) + O_{C,C_0,\varepsilon}(1). \tag{T20.4}$$

For fixed $\eta > 0$, triples admitting some cover with $\dim R_D \ge \eta L$ number $O_{C,C_0,\eta,\varepsilon}(N^{3/2+\varepsilon})$. On a prescribed family of planes as in Proposition T20.6, their number is $O(N^{1+\varepsilon})$ and their primitive mass is $O(N^{-1+\varepsilon})$.

Proof. For any fixed positive type (a,b,c) with a singleton, sufficiently many disjoint square products of that type have $O(N^{1+\varepsilon})$ primitive hosts. Fix the third centre and apply the degree-one part of Lemma T1.7 to the other two, putting the singleton in its linear variable. The specialized coefficients have fixed polynomial height. The original smooth coefficients and offsets have $N^{o(1)}$ choices; summing the third centre proves the count. Multiplication by p^2 gives mass $O(N^{-1+\varepsilon})$.

Apply this exclusion to the finitely many permutations of $(1,1,1)$, $(2,1,1)$, $(3,1,1)$ and $(2,2,1)$. Outside their union, the number s of free three-block components of size at most five is bounded, for every cover D. All other three-block components use at least six surviving vertices, and the small ones use at least three. Therefore

$$6c_3(D) - 3s \le 3B - |D|.$$

Lemma T20.7 proves (T20.4).

If $\dim R_D \ge \eta L$, the same lemma supplies a positive proportion of disjoint three-block square products. Select a fixed number of bounded-size ones. Fixing one centre and using the general two-parameter estimate of Lemma T1.7 gives $N^{3/2+\varepsilon}$ hosts after the same coefficient and offset enumeration. On the prescribed planes use instead Proposition T20.6. The covers are never enumerated: their existence supplies the disjoint sets, of which only a fixed number are selected for counting. □

T21 Domain dependence and polylogarithmic arithmetic cutoffs

Two features of the arithmetic estimates have uses beyond the dyadic law theorem. Changing the interval of starting points changes the dominant rational channel. Revealing more windows

changes the set of available prime coordinates, and the resulting loss must be measured by rank, not just by the number of exceptional positions. We record these facts, then explain how to cut off large ternary supports without assuming uniformity of a fixed-order counting theorem.

T21.1 The aligned profile on a macroscopic interval

For an interval of starts A, let $\sigma(x,y)$ be the dimension of its rational even-unit code and put

$$R_{\mathrm{rat}}(A,B)=\sum_{\substack{x,y\in A\\ |x-y|>L}}(2^{\sigma(x,y)}-1).$$

The rational code is zero unless there are at least two exact units. Let $R_{\mathrm{al}}(A,B)$ and $C_{\mathrm{al}}(A,L)$ be the sums over aligned ordered separated pairs of $2^{\rho(x,y)}-1$ and $\eta(x,y)2^{\rho(x,y)}-1$, respectively. Alignment has the definition in Section 3; $B=L+1$.

Proposition T21.1 (Profiles depend on the interval)**.** *In the dyadic critical band, writing $B=3k+r$, $0\le r<3$,*

$$R_{\mathrm{rat}}(I_N,B)\sim R_{\mathrm{al}}(I_N,B)\sim\frac{3+r}{18}NB2^k,\qquad R_{\mathrm{al}}(I_N,B)-R_{\mathrm{rat}}(I_N,B)=O_{C,\varepsilon}(N^{5/4+\varepsilon}).\tag{T21.1}$$

Fix $0<\delta<1$, let $A_{M,\delta}=[\lceil M^{\delta}\rceil,M)\cap\mathbb{Z}$, and suppose $|L-\log_2 M|\le C$. For $B=2k+r$, $0\le r<2$,

$$R_{\mathrm{rat}}(A_{M,\delta},B)\sim R_{\mathrm{al}}(A_{M,\delta},B)\sim\frac{2+r}{4}MB2^k,\tag{T21.2}$$

and their difference is $O_{C,\delta,\varepsilon}(M^{4/3+\varepsilon})$. For the signed profile, write instead $L=2m+s$, $0\le s<2$. Then

$$C_{\mathrm{al}}(A_{M,\delta},L)=(M-2M^{\delta})S_{1,2}(L)+O_{C,\delta,\varepsilon}(M^{4/3+\varepsilon}),\tag{T21.3}$$

$$S_{1,2}(L)=2^{m-1}\big[(s+1)m+s^2+s+4\big]-3L+1\qquad(m\ge1),\tag{T21.4}$$

$$C_{\mathrm{al}}(A_{M,\delta},L)\sim\frac{s+1}{4}ML2^m.\tag{T21.5}$$

Proof. In the channel $1:2$, each orientation permits $M/2-M^{\delta}+O(B)$ translations of the unit parameter. For $h=y-2x\in\{-1,\dots,B-2\}$, the exact units have the size of the congruence class of h modulo two in this interval. If these two sizes are m_0,m_1, their plateau contribution is

$$\sum_{j=0}^{1}m_j(2^{m_j-1}-1)=2^{k-1}\big[(2+r)k+2r\big]-B.$$

Outside the plateau, the unit count drops linearly with the distance, so the sum of the weights is $O(2^{B/2})$. Multiplying by both orientations proves the main term in (T21.2). The height-one channel has no exact unit for separated windows; its residual cost is $M^{1+o(1)}$. All other reduced heights are at least three and contribute at most $MB^{O(1)}2^{L/3+o(L)}$.

At a fixed height, the full kernel differs from the rational code only when the normalized windows contain a defective integer at a threshold proportional to B. There are $M^{1/2+o(1)}$ such hosts, by the square–smooth decomposition in Lemma T1.3. At height two their weight is $2^{L/2+o(L)}$, so the residual costs $M^{1+o(1)}$. The proof of Lemma T1.9 applies uniformly down to $x,y\ge M^{\delta}$: its Runge distance becomes $\gg_\delta \log M/\log B$, while its redundancy remains $O(B/\log B)$. Thus the other aligned heights have the stated $M^{4/3+\varepsilon}$ total bound, including all residual relations. This proves the comparison of the two positive sums.

Retaining the predecessor sign replaces each placement factor $L+1-D$ by $g_L(D)=L-1-D$. Hence the signed contribution per translation is $S_{1,2}(L)$; the identity (3.9) gives (T21.4). The

endpoints cost $O(B^3 2^{L/2})$ and the other channels and residual kernels have the same bounds in absolute value as above. This proves (T21.3) and its equivalent.

For the dyadic assertion, the height-two channel is confined to the endpoints. The channel $2:3$ has translation count $N/3+O(B)$ over the two orientations, and three residue classes of total size $B=3k+r$. Its plateau weight is $2^{k-1}[(3+r)k+2r]-B$. The height-one residual costs $N^{1+o(1)}$; the remaining heights are at least four and cost $N^{5/4+o(1)}$. The fixed-height residual costs $N^{5/6+o(1)}$. These estimates prove (T21.1). □

The residue classes of B in the positive formulas and of L in the signed formulas are different. They cannot be interchanged. With $\lambda_M = M2^{-L}$, the last formula gives

$$p^2 C_{\mathrm{al}}(A_{M,\delta}, L) \sim \frac{s+1}{4\,2^{s/2}} \lambda_M^{3/2} L M^{-1/2}.$$

This is an aligned-pair statement. It does not assert a macroscopic analogue of Theorem 1.1. The actual mean parameter for this interval would be $|A_{M,\delta}|p$, whereas λ_M expresses the ambient critical scale.

T21.2 How exceptional vertices bound the conditional rank

Fix j old windows and let W be the square-class span of all their raw values. Let $\mathcal{P}$ contain the primes at most B and every prime larger than B dividing an old value. Put

$$\mathcal{S}(\mathcal{P}) = \{n \le 3N : \mathrm{sf}(n) \text{ is supported on } \mathcal{P}\}, \qquad t_x = |V_x \cap \mathcal{S}(\mathcal{P})|.$$

Lemma T21.2 (Conditional rank loss)**.** *Let ρ_x be the dimension of the kernel of the map from E_x to square classes modulo W. Then*

$$\rho_x \le (t_x - 1)_+.$$

Conditional on any realizable vector of the old raw signs, the probability of a new start is either zero or $p2^{\rho_x}$. It equals p if $t_x \le 1$.

Let O be the set of starts at distance at most L from an old start, and put $D_T = \#\{x \in I_N \setminus O : t_x > T\}$ for $1 \le T \le L$. Uniformly over the old signs,

$$\mathbb{E}[Z_N \mid \mathcal{F}_{\mathrm{old}}] \le \lambda_N + |O| + p2^{T-1} B |\mathcal{S}(\mathcal{P})| + D_T. \tag{T21.6}$$

For fixed j and $T = \lfloor (1-\delta)L \rfloor$, $0 < \delta < 1$, the third term is $N^{1/2-\delta+o_{C,j}(1)}$.

Proof. A nonzero element of E_x is an even nonempty selection of values. If its square class lies in W, every prime larger than B with odd valuation in a selected value must divide an old value: it cannot divide a second value of V_x. Thus every selected value lies in $\mathcal{S}(\mathcal{P})$. The kernel injects into the even-subset space on these t_x values, of dimension $(t_x-1)_+$. The conditional distribution of the new relative signs is uniform on an affine image of dimension $L-\rho_x$, proving the probability statement.

Use probability at most one on O and on $\{t_x > T\}$. On $\{t_x \le 1\}$ the probability is p; on $\{2 \le t_x \le T\}$ it is at most $p2^{T-1}$. Moreover $\#\{x : t_x \ge 2\} \le \sum_x t_x \le B|\mathcal{S}(\mathcal{P})|$. This proves (T21.6). The Euler-product bound in Lemma T19.4 gives $|\mathcal{S}(\mathcal{P})| = N^{1/2+o_{C,j}(1)}$ when j is fixed; substituting T proves the final assertion. □

For example, if $Q(\mathcal{P})$ counts pairs $n < m$ in $\mathcal{S}(\mathcal{P})$ with $m - n \le L$, double counting gives

$$D_T \le \frac{BQ(\mathcal{P})}{\binom{T+1}{2}}.$$

No independence of the two exceptional integers is assumed. The rank factor explains why merely counting dependent windows throws away useful conditional information.

T21.3 Cutting off large primes and large support groups

Use the triples of blocks and the groups G_u from Proposition T1.16. The offsets and width constants are fixed; a polynomial number in B of such patterns is allowed. Let $\mathcal{Q}$ be the set of primes greater than the diameter threshold that occur with odd valuation in all three blocks. A positive mass below means a sum of nonnegative host weights, each at most $N^{-\beta+o(1)}$, with β fixed before N and uniform $o(1)$. For an original primitive triple one can take $\beta = 2$: two of its window spaces are in direct sum.

Lemma T21.3 (A Rankin bound for offset patterns)**.** *For fixed $c, D_0 > 0$, $d \in \{2, 3\}$ and $X \geq 1$, put $s = 1 + (d-1)/c$. Then*

$$\sum_{\substack{Q \leq X \text{ squarefree} \\ P^-(Q) > B^c}} (D_0 B)^{d\omega(Q)} \leq X^s \exp\{O_{c,D_0}(B/\log B)\}. \tag{T21.7}$$

If instead the factors are arbitrary distinct integers $g > B^c$, then, for every fixed $0 < \zeta < 1$ and $s_\zeta = 1 + (d-1+\zeta)/c$,

$$\sum_{\substack{A \subset \mathbb{N} \text{ finite} \\ g > B^c \ (g \in A),\ \prod_{g \in A} g \leq X}} (D_0 B)^{d|A|} \leq X^{s_\zeta} \exp\{O_{c,d,\zeta,D_0}(B^{1-\zeta})\}, \tag{T21.8}$$

where D_0 is fixed.

Proof. Rankin's inequality bounds the first sum by $X^s \prod_{q > B^c}(1 + (D_0 B)^d q^{-s})$. Chebyshev's prime estimate and partial summation give $\sum_{q>Y} q^{-s} \ll_s Y^{1-s}/\log Y$. Since $d + c(1-s) = 1$, taking the logarithm proves (T21.7). For integer factors the analogous product is $\prod_{g > B^c}(1 + (D_0 B)^d g^{-s_\zeta})$. Bound its logarithm by the corresponding sum over integers; the exponent of B is now $1 - \zeta$. □

Proposition T21.4 (Polylogarithmic confinement)**.** *For every fixed $c > 0$ and $\varepsilon > 0$, there is a fixed integer k such that, outside a mass*

$$O_{c,\beta,\varepsilon}\left(N^{-\min\{\beta - 6/5,\, \beta - 1 - 2/c\} + \varepsilon}\right), \tag{T21.9}$$

one has

$$\#\{u : G_u > B^c\} \leq k - 1 + \left\lfloor \frac{\log N}{c \log B} \right\rfloor = o(L). \tag{T21.10}$$

The same assertion holds for $\#\{q \in \mathcal{Q} : q > B^c\}$. On a family of $B^{O(1)}$ fixed affine planes for the centers, replace $2/c$ in the exceptional mass by $1/c$. For original triples, $c > 6$ is therefore sufficient for an exceptional mass $o(LN^{-2/3})$; on these planes $c > 3$ suffices. These are sufficient thresholds, not optimality claims.

Proof. First, the fixed-k budget of Proposition T1.16 gives a mass $O(N^{6/5-\beta+\varepsilon/3})$ for $\mathcal{G}_k > N$. The identical budget holds for the product of the k largest individual primes. Indeed CRT fixes each center modulo their product, exactly as in (T1.9). In the three-prime tail, repeated offset supports cannot cause a new case: if all supports coincide the product divides one value; if exactly two coincide, or three are collinear, it divides a nonzero coordinate of $d \times (X + u)$, of size $O(NB)$. A zero cross product would give a forbidden pair relation, as in the proof of the group budget. The large product therefore forces a genuine plane. The CRT tail and the induction on fixed k apply unchanged.

Outside this first exception at most $k - 1$ factors exceed $N^{1/k}$. If the product of the remaining factors in $(B^c, N^{1/k}]$ exceeds N, multiply them in a deterministic order until their product first exceeds N. The resulting Q satisfies $N < Q \leq X = N^{1+1/k}$. For individual primes, choosing their

three offsets costs $(D_0B)^{3\omega(Q)}$ for a fixed width constant D_0. CRT gives $O(1)$ choices for each center. Equation (T21.7) bounds the weighted mass by

$$N^{-\beta+(1+1/k)(1+2/c)+o(1)}.$$

For groups, each factor has one support, costing at most $(D_0B)^3$. Their pairwise coprimality is used in CRT first; it may then be dropped in the positive sum over factor sets. Equation (T21.8) gives instead

$$N^{-\beta+(1+1/k)(1+(2+\zeta)/c)+o(1)}.$$

Choose ζ sufficiently small and then k sufficiently large, both before N, to absorb the extra exponent into ε. If the centers lie on one of the specified planes, enumerate only the two free coordinates and their offsets; the third is fixed by the plane. This uses $d = 2$ and gives $1/c$ in place of $2/c$. In the remaining configurations the product is at most N, so the number of factors greater than B^c is at most $\lfloor \log N/(c \log B)\rfloor$. □

The cutoff applies to whole support groups, even if a group contains many primes. It can therefore be imposed before a coordinate regularization. Reversing that order could merge support labels and invalidate a bound stated only for the original labels.

T22 Further uniform consequences of the arithmetic estimates

The following consequences record uniformities useful when the counting argument is applied to selected subfamilies or to conditional prime-sign systems. Their parameters and the quantities being summed are kept explicit.

T22.1 Full-support relations at all cardinalities

For $r \geq 2$, let $\mathcal{H}_{r,N}$ be the ordered lists of distinct, mutually separated regular windows for which

$$s_i \in S_{x_i} \setminus \{0\}, \qquad s_1 + \cdots + s_r = 0$$

has a solution. Minimality and affine compatibility are not required.

Proposition T22.1 (A weighted count of every full support)**.** *For fixed $s \geq 2$, $u > 0$, and $\varepsilon > 0$,*

$$\sum_{r=s}^{|G|} \frac{u^r}{r!} \sum_{\boldsymbol{x} \in \mathcal{H}_{r,N}} p^r \ll_{s,u,C,\varepsilon} N^{-2s/(2s+1)+\varepsilon}.$$

Proof. A full-support relation selects at least two true values in every window, since selections are even and nonzero. Consequently the moderate certificates and the saturation graph of Section 7 apply to these families, without any primitivity assumption.

Use the parameters in (T14.7). The total independent weight of moderate certificates is $N^{-2\beta+4/A_0+o(1)} = N^{-91/50+o(1)}$, by Lemma 7.2. If a family has no moderate certificate, every unmarked vertex has a neighbour in the saturation graph. The tilted generating function of Lemma 7.4, now with no factor for rank loss, bounds its weight beyond size s_0 by $O_u(N^{-s_0\delta/2})$. Choose a fixed $s_0 > \max(s, 5000)$. Both large-size contributions are smaller than $N^{-1+\varepsilon}$.

For each of the finitely many $s \leq r < s_0$, Proposition T1.14 gives

$$|\mathcal{H}_{r,N}| \ll_{r,C,\varepsilon} N^{r-2r/(2r+1)+\varepsilon}.$$

Multiplying by $(up)^r/r!$, and using $2r/(2r+1) \geq 2s/(2s+1)$, completes the proof. The factor $1/r!$ counts unordered sets; it is not a bound for the growth of an arithmetic constant in r. □

Lemma T22.2 (When no relation uses every block)**.** *If a family I of r regular windows has no relative relation meeting every block, then*

$$q(I) \le p \sum_{x \in I} q(I \setminus \{x\}).$$

Summed over any subfamily of ordered distinct r-tuples, the resulting upper bound is at most $rNp\,\mathbb{E}(Z_G)_{r-1}$.

Proof. The relation space is covered by the kernels of its r coordinate-block projections. Hence $2^{\rho(I)} \le \sum_x 2^{\rho(I\setminus\{x\})}$. If the full affine event is compatible, every subevent is compatible and multiplication by p^r proves the claim. If it is incompatible, the left side is zero. Summing over the omitted centre gives at most N choices, for each of the r omitted positions. □

An elementary companion to such support bounds is the following weighted triangle inequality. If (w_{xy}) is a finite symmetric matrix of nonnegative integers and $W = \sum_{x,y} w_{xy}$, then

$$\sum_{x,y,z} \min(w_{xy}, w_{xz}, w_{yz}) \le W^{3/2}.$$

Indeed, for each integer $h \ge 1$, let A_h be the symmetric zero-one matrix of entries at least h, and M_h its number of ones. Its triangle sum is $\operatorname{tr}(A_h^3) \le (\operatorname{tr}(A_h^2))^{3/2} = M_h^{3/2}$. Sum in h and use $\sum_h M_h^{3/2} \le (\sum_h M_h)^{3/2}$.

T22.2 Old square classes and a growing number of sources

Let $\mathcal{P}$ be a finite set of primes, and

$$A(X;\mathcal{P}) = \#\{n \le X : \mathrm{sf}(n) \text{ is supported on } \mathcal{P}\}.$$

Lemma T22.3 (Rankin under a logarithmic prime budget)**.** *For $1/2 < \sigma < 1$,*

$$A(X;\mathcal{P}) \le X^{\sigma}\zeta(2\sigma) \prod_{\ell \in \mathcal{P}} (1 + \ell^{-\sigma}).$$

Uniformly for $3/5 \le \sigma \le 3/4$, $Y \ge 3$, and $\sum_{\ell\in\mathcal{P}} \log \ell \le Y$,

$$\log A(X;\mathcal{P}) \le \sigma \log X + O\left(1 + \frac{Y^{1-\sigma}}{\log Y}\right).$$

Proof. Write $n = du^2$, where d is squarefree and supported on $\mathcal{P}$, and sum $(X/(du^2))^{\sigma}$. For the second assertion, the prime-counting upper bound and partial summation give $\sum_{\ell \le Y} \ell^{-\sigma} \ll Y^{1-\sigma}/\log Y$ uniformly in the stated compact interval. There are at most $Y/\log Y$ members of $\mathcal{P}$ exceeding Y; their contribution satisfies the same bound. Finally $\zeta(2\sigma)$ is uniformly bounded. □

Proposition T22.4 (Independent extensions at the critical order)**.** *Put $H = \log N$, $h = \log H$, and fix $K > 0$. For every $j \le KH/h$ source windows, there is an exceptional set of new centres of size at most D_N^*, where*

$$\log D_N^* \le \frac{2}{3}H - \frac{H \log h}{9h} + O_{K,C}(H/h), \tag{T22.1}$$

such that outside it the L new relative bits are exactly uniform and independent of all previously exposed values.

Proof. Include in $\mathcal{P}$ all primes at most B and all primes dividing one of the jB old values. Then

$$\sum_{\ell\in\mathcal{P}} \log \ell \ll_C B + jB\log(3N) \ll_{K,C} H^3/h.$$

Take $Y = C_{K,C}H^3/h$ and $\sigma = 1-h/\log Y$. Eventually $\sigma \in [3/5, 3/4]$ and $Y^{1-\sigma} = H$. Lemma T22.3 gives

$$\log A(3N;\mathcal{P}) \le H - \frac{Hh}{\log Y} + O_{K,C}(H/\log Y).$$

Since $\log Y = 3h - \log h + O_{K,C}(1)$, the right side equals the bound in (T22.1).

Exclude centres whose windows overlap an old window, or contain a value whose square class is supported on $\mathcal{P}$. This costs at most $j(2L+1) + BA(3N;\mathcal{P})$, with the additional polylogarithmic factors absorbed in the error term. Every value in a remaining window has an odd prime outside $\mathcal{P}$. Such a prime exceeds the diameter and occurs nowhere else in the window. These private coordinates form an identity minor in the new valuation matrix and are independent of all old values. This proves the exact independence assertion. □

Corollary T22.5 (A single pair at all critical orders)**.** *Let $A_N = 2\sum_{e\in E_N} q_e$. Construct ordered r-tuples by choosing the two positions of a good pair and then adding windows in position order, always outside the exceptional set of Proposition T22.4. Their total positive mass $B_{r,N}$ satisfies, uniformly for $2 \le r \le m \le KH/h$,*

$$B_{r,N} = \binom{r}{2}\lambda_N^{r-2}A_N\{1 + O_{K,C}(rD_N^*/N)\}.$$

Furthermore,

$$\sum_{r=2}^{m} \frac{2^{r-1}}{r!}\left|B_{r,N} - \binom{r}{2}\mu_N^{r-2}\alpha_N\right| = o_C(a_N).$$

Proof. For a fixed pair each added window contributes exactly p. The number of extensions is between $(N-D_N^*)^{r-2}$ and N^{r-2}; $mD_N^*/N \to 0$. The pair is the unique dependent pair in the resulting tuple, so its choice of positions is unique. This proves the first formula. The weighted branching error sums to

$$O_{K,C}\left(A_N\frac{D_N^*}{N}\sum_{r\ge2}\frac{2^{r-1}}{r!}r\binom{r}{2}\Lambda^{r-2}\right) = o(a_N),$$

where Λ bounds the means. The pair and mean estimates replace A_N by α_N, and λ_N by μ_N, at the same aggregate precision. □

This corollary controls a selected positive family. Its independence guarantee is uniform at growing orders, but it does not by itself estimate the full factorial moments at those orders.

T22.3 Exact moments of the modified count

Let E_N' be the event in Lemma T15.2, let $A = \{x \in G : J_x = 1\}$, and put $\widetilde{Z} = Z_G\mathbf{1}_{(E_N')^c}$. For $r \ge 1$, define

$$D_r(A) = \#\{I \subseteq A : |I| = r,\ I \text{ direct}\}, \qquad L_r' = \mathbb{E}[D_r(A); E_N'].$$

For $r \ge 2$, also put

$$P_r(A) = \sum_{e\in E_N} J_e\#\{U \subseteq A\setminus e : U \in \Gamma_e(r-2)\}, \qquad H_r = \mathbb{E}[P_r(A); E_N'],$$

and set $H_1 = 0$. Here Γ_e, the independent deficits F_r, G_r, and $\beta_N = \sum_e q_e$ are those used in the stopped expansion.

Proposition T22.6 (Moment decomposition and pair leakage)**.** *For every integer $r \geq 1$,*

$$\mathbb{E}\binom{\widetilde{Z}}{r} = \binom{n}{r}p^r - F_r - L'_r + \beta_N\binom{n-2}{r-2}p^{r-2} - G_r - H_r, \tag{T22.2}$$

where the last three terms are zero for $r = 1$. All four deficits F_r, G_r, L'_r, H_r are nonnegative. Moreover,

$$\sum_{r\geq 2} 2^r G_r \ll_\varepsilon N^{-1+\varepsilon}, \qquad \sum_{r\geq 2} 2^r H_r = o(a_N).$$

Proof. Outside E'_N, an active family is either direct or contains the unique active good pair and is direct after its compression. Thus

$$\binom{Z_G}{r} = D_r(A) + P_r(A) \qquad \text{on } (E'_N)^c.$$

The expectations without restriction to this event are, respectively,

$$\mathbb{E}D_r(A) = \binom{n}{r}p^r - F_r, \qquad \mathbb{E}P_r(A) = \beta_N\binom{n-2}{r-2}p^{r-2} - G_r.$$

Subtract the two contributions on E'_N to obtain (T22.2). The bound for G_r was proved in Lemma T17.1.

For each fixed r, $P_r(A) \leq C_r Z_G^r$. Hölder's inequality and the fixed moments give

$$H_r \leq C_r(\mathbb{E}Z_G^{8r})^{1/8}\mathbb{P}(E'_N)^{7/8} \ll_{r,\varepsilon} N^{-7/10+7\varepsilon/8} = o(a_N)$$

when the margin is sufficiently small. To sum all r, use directness before taking expectations:

$$\sum_{r\geq 2} 4^r H_r \leq \sum_{r\geq 2} 4^r \mathbb{E}P_r(A) \leq 16\beta_N(1+4p)^{n-2} \ll_C a_N.$$

For any fixed K, the terms $2 \leq r \leq K$ are $o(a_N)$, and the remaining 2^r-weighted sum is $O_C(2^{-K}a_N)$. Take first $N \to \infty$, then $K \to \infty$. This proves the asserted small o, without giving it an additional polynomial rate. □

The direct leakage L'_r and the deterministic deficit F_r are different quantities. The small probability of E'_N gives a bound for L'_r at each fixed order, but does not on its own control their exponentially weighted sum. The stopped expansion avoids paying that whole sum: it keeps the suffix inside a positive kernel at the first excluded prefix.

T22.4 Replicas with common small-prime signs

Fix a threshold Y. Keep the signs at primes $\ell \leq Y$ common, and give each member of a partition π an independent copy of the signs at primes $\ell > Y$. Occurrences may have repeated centres; their formal coordinates remain distinct. For the resulting valuation map let K_π be its relative kernel and let $\eta_\pi \in \{0,1\}$ denote affine compatibility.

Proposition T22.7 (The replica formula and positive propagation)**.** *For r prescribed occurrences,*

$$\mathbb{E}\prod_{A\in\pi}\mathbb{P}(J_{x_i} = 1 \ (i \in A) \mid \mathcal{F}_Y) = p^r\eta_\pi 2^{\dim K_\pi}. \tag{T22.3}$$

After any fixed number of compatible colored sources, the sum of the conditional probabilities of targets is $O_\varepsilon(N^\varepsilon)$, uniformly in Y and the coloring. Here conditioning means prescribing these

source events in the replicated space; the common small-prime signs are still averaged. Thus a positive random sum

$$R = \sum_{\boldsymbol{x}\in\mathcal{A}} w_{\boldsymbol{x}} \prod_{A\in\pi} \mathbb{P}(J_{x_i} = 1 \ (i \in A) \mid \mathcal{F}_Y),$$

with fixed tuple length, fixed partition π, and nonnegative deterministic weights bounded by a fixed power of $\log N$, satisfies for every fixed integer $q \geq 1$

$$\mathbb{E}R^q \ll_\varepsilon N^\varepsilon \mathbb{E}R, \tag{T22.4}$$

when its index set is a subfamily of all ordered tuples of the stated fixed length.

Proof. Conditionally on the common small-prime signs, the independent large-prime copies realize the product of conditional probabilities on the left of (T22.3). Fourier inversion in this larger binary space gives its right side, with its compatibility factor retained.

Merging colors adds prime-parity equations together, so a relation in the colored system is also a relation in the merged one. The projection of its kernel to the target block is contained in the corresponding merged projection. Consequently the target's colored rank loss is no larger than its uncolored rank loss. Proposition 6.2 and the proof of (6.4) apply. Targets whose centre repeats a source account for only a fixed number of choices and have probability at most one.

Expand R^q positively, retain the first replica product as a seed, and sum all further occurrences successively. Each step has the conditional bound just proved. Repeated centres are treated as above. There are only a fixed number of steps; splitting ε among them and absorbing the fixed powers of $\log N$ proves (T22.4). □

The same reasoning permits measurable gates in positive estimates. For example, suppose $\mathbb{E}R \ll_\delta N^{-\eta+\delta}$ for every $\delta > 0$, and let $\mathcal{A} \in \mathcal{F}_Y$. For an integer $k \geq 1$, interpolation gives

$$\mathbb{E}R^{1+1/k} \leq (\mathbb{E}R)^{1-1/k}(\mathbb{E}R^2)^{1/k} \ll_\delta N^\delta \mathbb{E}R.$$

Hölder's inequality with conjugate exponents $1 + 1/k$ and $k + 1$, followed by the fixed moment of order $d(k + 1)$, gives $\mathbb{E}[RZ_G^d] \ll_\delta N^{-\eta k/(k+1)+O(\delta)}$. Choose k after the desired error margin. Since the gate is at most one, for fixed d this yields

$$\mathbb{E}[R\, Z_G^d\, \mathbf{1}_\mathcal{A}] \ll_\varepsilon N^{-\eta+\varepsilon}.$$

Here (T22.4) supplies the second moment of R. The estimate is an unconditional weighted expectation; it makes no division by $\mathbb{P}(\mathcal{A})$.

T23 Fixed moments do not give a polynomial rate

This example isolates the logical role of the growing-order estimate. It is a family of probability laws, with no claimed multiplicative realization.

Proposition T23.1 (A perturbation invisible at each fixed order)**.** *For $0 \leq a \leq 1$ put*

$$Q_a = \Pi_1 + \frac{a}{2}\Pi_1 * \Delta^{*2}.$$

For every integer $m \geq 3$, define

$$\epsilon_{m,a} = \frac{(1 - a/2)e^{-1}}{2m!}, \qquad P_{m,a} = Q_a + \epsilon_{m,a}\Delta^{*m}.$$

Then $P_{m,a}$ is a probability measure satisfying $\frac{1}{2}Q_a \leq P_{m,a} \leq \frac{3}{2}Q_a$ pointwise. Its factorial moments agree exactly with those of Q_a at every order other than m, and

$$d_{\mathrm{TV}}(P_{m,a}, Q_a) = \frac{(1 - a/2)2^{m-2}}{e\, m!}.$$

For every fixed $t \geq 0$, its exponential moment $\mathbb{E}_{P_{m,a}} e^{tX}$ is bounded uniformly in m, a.

Proof. The second Charlier polynomial at mean one is $k^2 - 3k + 1$, whose minimum on $\mathbb{N}$ is -1. Thus

$$Q_a\{k\} \geq (1 - a/2)e^{-1}/k!.$$

For $0 \leq k \leq m$,

$$\frac{\epsilon_{m,a}\binom{m}{k}}{Q_a\{k\}} \leq \frac{1}{2(m-k)!} \leq \frac{1}{2}.$$

Outside this range the perturbation is zero. Positivity and the two pointwise bounds follow. The perturbation has total mass zero, so $P_{m,a}$ has mass one.

The probability generating function of Δ^{*m} is $(z-1)^m$. Its derivatives at $z = 1$ vanish at every order other than m, where the derivative is $m!$. This proves the factorial-moment statement. Its ℓ^1 norm is 2^m, yielding the exact total variation distance. Finally,

$$\mathbb{E}_{P_{m,a}} e^{tX} \leq \frac{3}{2} \exp(e^t - 1) \left(1 + \frac{1}{2}(e^t - 1)^2\right),$$

which is independent of m, a. □

Take $m_N = \lfloor \sqrt{\log N} \rfloor$ and $a_N^\circ = (\log N) N^{-2/3}$, for N sufficiently large. At every fixed order r, the factorial moments of P_{m_N, a_N° eventually equal $1 + \binom{r}{2} a_N^\circ$ exactly. Nevertheless, Stirling's formula gives

$$\log d_{\mathrm{TV}}(P_{m_N, a_N^\circ}, Q_{a_N^\circ}) = -\tfrac{1}{2}\sqrt{\log N} \log\log N + O(\sqrt{\log N}).$$

This distance exceeds $N^{-\gamma}$ for every fixed $\gamma > 0$, eventually. Fixed-order information, even with the displayed exponential-moment control, therefore cannot replace the uniform estimate on the boundary of the stopped expansion.

Statement on AI assistance

ChatGPT (GPT-6 Astra) was used to organize the research material, develop and rewrite mathematical arguments, inspect references, and prepare both manuscripts and their sources. Parallel AI-assisted reviews and exact finite computations examined selected identities, constants, and proof dependencies. These checks do not constitute independent human verification or formal certification of the asymptotic theorems. The author remains responsible for the claims and any submission.

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